Z + b-tagged jet in Summer22: three taggers at the Loose working point

ParticleNet, DeepJet and RobustParT, each with its own BTV SFs. All POG corrections, including the muon changes of 8 November.

8 November 2026

Contents

Selection: Z→ee or Z→μμ + at least 1 jet, leading jet b-tagged at Loose. Every tagger has the same 19 plots in the same order.

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What goes into these plots

Corrections and SFs (POG recipes)

Jets: JEC, JER, jet veto map, type-1 PUPPI MET. Leptons: scale and smearing (muons: the Run 3 replacement of Rochester), ID, isolation and reco SFs, trigger SFs. Events: golden JSON, pileup weights, MET filters. DY Z pT weights off.

New since the 237-page deck (8 November)

Muon trigger SF from both muons (IsoMu24 fires if either muon fires) and both muons pT > 26 GeV: Z→μμ data/MC 0.943, Z→ee 0.949. MET shown without the MET cut up to 200 GeV, zoomed plots, same legend order in both channels.

tagger	Loose	Medium	Tight	SFs (BTV, fixed WP)
PNet	0.047	0.245	0.6734	particleNet_comb (b, c), particleNet_light
DeepJet	0.0583	0.3086	0.7183	deepJet_comb (b, c), deepJet_light
RobustParT	0.0849	0.4319	0.8482	robustParticleTransformer_comb (b, c), robustParticleTransformer_light

B-score cut on the leading jet. Only Loose is shown in this deck. In MC the tagged jet gets the SF for its true flavour.

Yields and tagging SFs for the three taggers

Flavour shares are of all MC (DY + tt + VV). Tagged = fraction of Z + jet events with a tagged leading jet. Tag ratio = tagged in data / tagged in MC (the overall normalisation cancels).

	data/MC	DY+b	DY+c	DY+light	tagged MC	tagged data	tag ratio
no tag, Z→ee	0.949	4%	8%	87%	–	–	–
no tag, Z→μμ	0.943	4%	8%	87%	–	–	–
PNet, Z→ee	0.947 ± 0.031	21%	23%	54%	16.9%	16.9%	0.998 ± 0.031
PNet, Z→μμ	0.925 ± 0.021	22%	21%	55%	17.9%	17.6%	0.980 ± 0.022
DeepJet, Z→ee	0.926 ± 0.029	18%	22%	58%	18.9%	18.4%	0.976 ± 0.029
DeepJet, Z→μμ	0.924 ± 0.021	20%	21%	57%	19.3%	18.9%	0.979 ± 0.022
RobustParT, Z→ee	0.962 ± 0.031	20%	22%	55%	17.2%	17.4%	1.013 ± 0.031
RobustParT, Z→μμ	0.981 ± 0.023	22%	21%	54%	17.2%	17.9%	1.040 ± 0.024

mean SF of the tagged jet (DY MC)	b jets	c jets	light jets	also c-tag L, MC	also c-tag L, data μμ
PNet	0.95	0.94	1.21	77%	83%
DeepJet	0.95	0.95	1.19	80%	83%
RobustParT	0.96	0.96	1.18	78%	84%

Mean SF weighted by the number of tagged DY MC jets of each flavour, both channels.

PNet, Loose b-tag

Leading jet PNet B > 0.047, with the particleNet BTV SFs

	data/MC	DY+b	DY+c	DY+light	tt	tagged MC	tagged data
Z→ee	0.947 ± 0.031	21%	23%	54%	2%	16.9%	16.9%
Z→μμ	0.925 ± 0.021	22%	21%	55%	2%	17.9%	17.6%

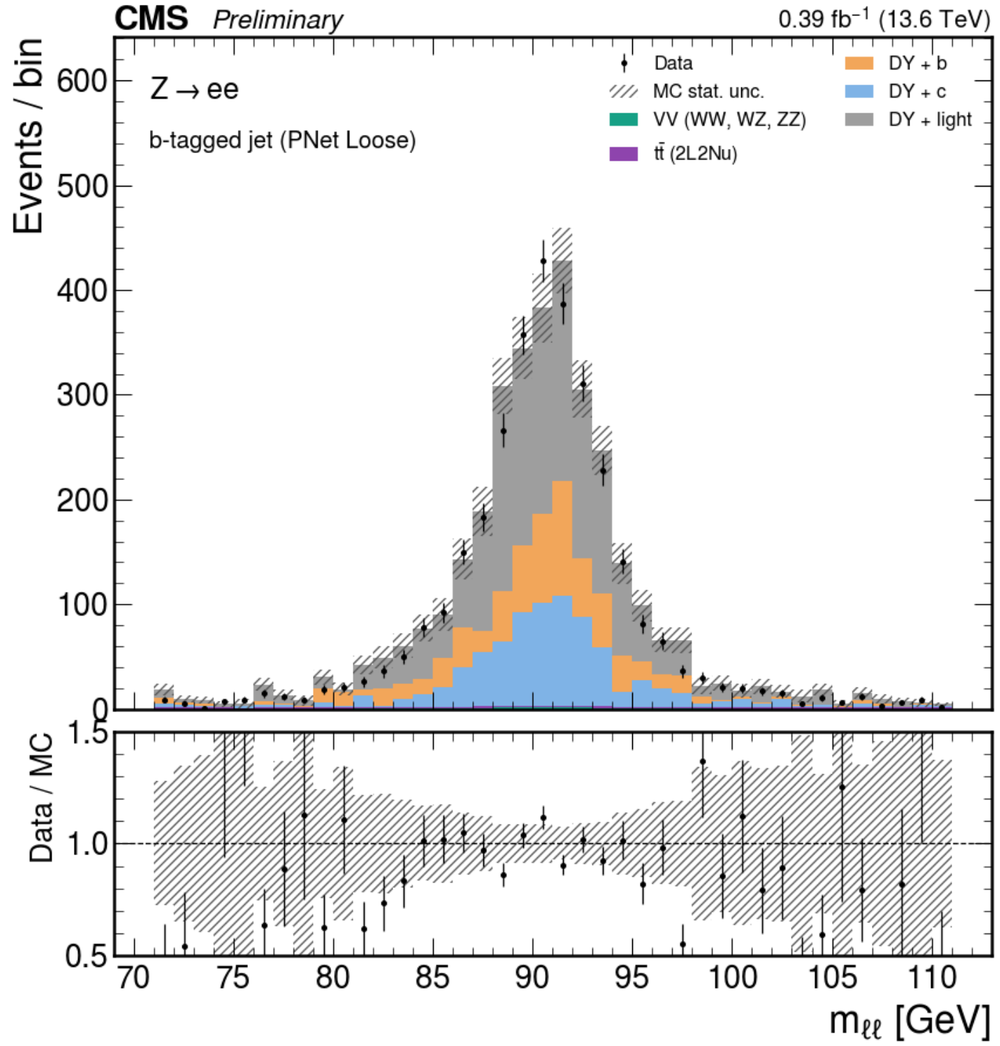
Flavour shares are of all MC. Tagged = fraction of Z + jet events with a tagged leading jet. Errors are statistical only.

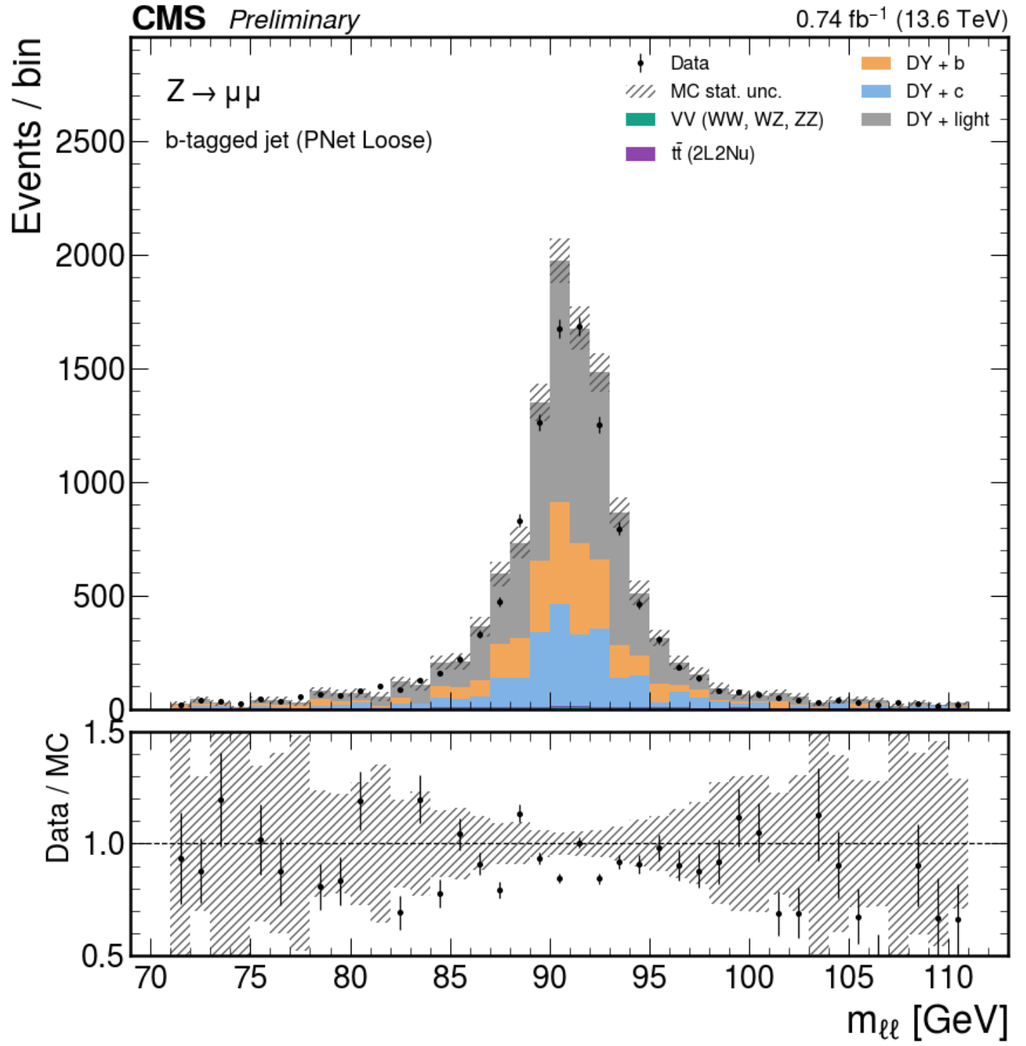
19 plots follow. Left Z→ee, right Z→μμ.

PNet Loose b-tag (PNet B > 0.047)

Dilepton mass

71 < m(ℓℓ) < 111 GeV, linear scale.

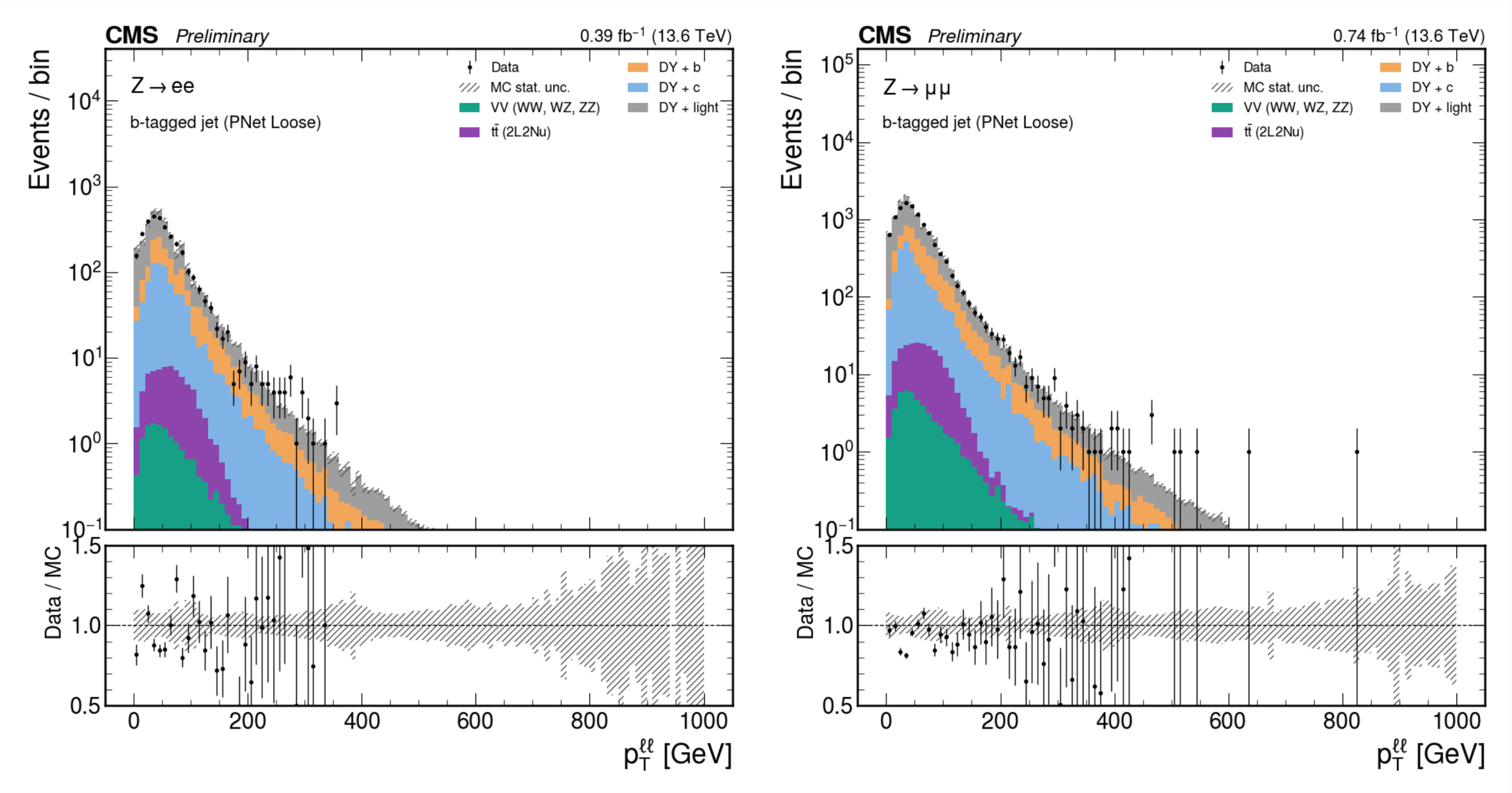




PNet Loose b-tag (PNet B > 0.047)

Z pT

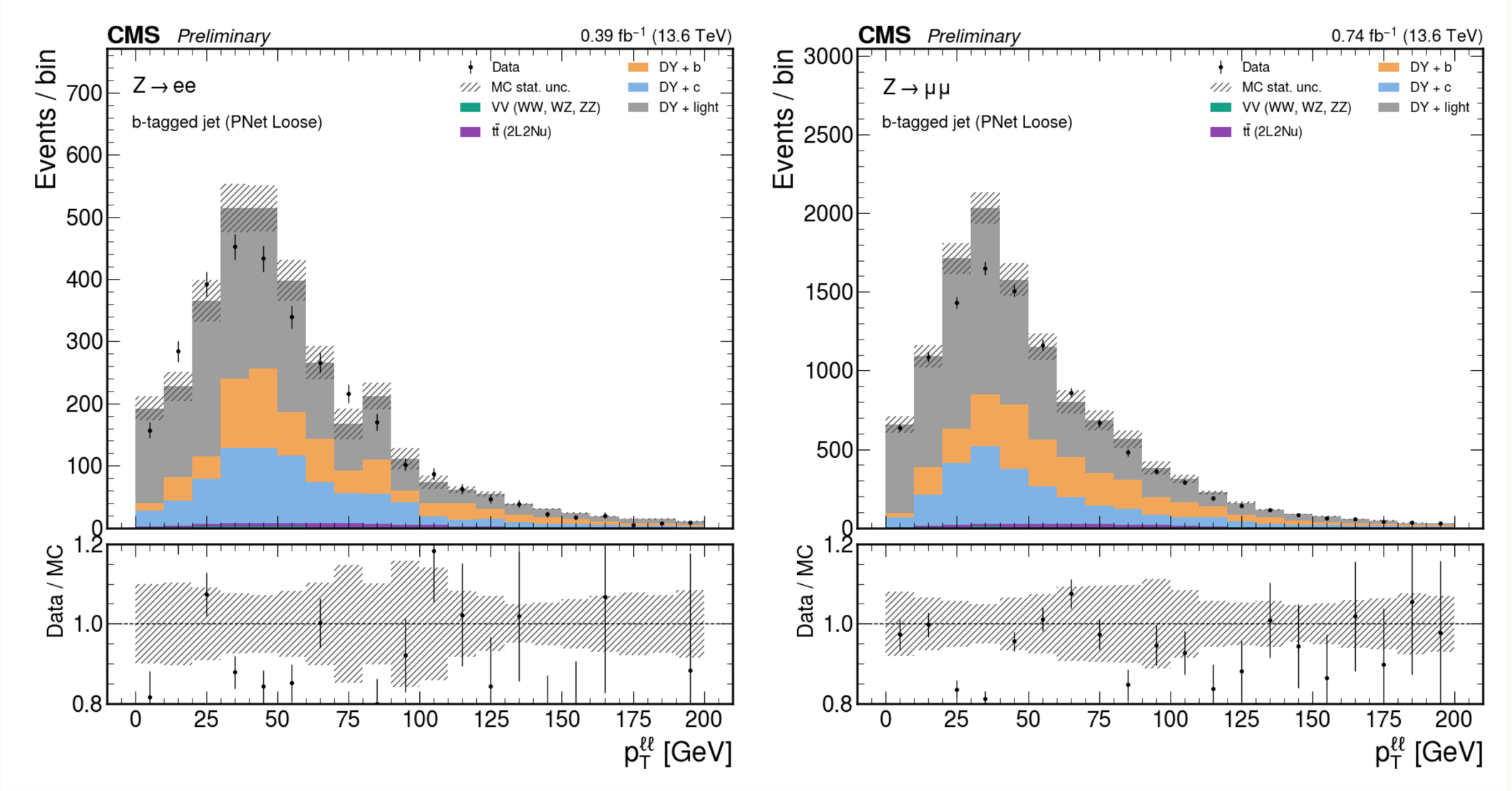
pT of the lepton pair.



PNet Loose b-tag (PNet B > 0.047)

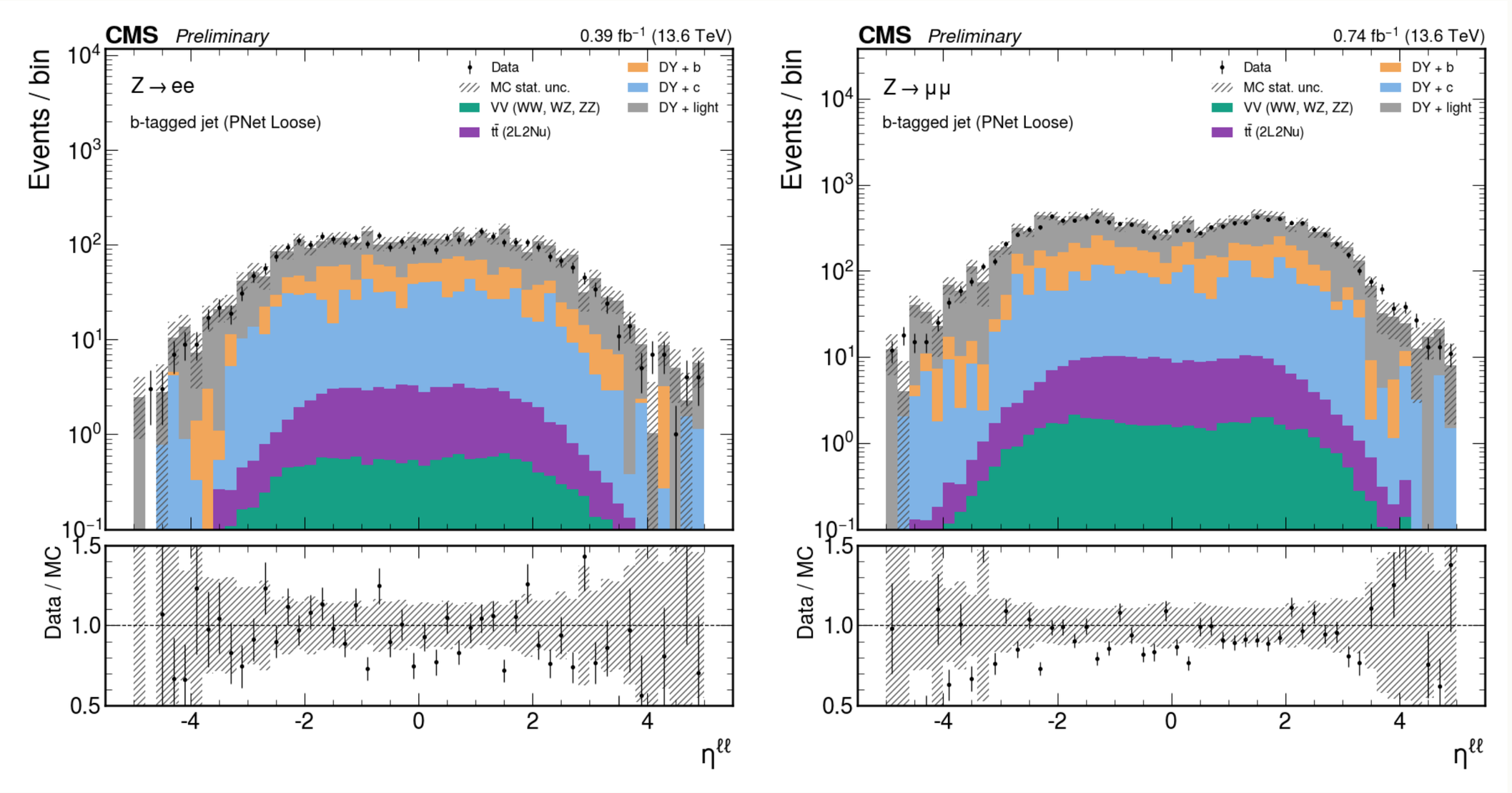
Z pT, 0–200 GeV

Zoom: linear scale, ratio 0.8–1.2.



PNet Loose b-tag (PNet B > 0.047)

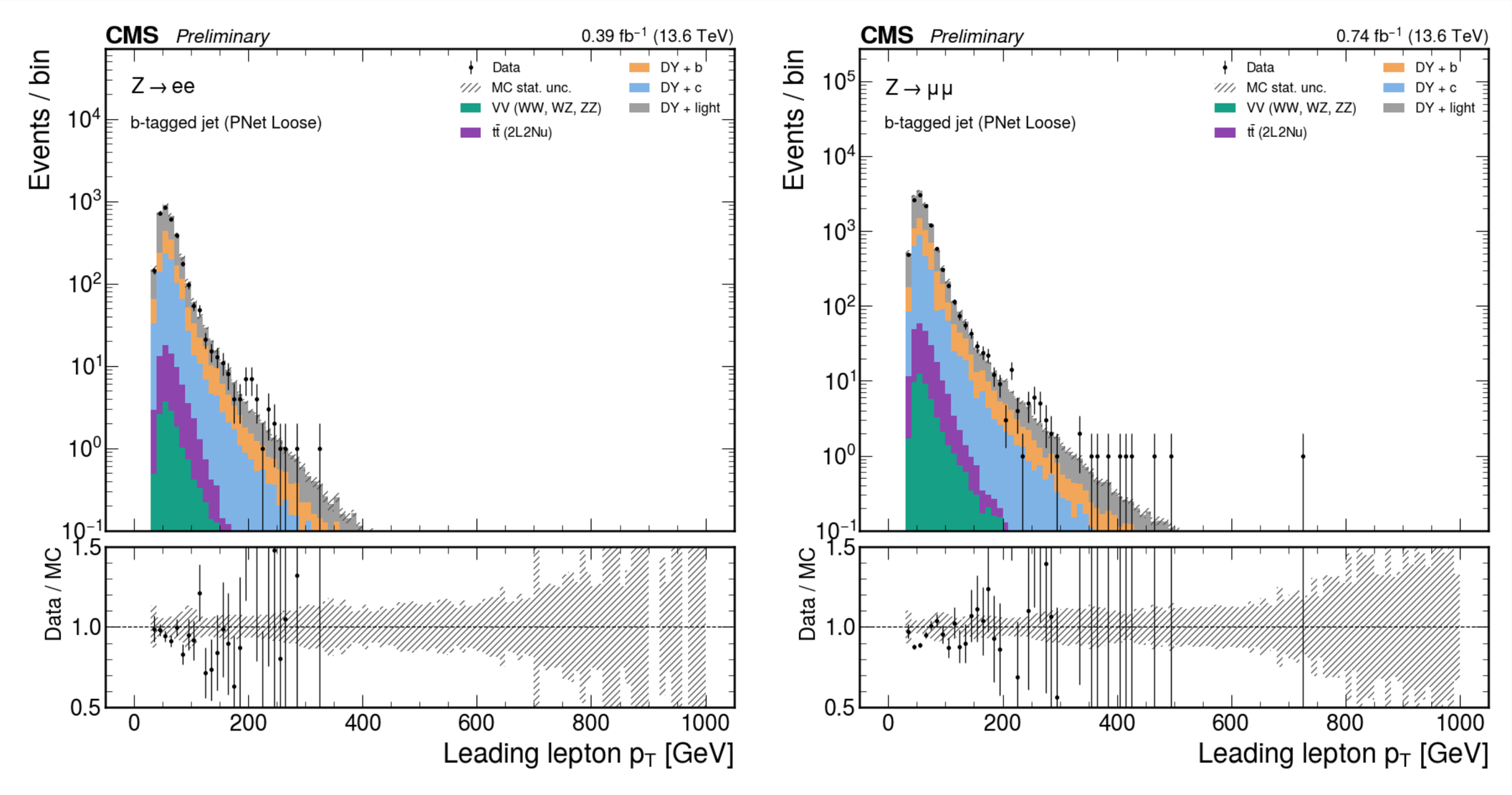
Z η



PNet Loose b-tag (PNet B > 0.047)

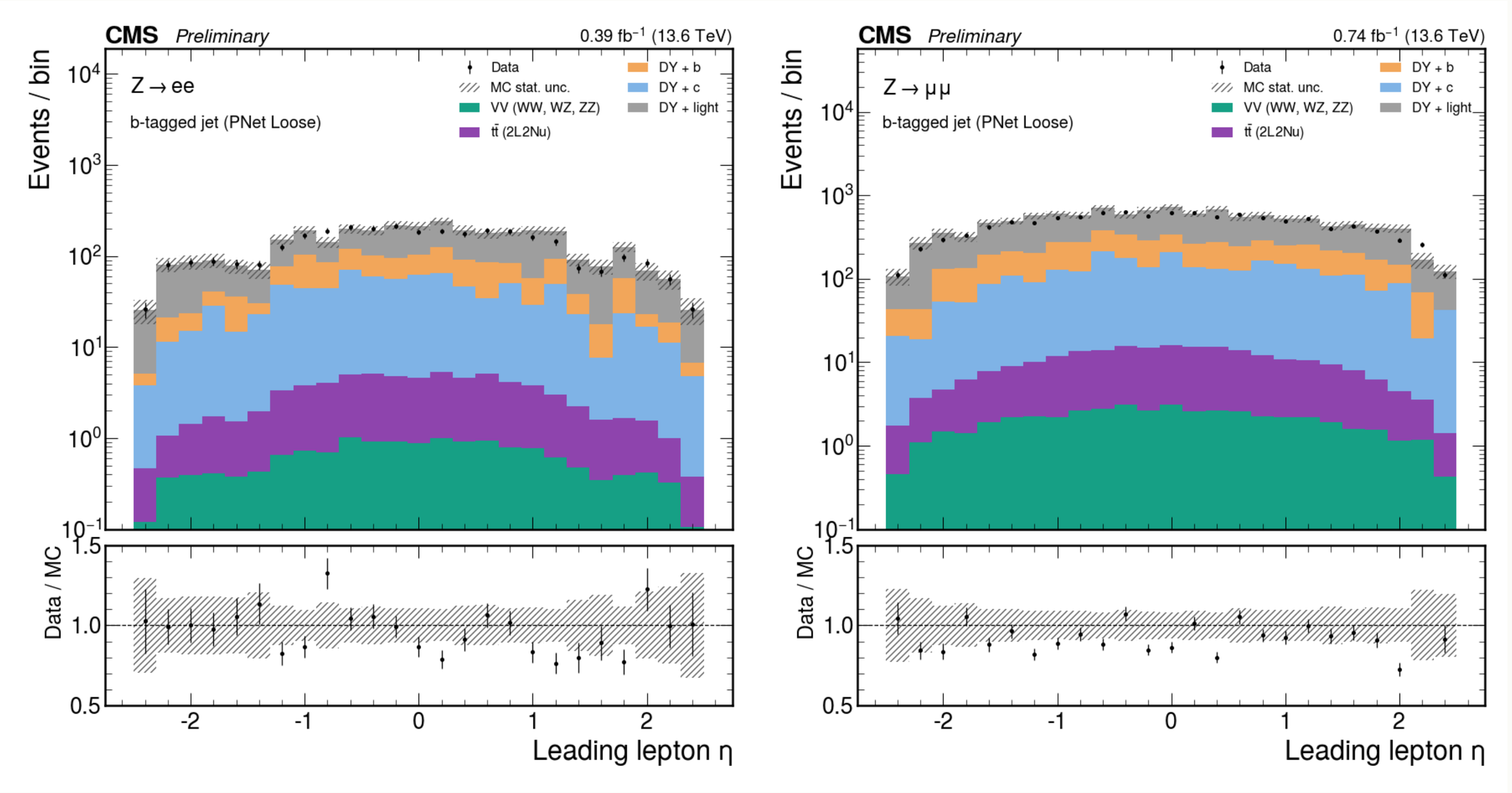
Leading-lepton pT

Leading lepton pT > 35 GeV.



PNet Loose b-tag (PNet B > 0.047)

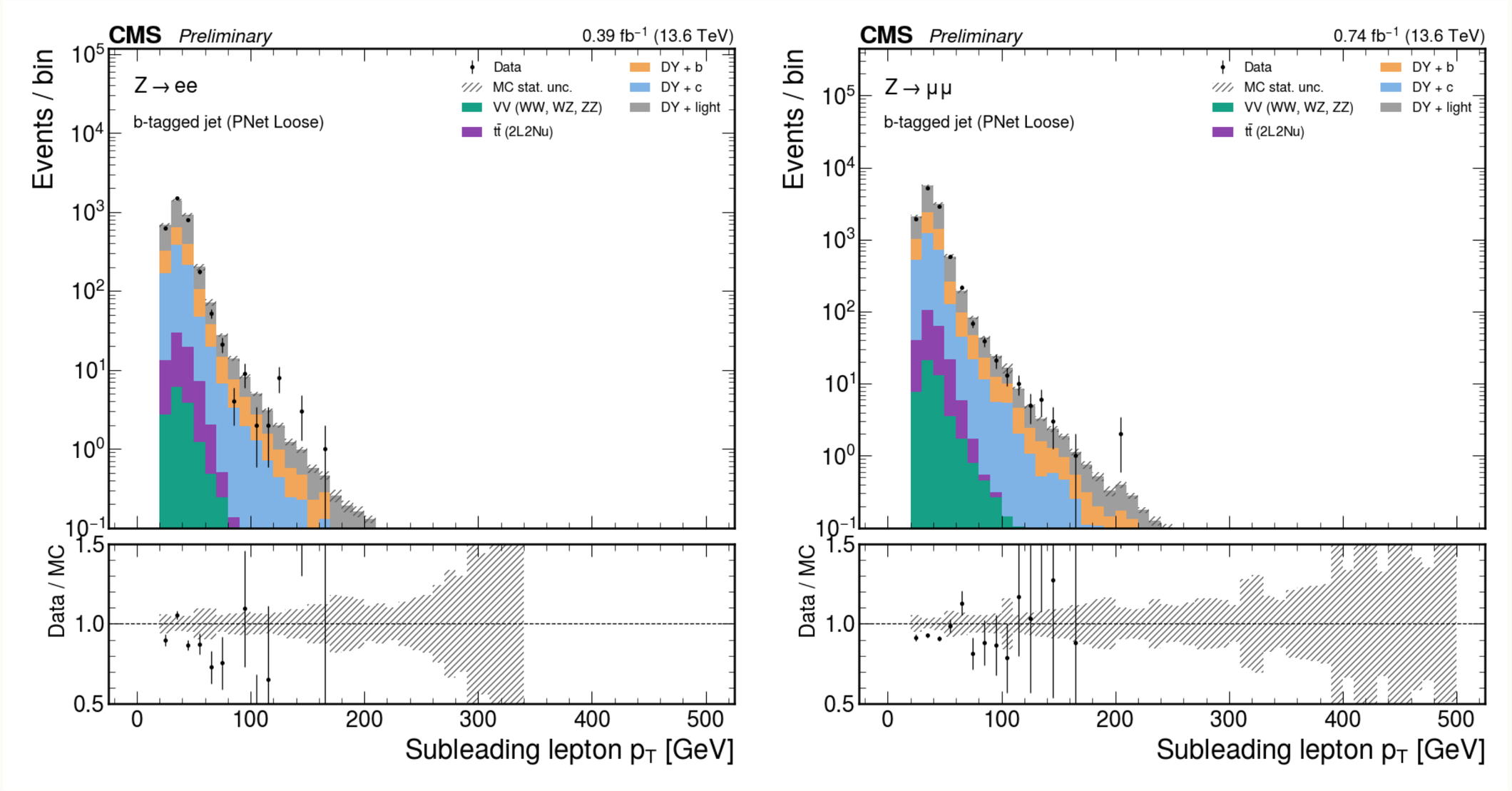
Leading-lepton η



PNet Loose b-tag (PNet B > 0.047)

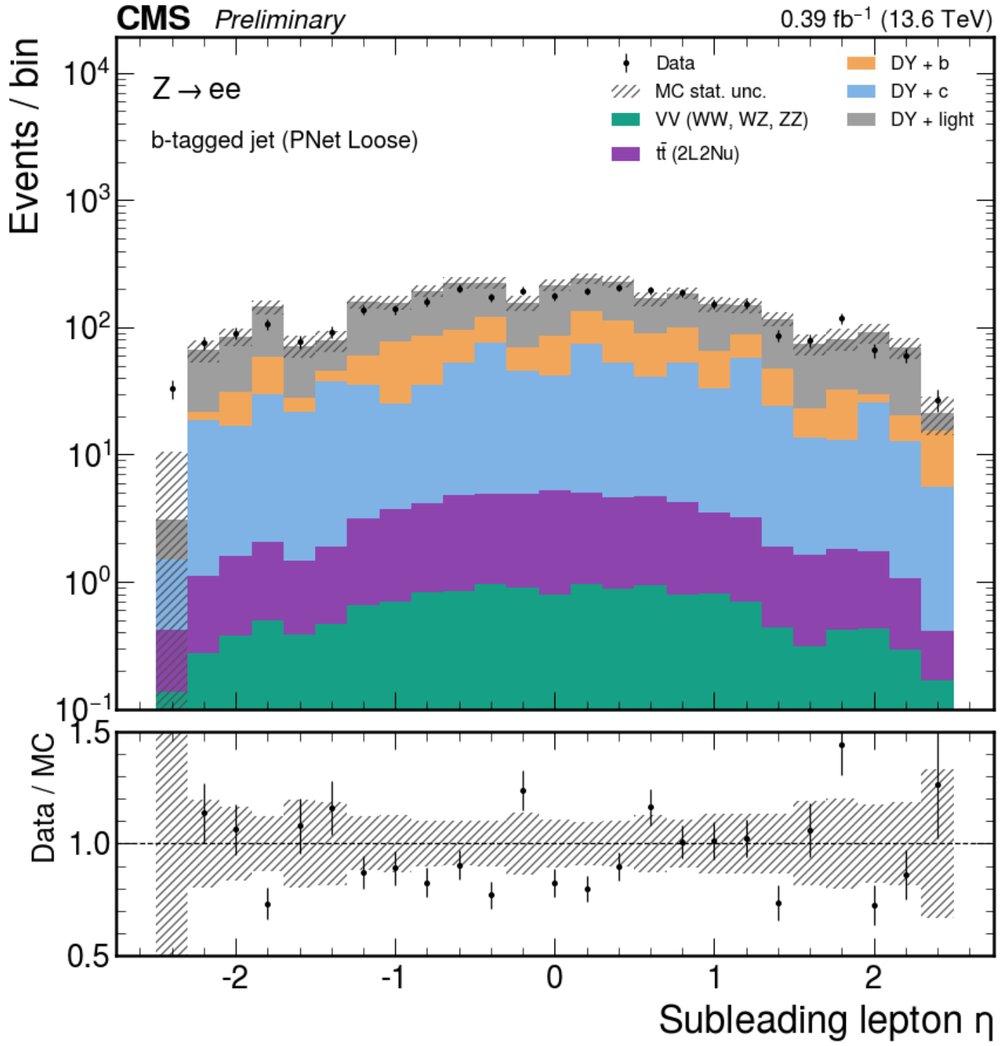
Second-lepton pT

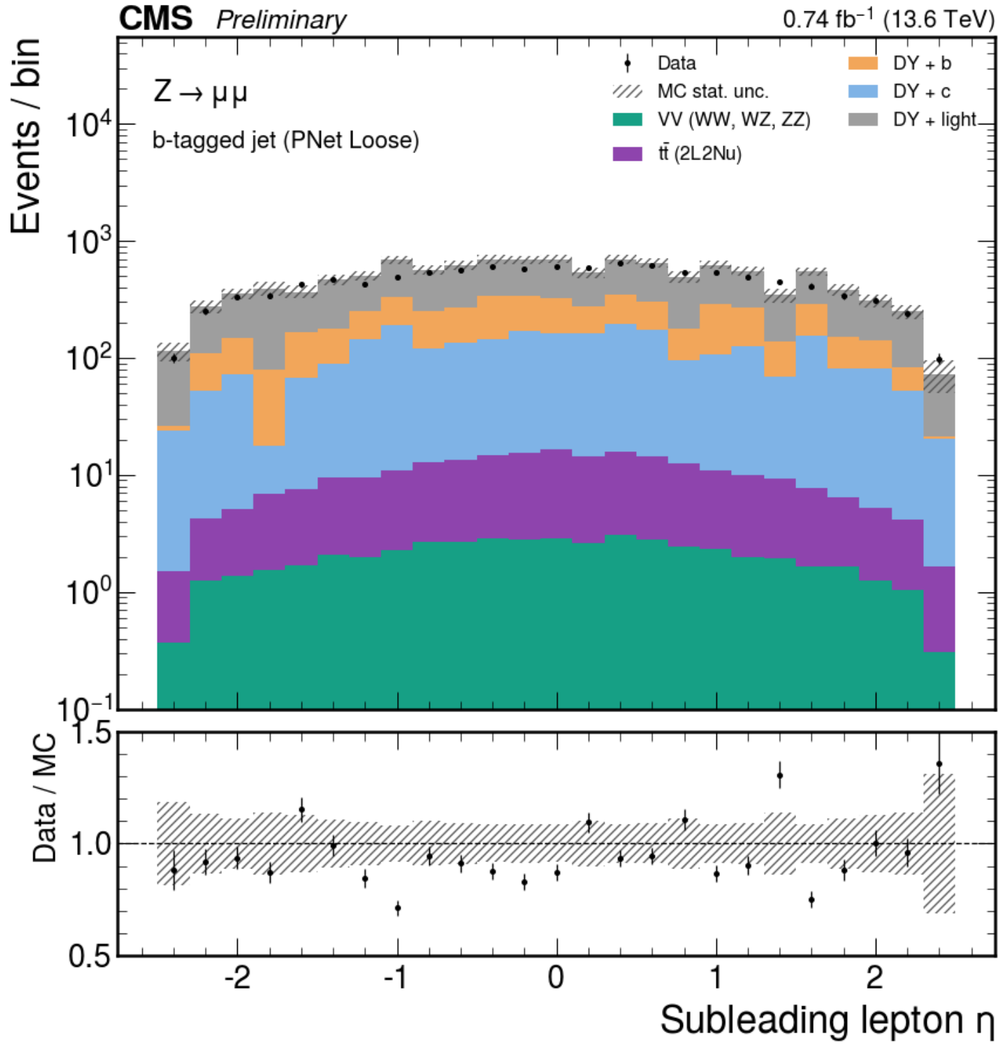
Second lepton pT > 25 GeV (electrons), > 26 GeV (muons, new: trigger SF range).



PNet Loose b-tag (PNet B > 0.047)

Second-lepton η

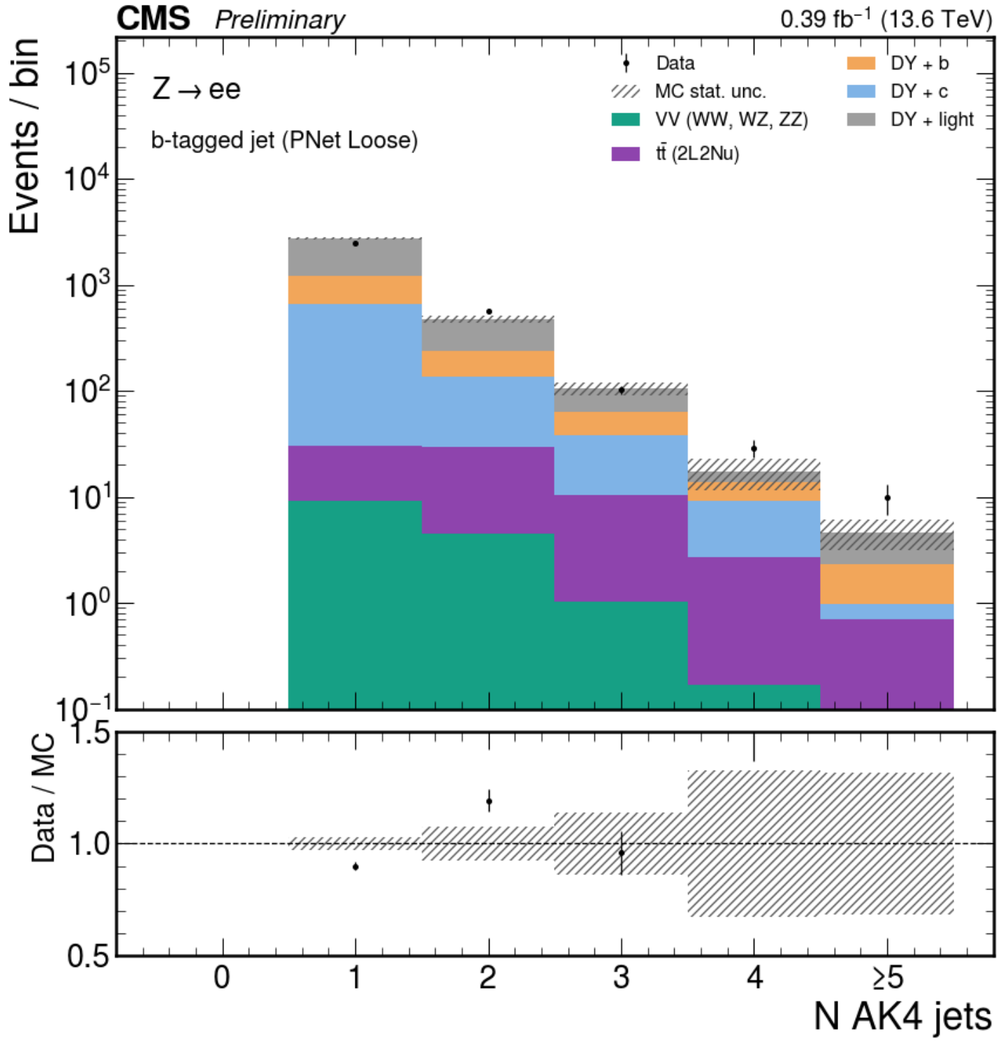


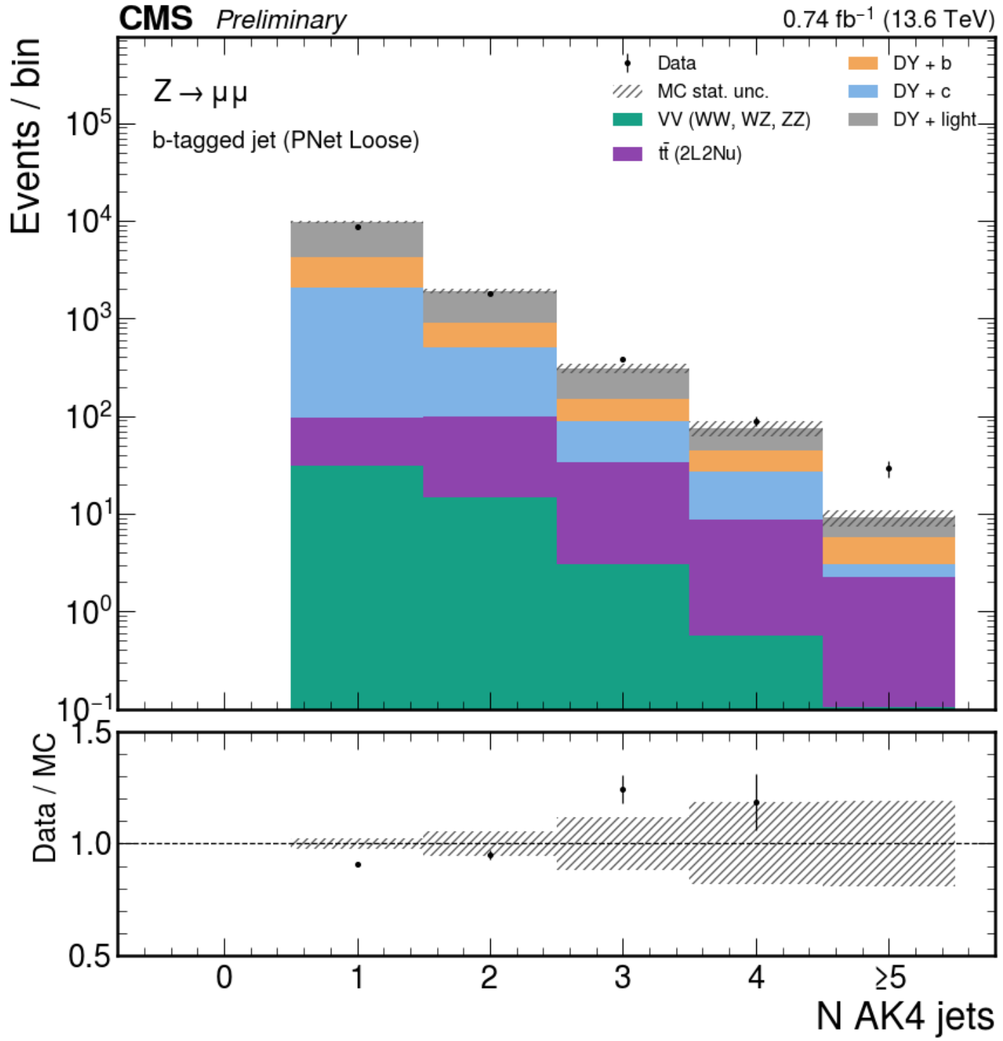


PNet Loose b-tag (PNet B > 0.047)

Number of jets

AK4 jets with pT > 30 GeV, |η| < 2.5, away from the leptons.

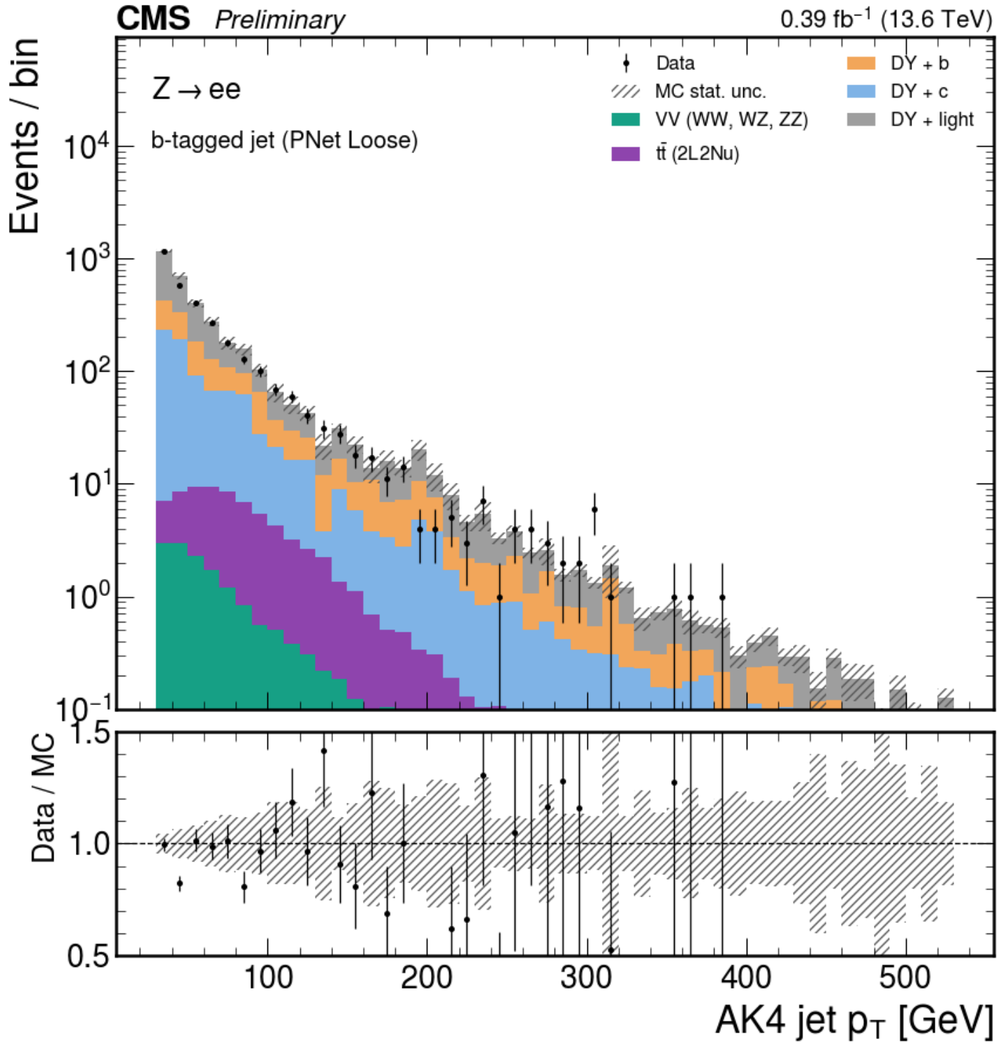


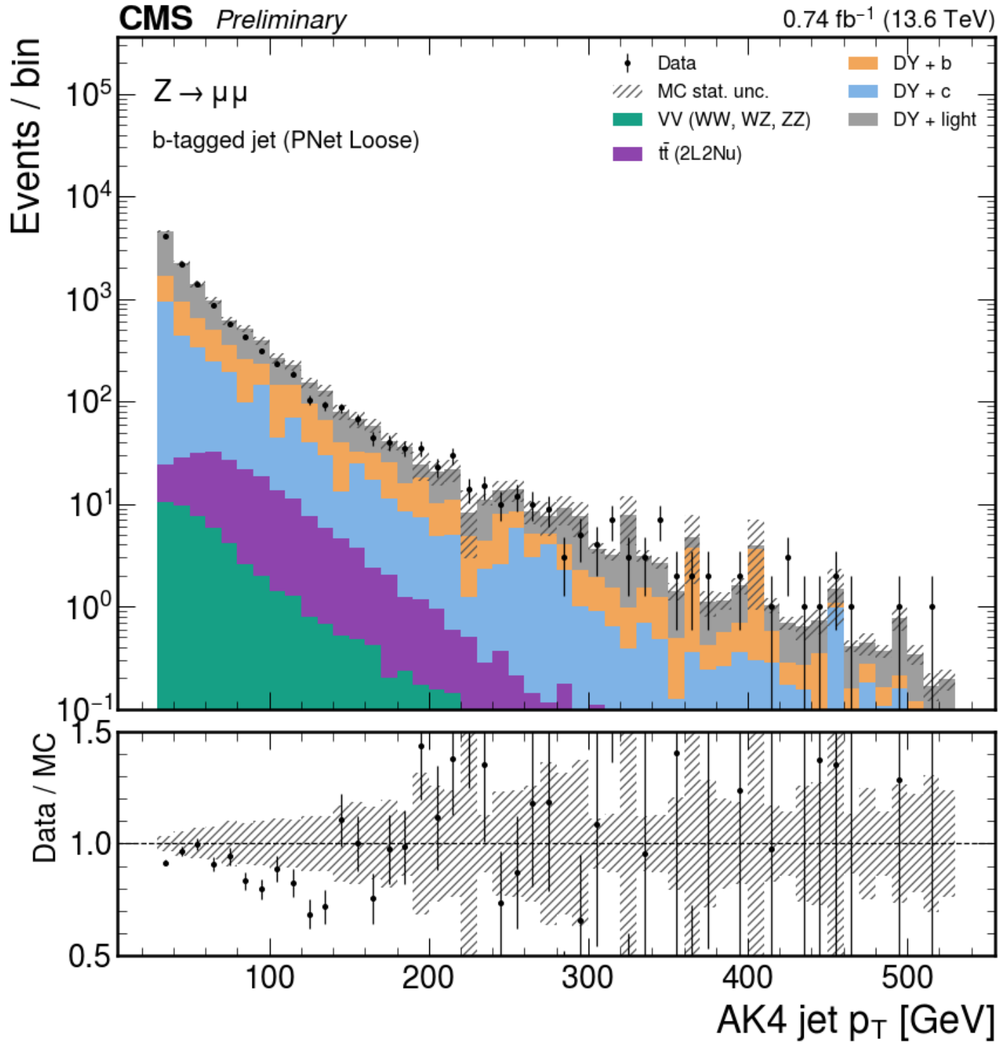


PNet Loose b-tag (PNet B > 0.047)

Leading-jet pT

The b-tagged jet.

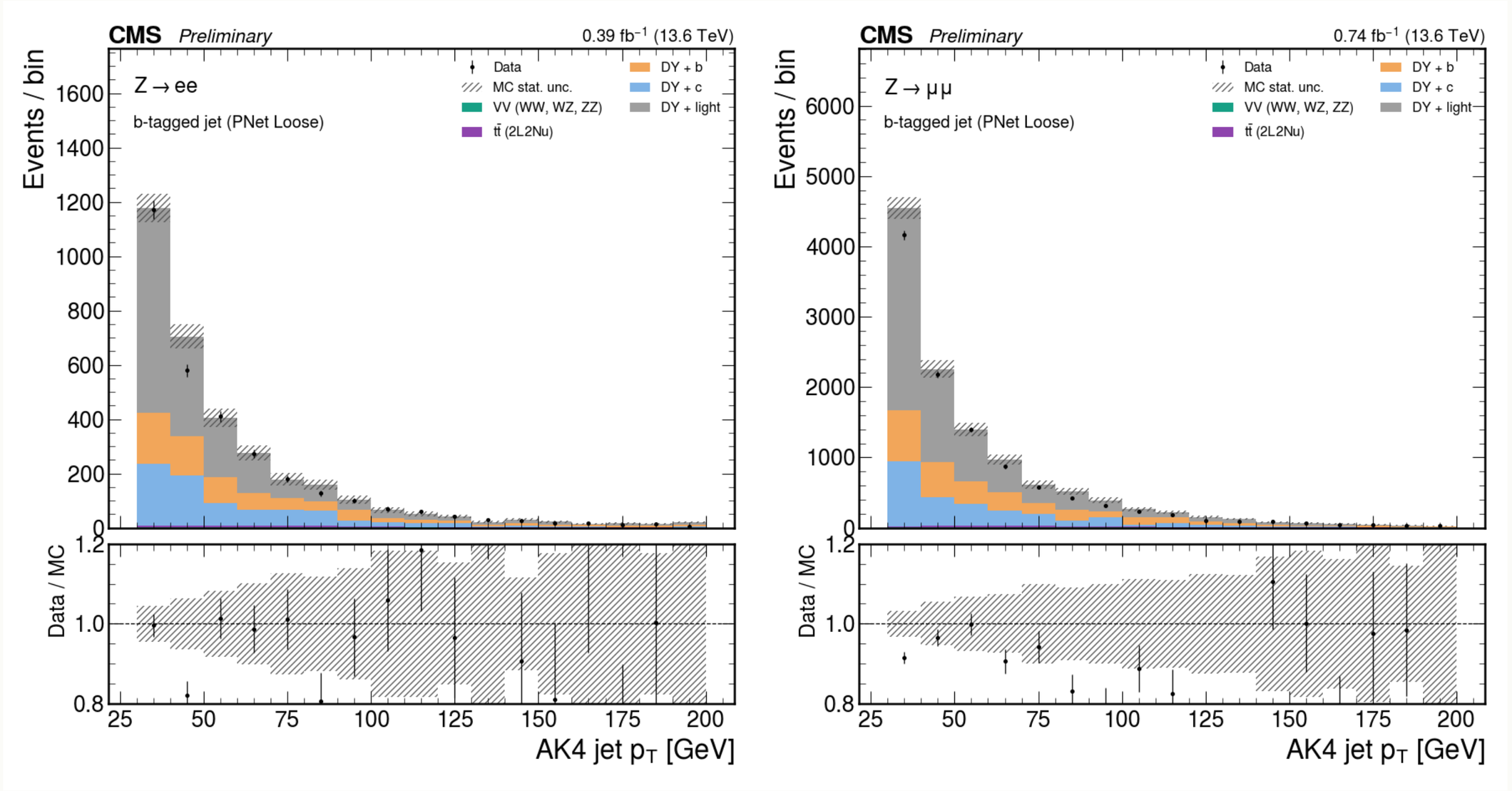




PNet Loose b-tag (PNet B > 0.047)

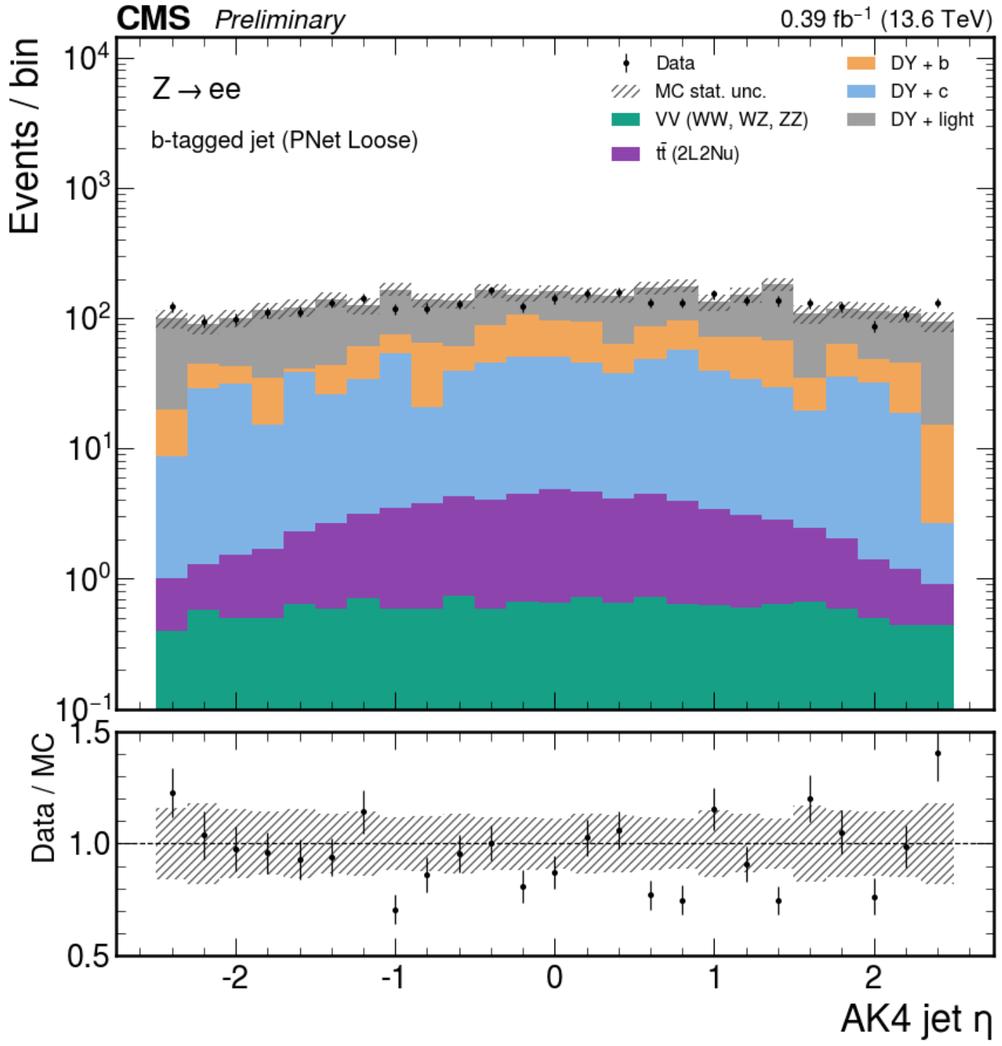
Leading-jet pT, 30–200 GeV

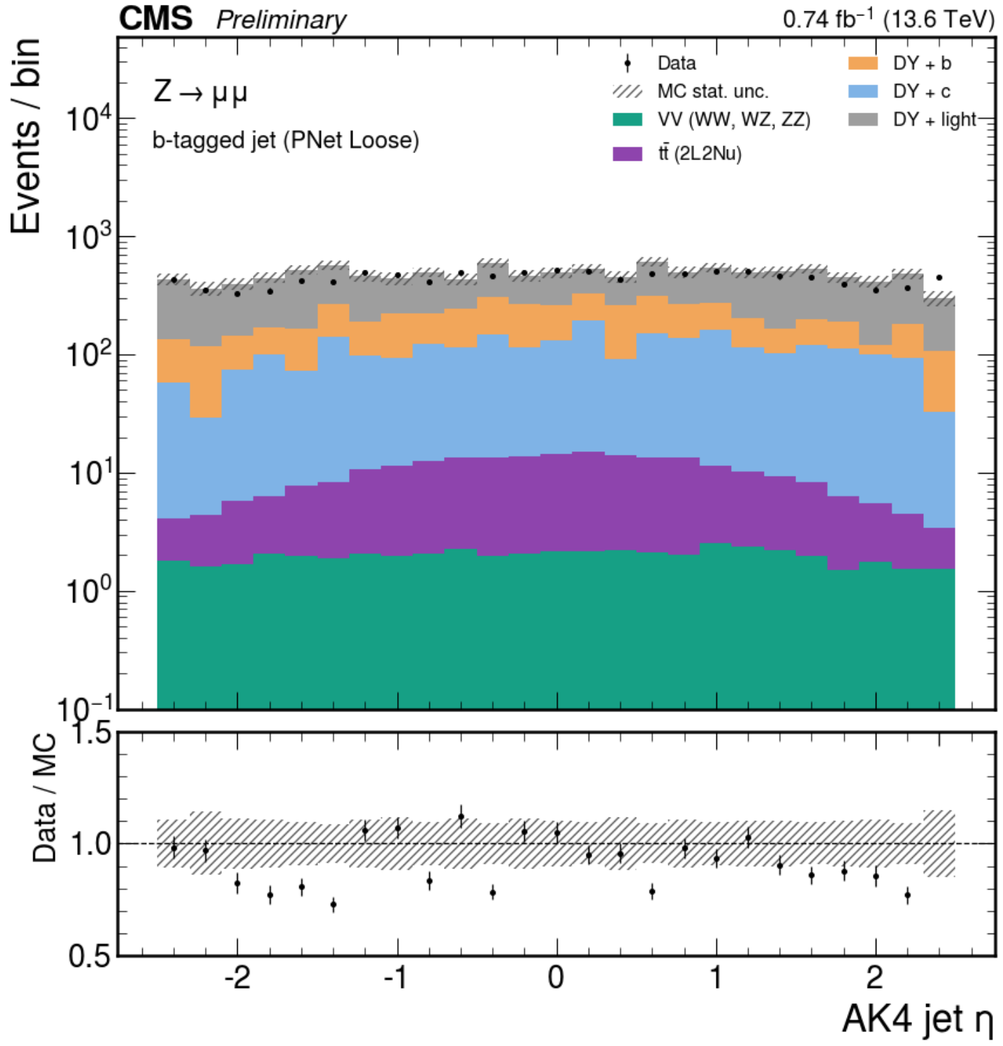
Zoom: linear scale, ratio 0.8–1.2.



PNet Loose b-tag (PNet B > 0.047)

Leading-jet η

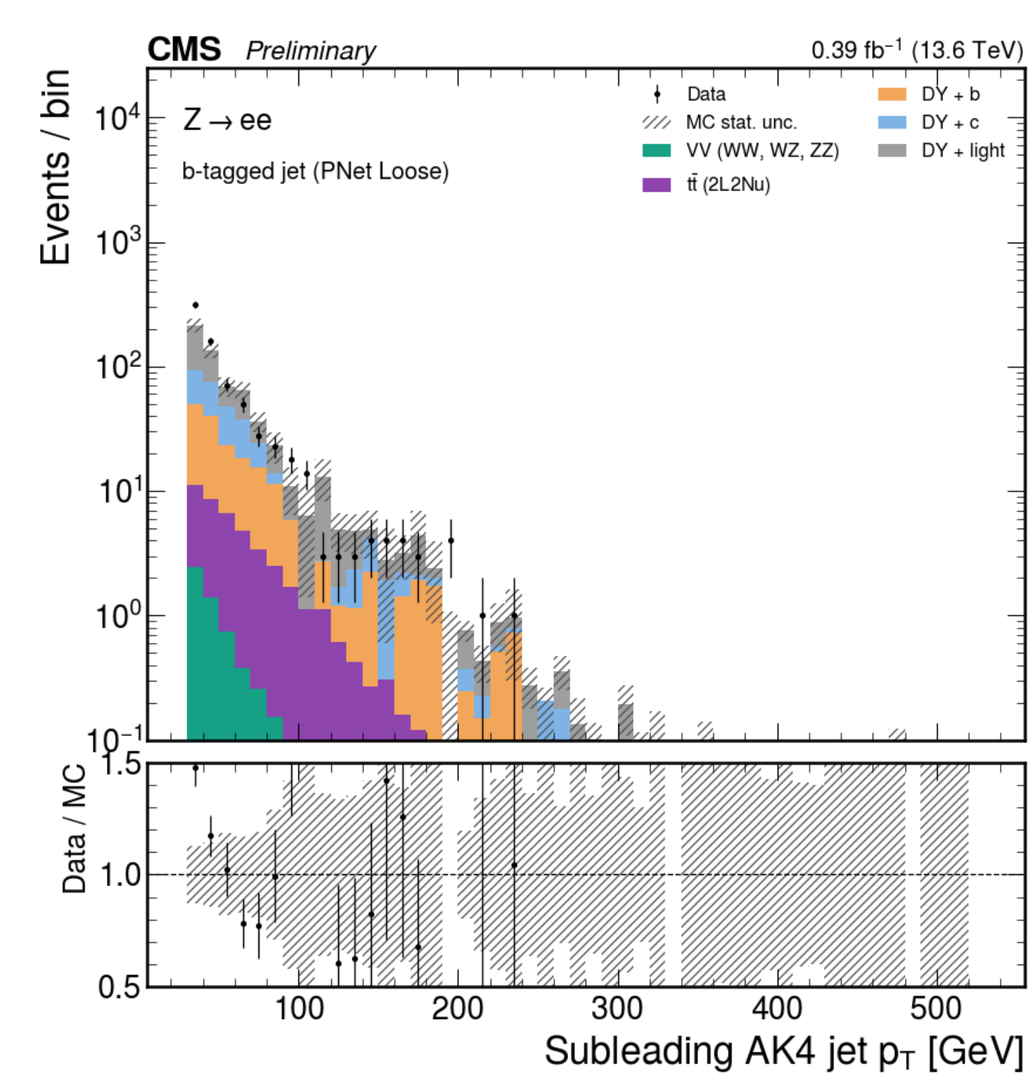


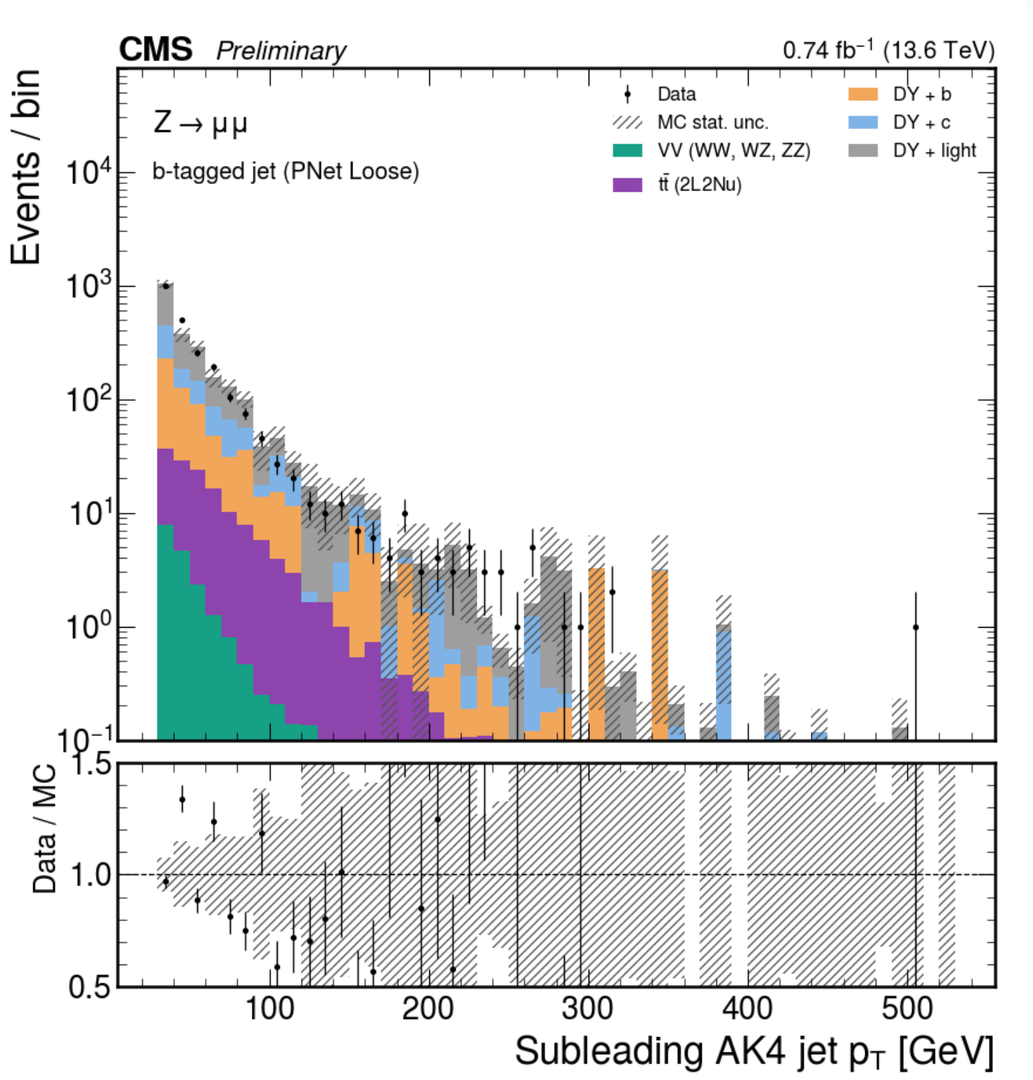


PNet Loose b-tag (PNet B > 0.047)

Second-jet pT

Events with at least 2 jets.

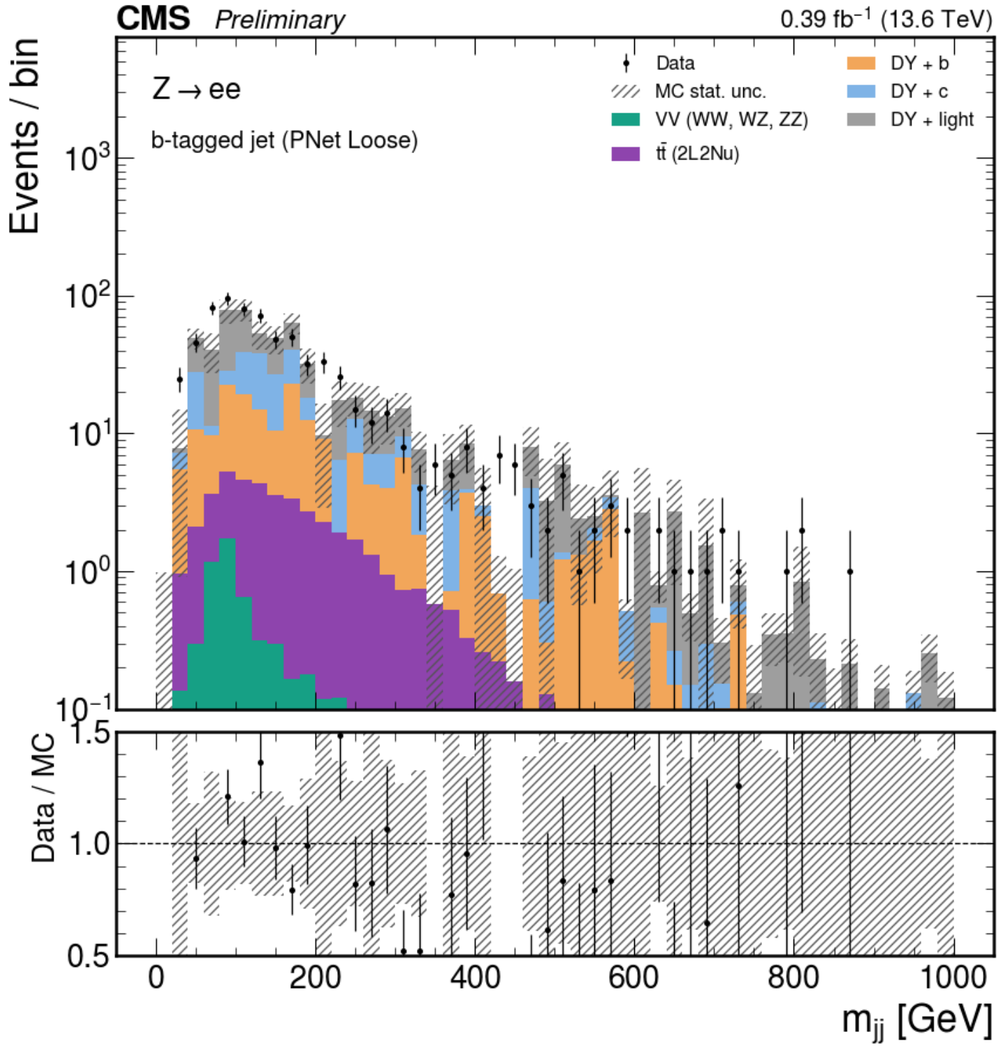


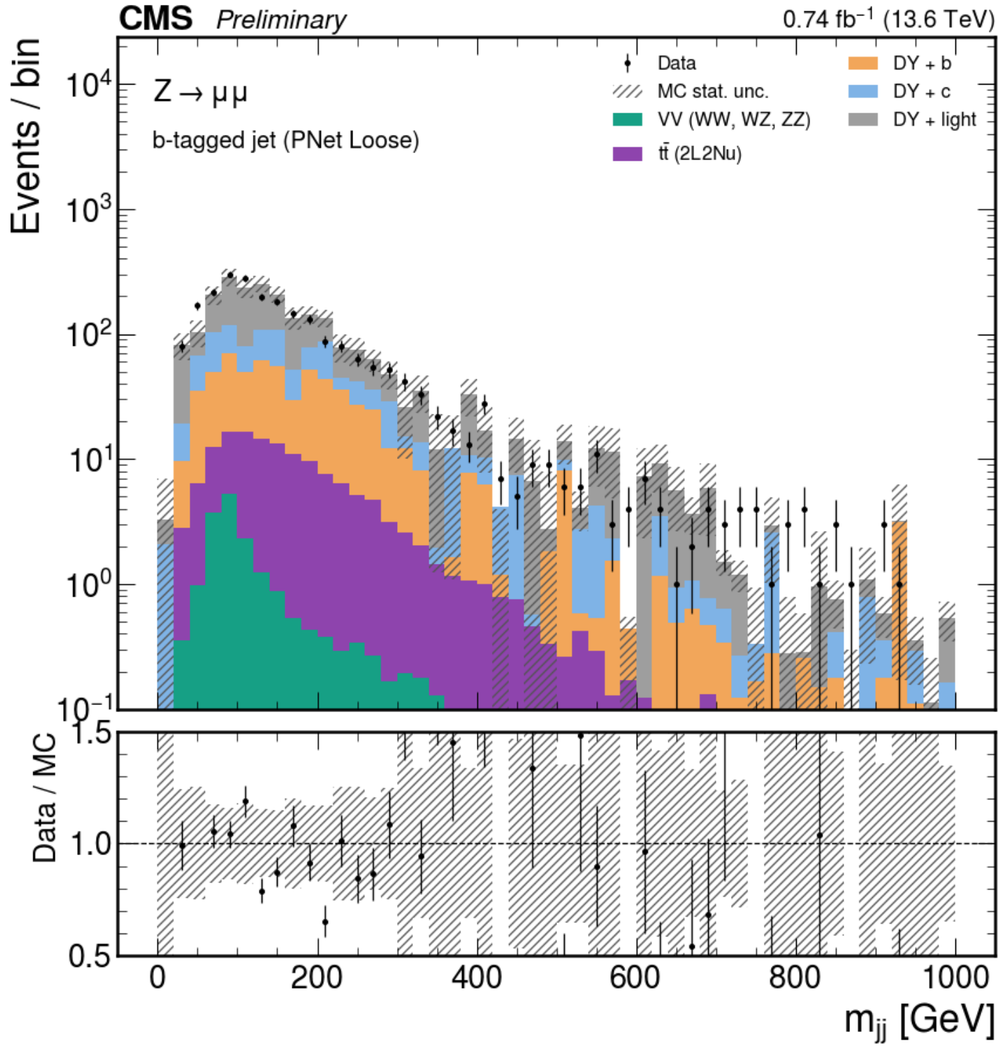


PNet Loose b-tag (PNet B > 0.047)

Dijet mass

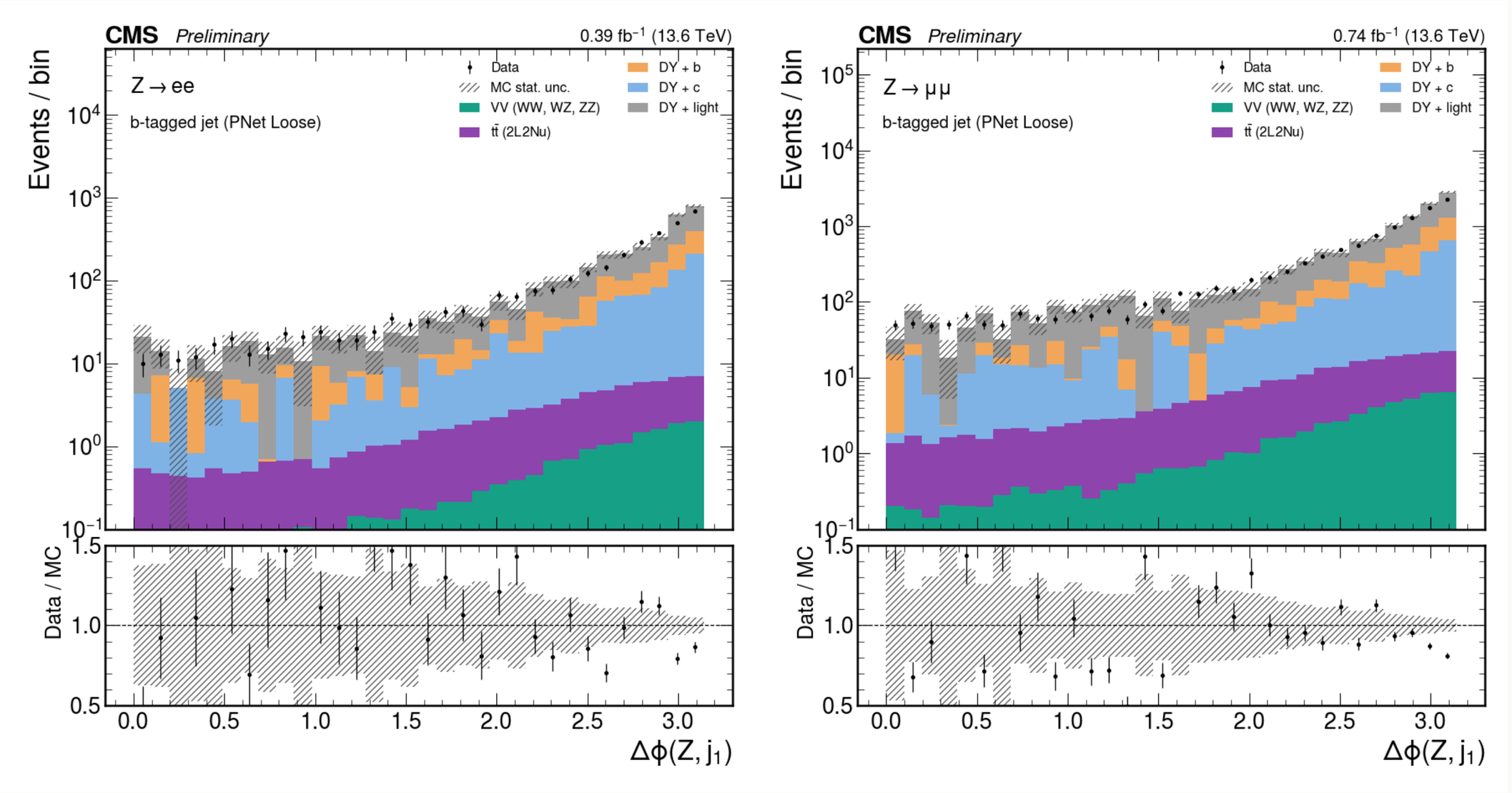
Events with at least 2 jets.





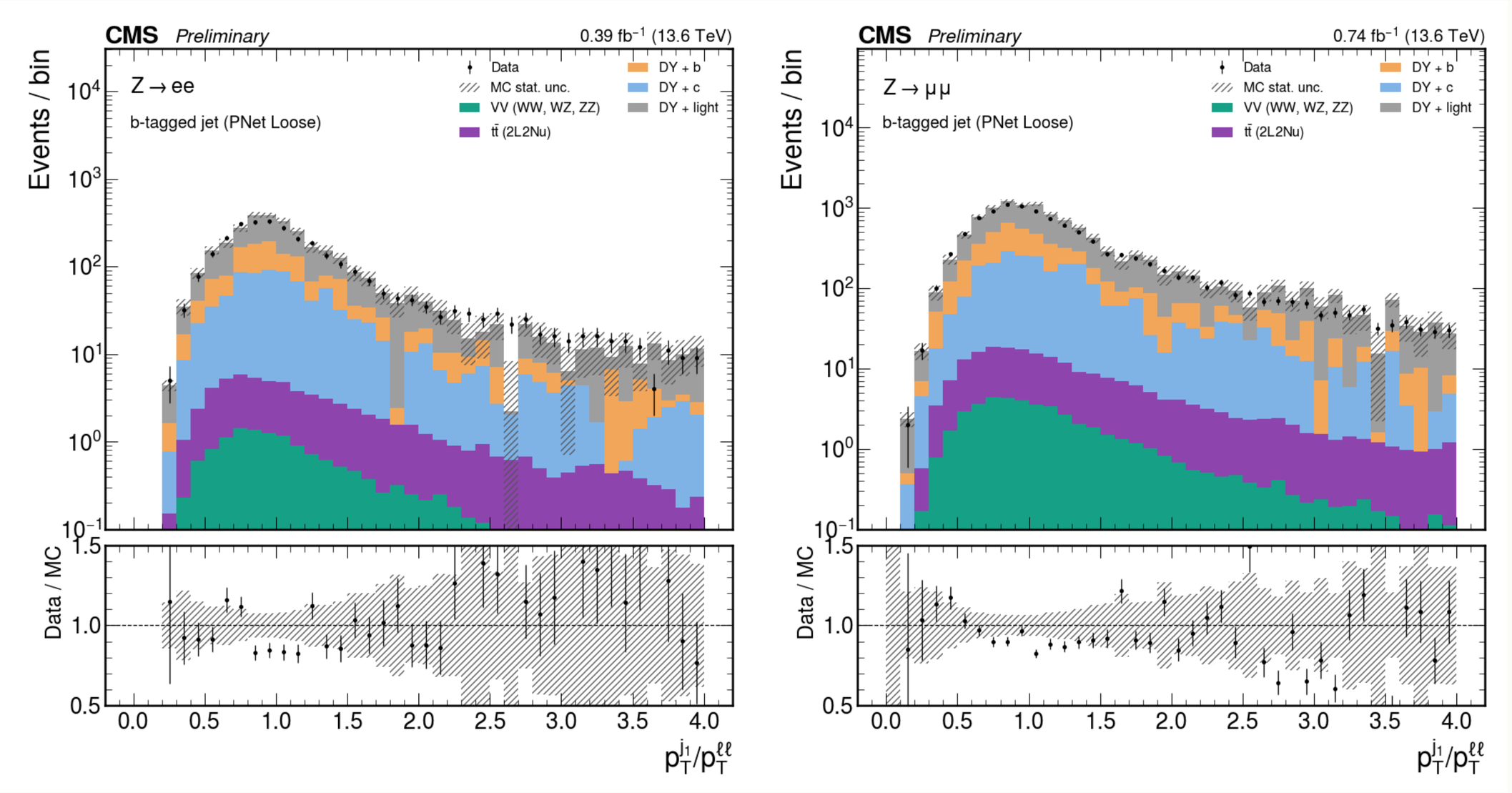
PNet Loose b-tag (PNet B > 0.047)

Δφ(Z, leading jet)



PNet Loose b-tag (PNet B > 0.047)

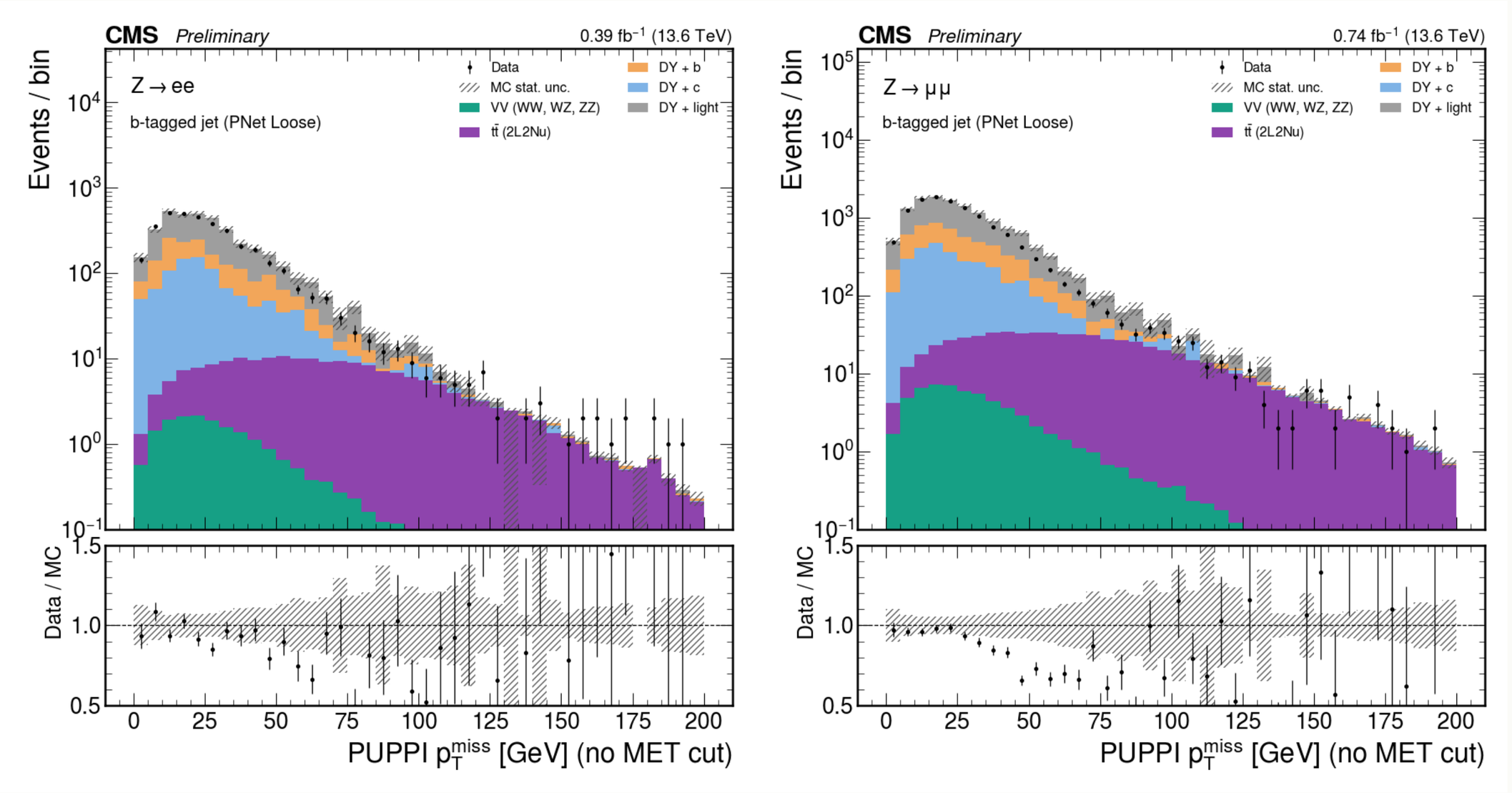
pT balance: leading-jet pT / Z pT



PNet Loose b-tag (PNet B > 0.047)

PUPPI MET without the MET cut, up to 200 GeV

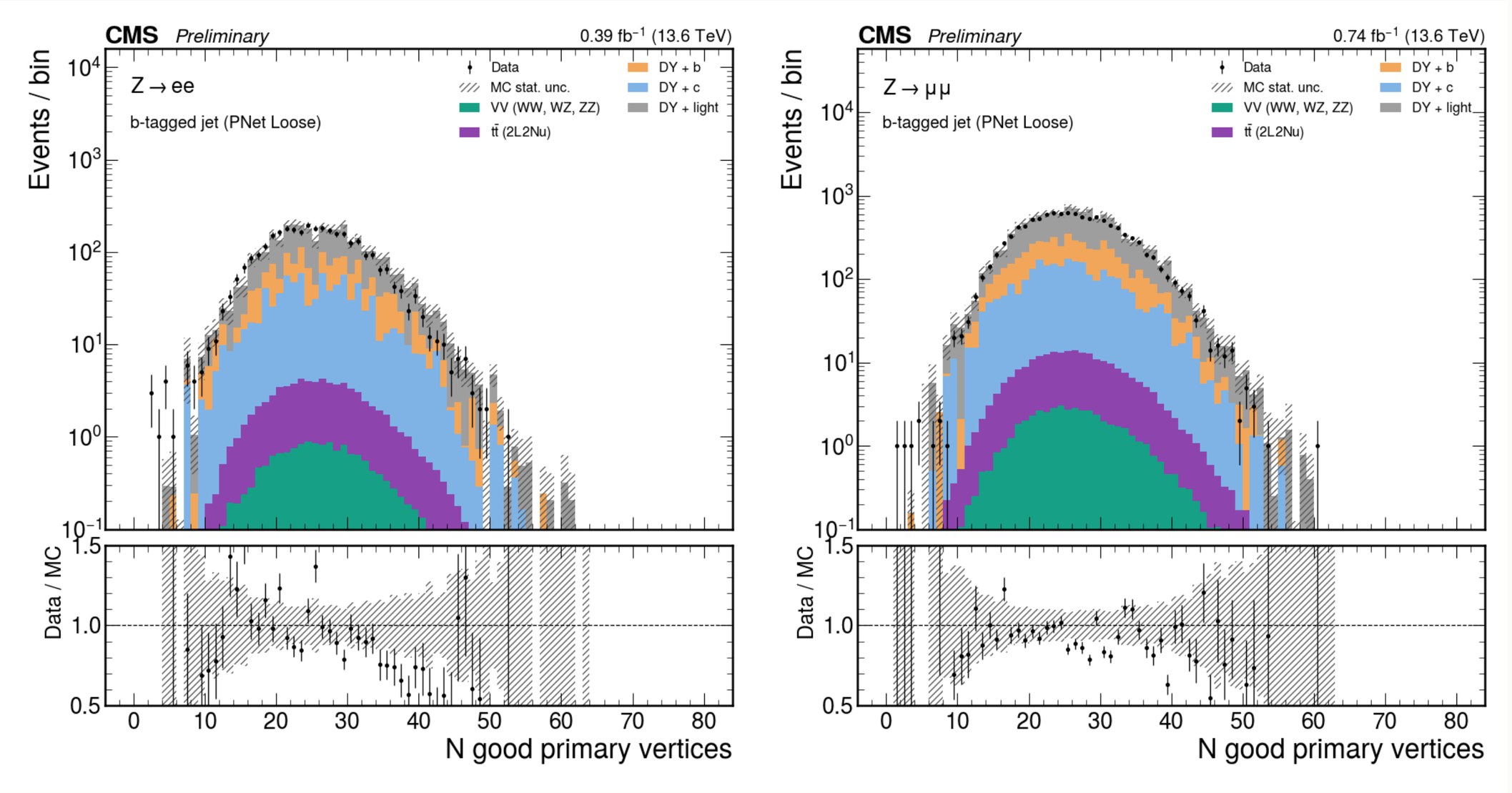
All other cuts applied; the selection cuts at 50 GeV.



PNet Loose b-tag (PNet B > 0.047)

Number of good primary vertices

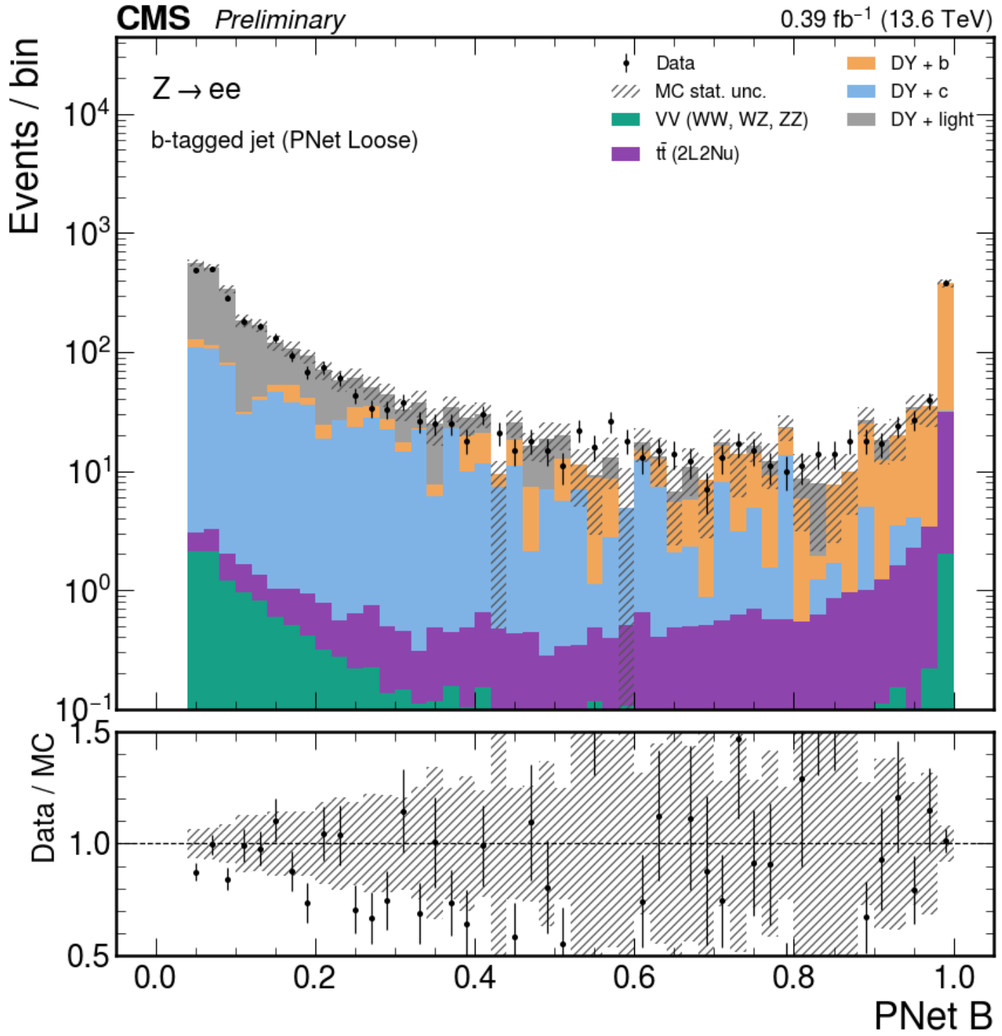
After pileup weights.

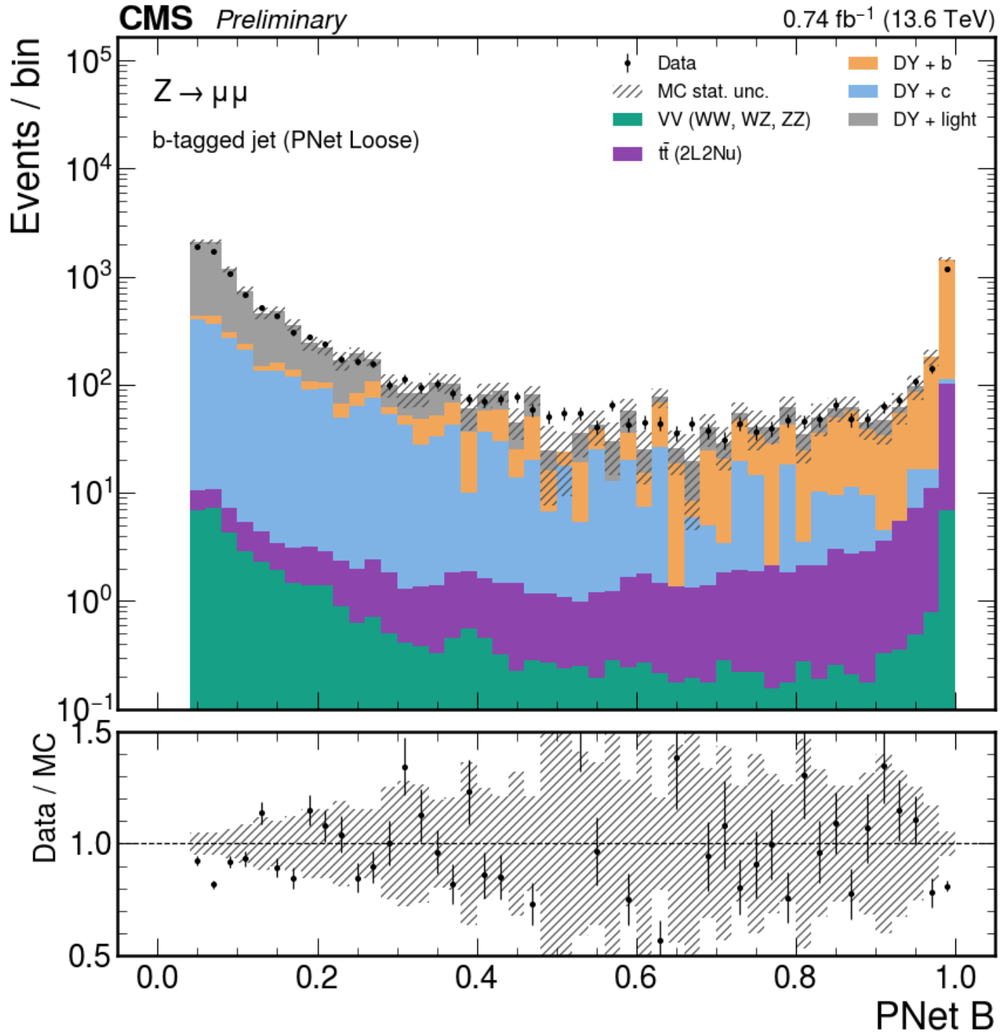


PNet Loose b-tag (PNet B > 0.047)

Leading-jet PNet B score

Only jets above 0.047 enter this region.





DeepJet, Loose b-tag

Leading jet DeepJet B > 0.0583, with the deepJet BTV SFs

	data/MC	DY+b	DY+c	DY+light	tt	tagged MC	tagged data
Z→ee	0.926 ± 0.029	18%	22%	58%	2%	18.9%	18.4%
Z→μμ	0.924 ± 0.021	20%	21%	57%	1%	19.3%	18.9%

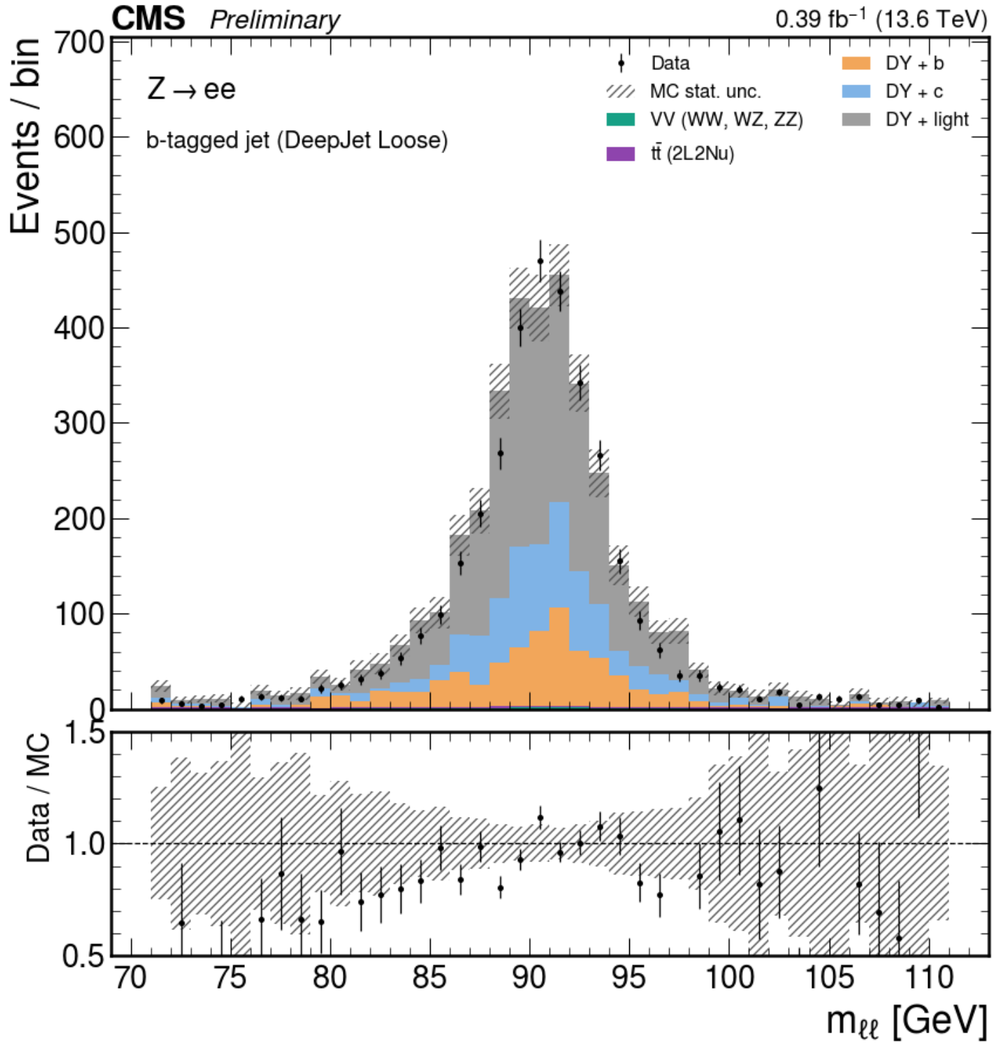
Flavour shares are of all MC. Tagged = fraction of Z + jet events with a tagged leading jet. Errors are statistical only.

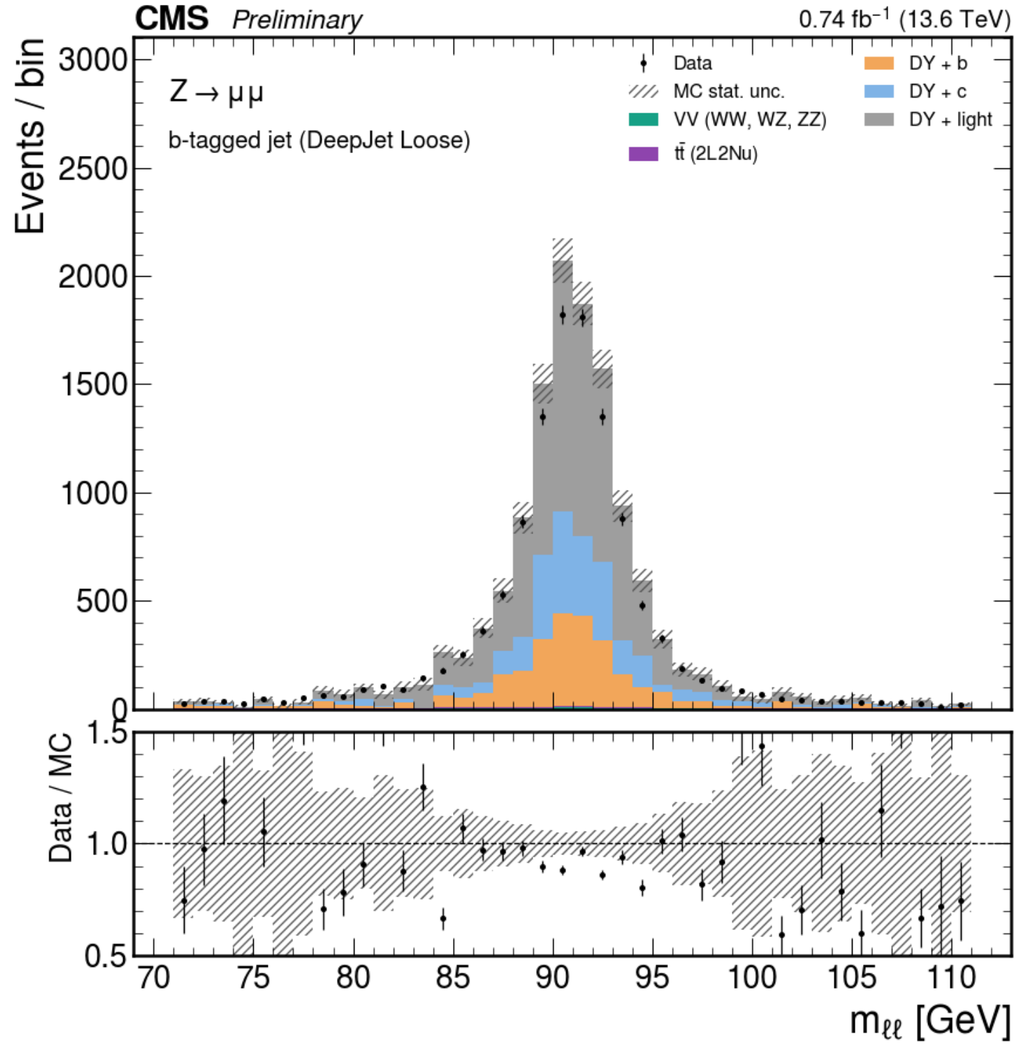
19 plots follow. Left Z→ee, right Z→μμ.

DeepJet Loose b-tag (DeepJet B > 0.0583)

Dilepton mass

71 < m(ℓℓ) < 111 GeV, linear scale.

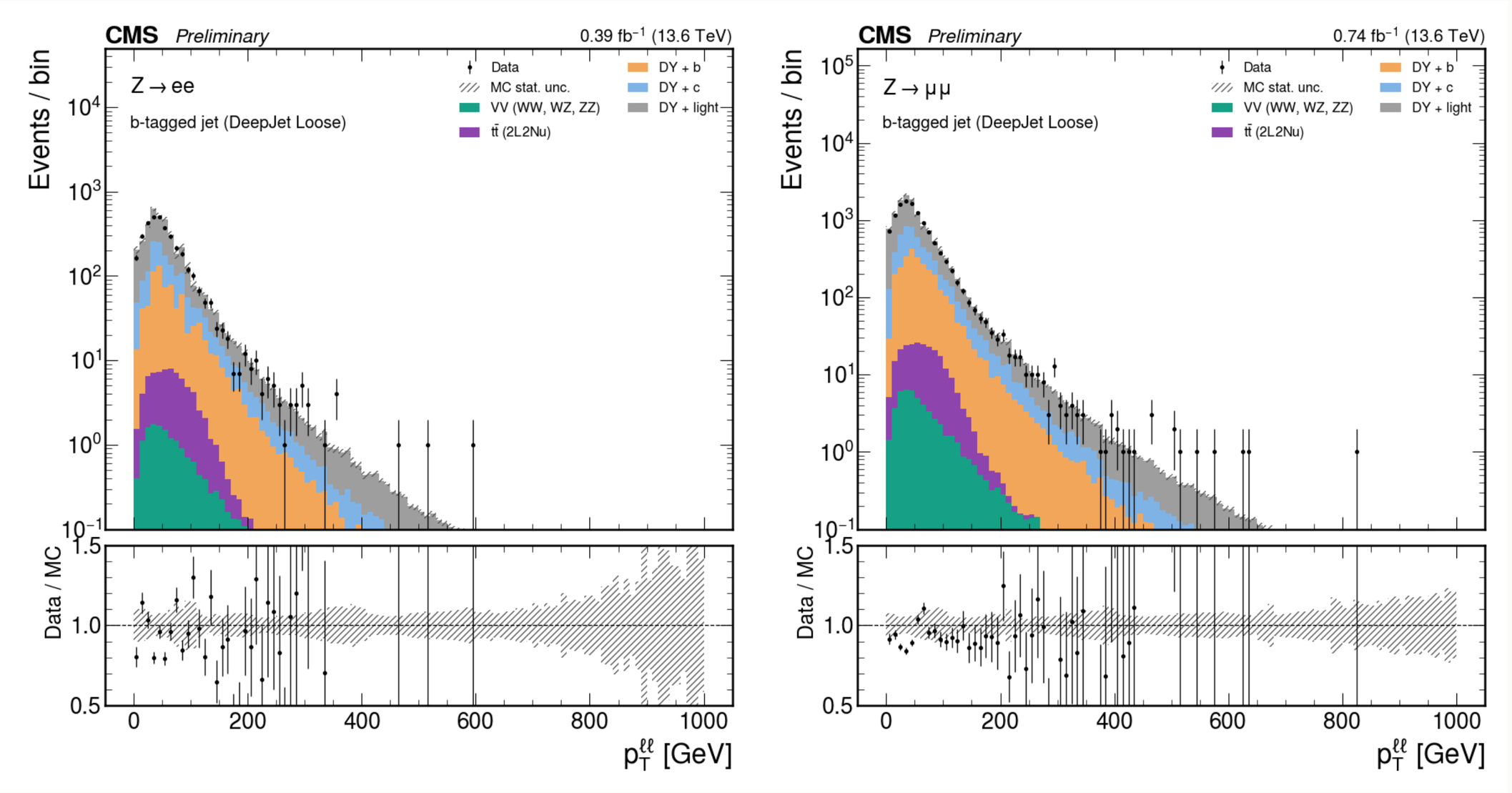




DeepJet Loose b-tag (DeepJet B > 0.0583)

Z pT

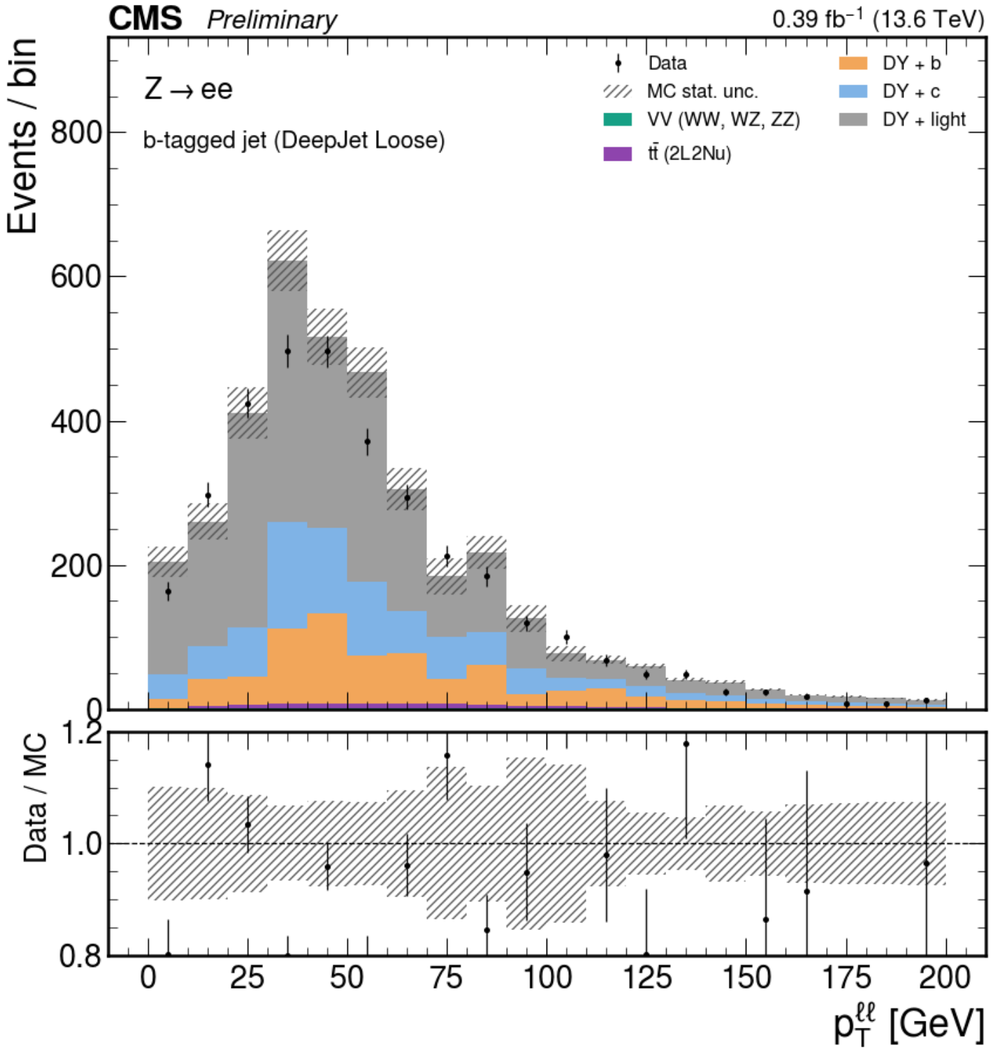
pT of the lepton pair.

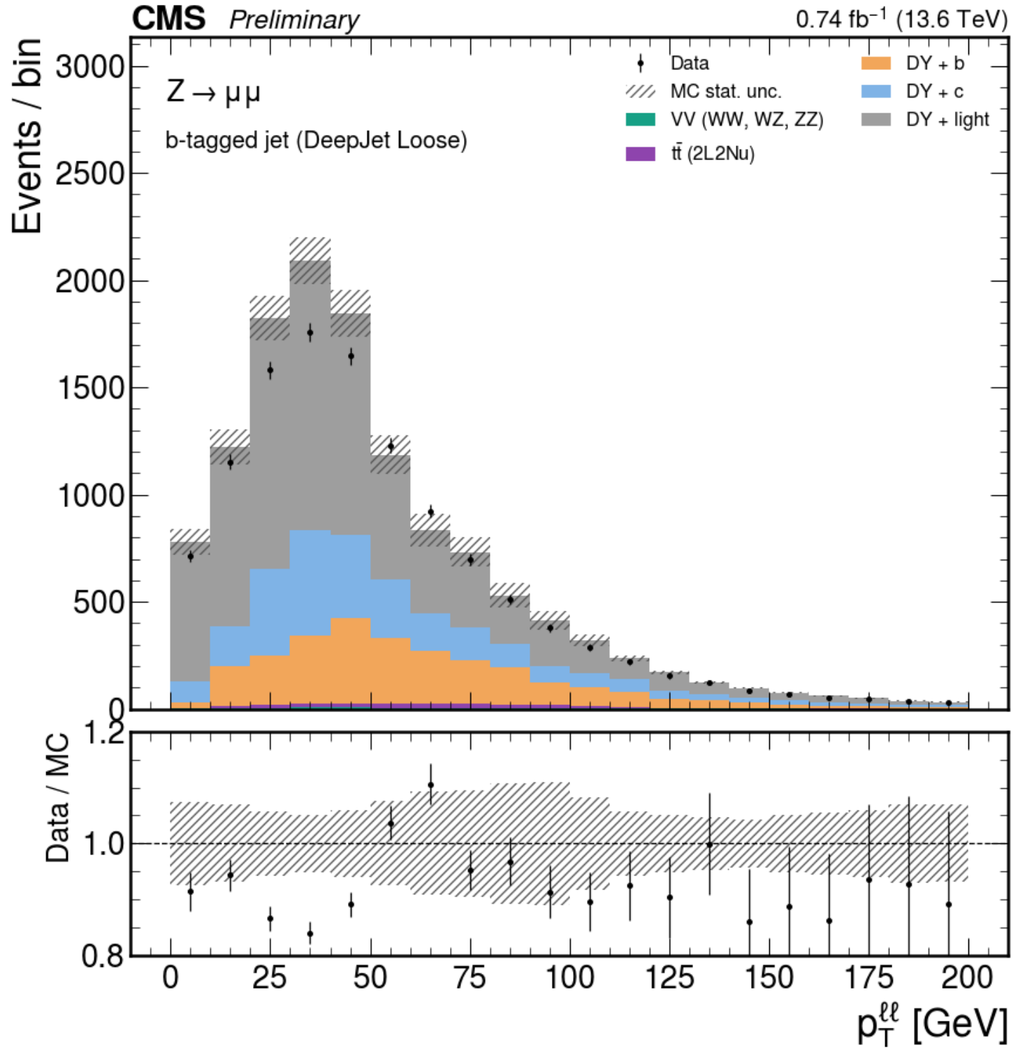


DeepJet Loose b-tag (DeepJet B > 0.0583)

Z pT, 0–200 GeV

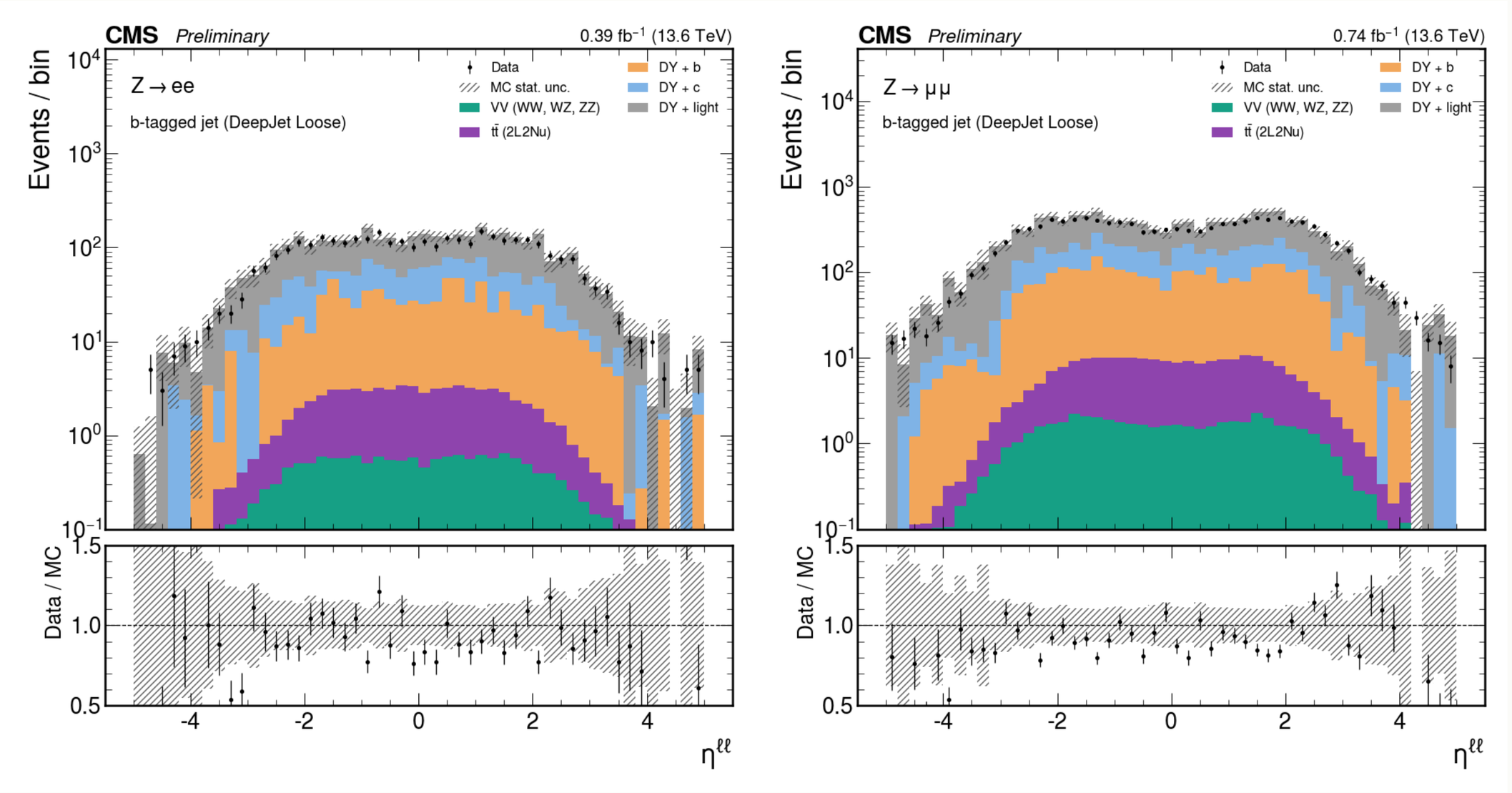
Zoom: linear scale, ratio 0.8–1.2.





DeepJet Loose b-tag (DeepJet B > 0.0583)

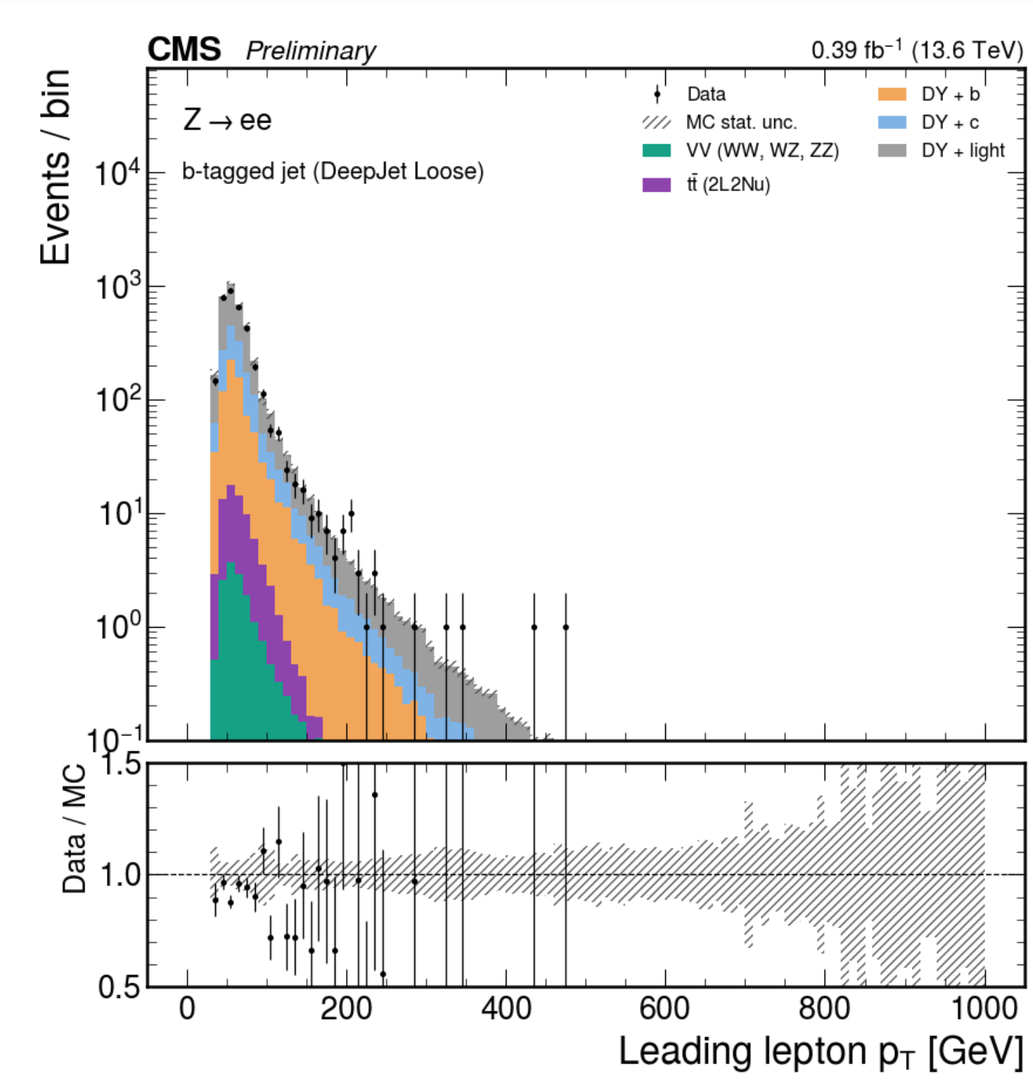
Z η

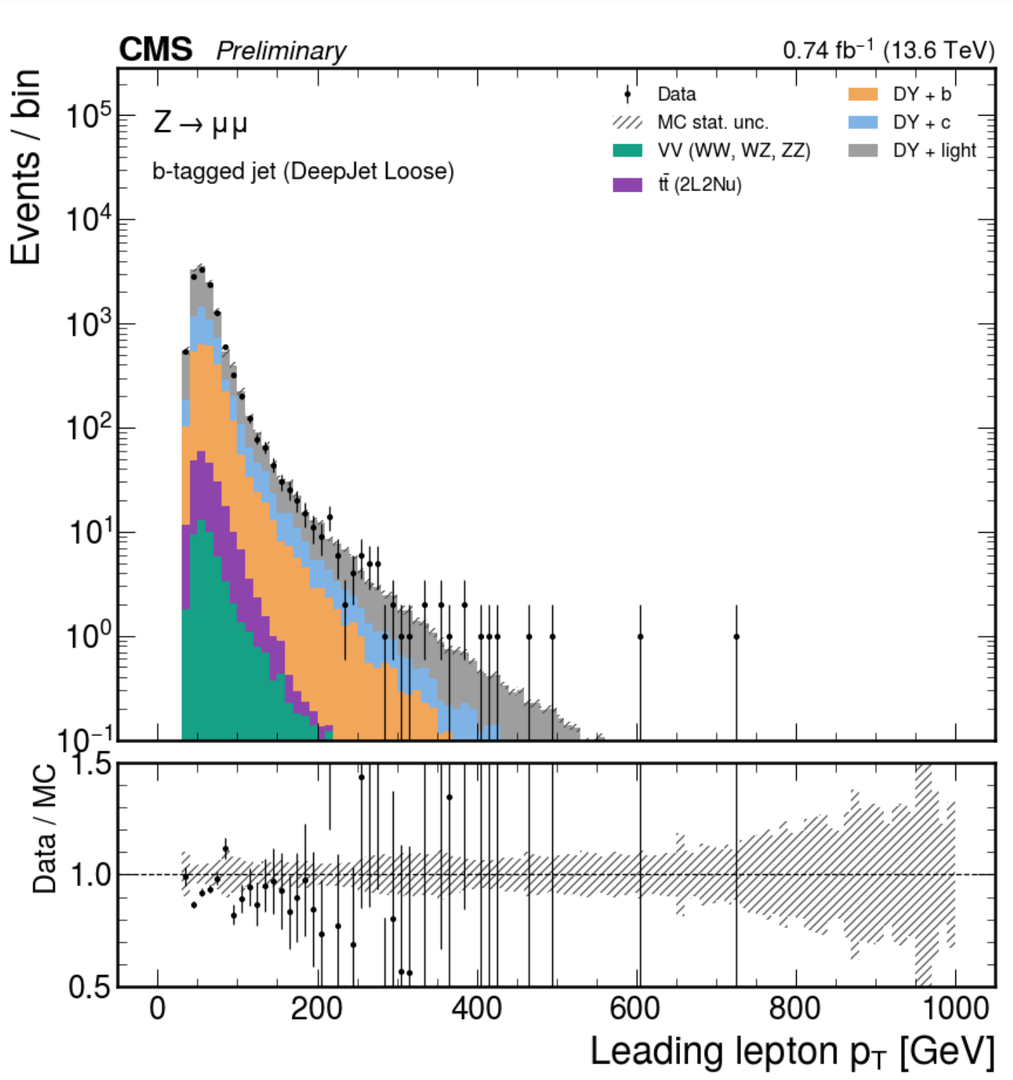


DeepJet Loose b-tag (DeepJet B > 0.0583)

Leading-lepton pT

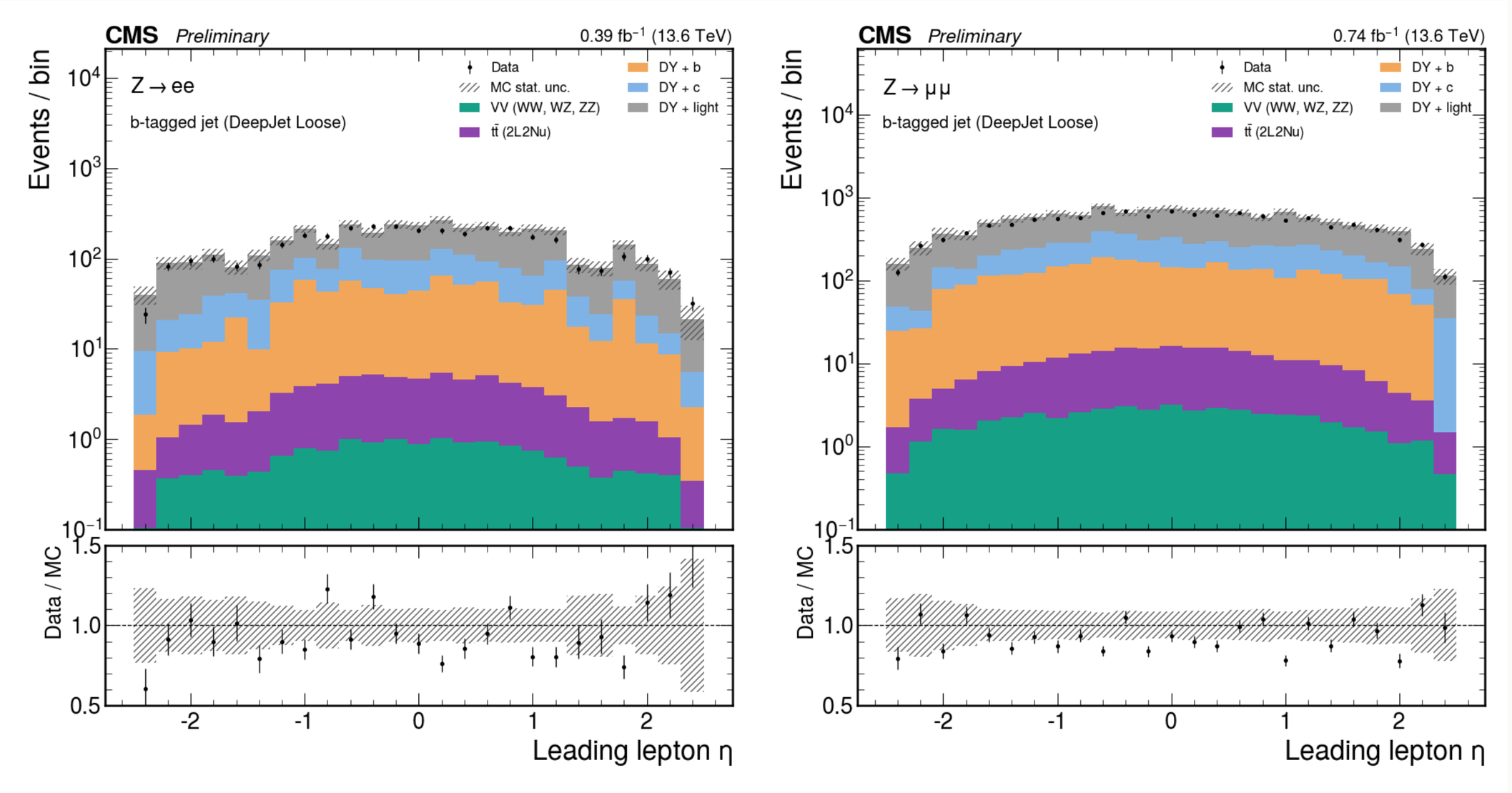
Leading lepton pT > 35 GeV.





DeepJet Loose b-tag (DeepJet B > 0.0583)

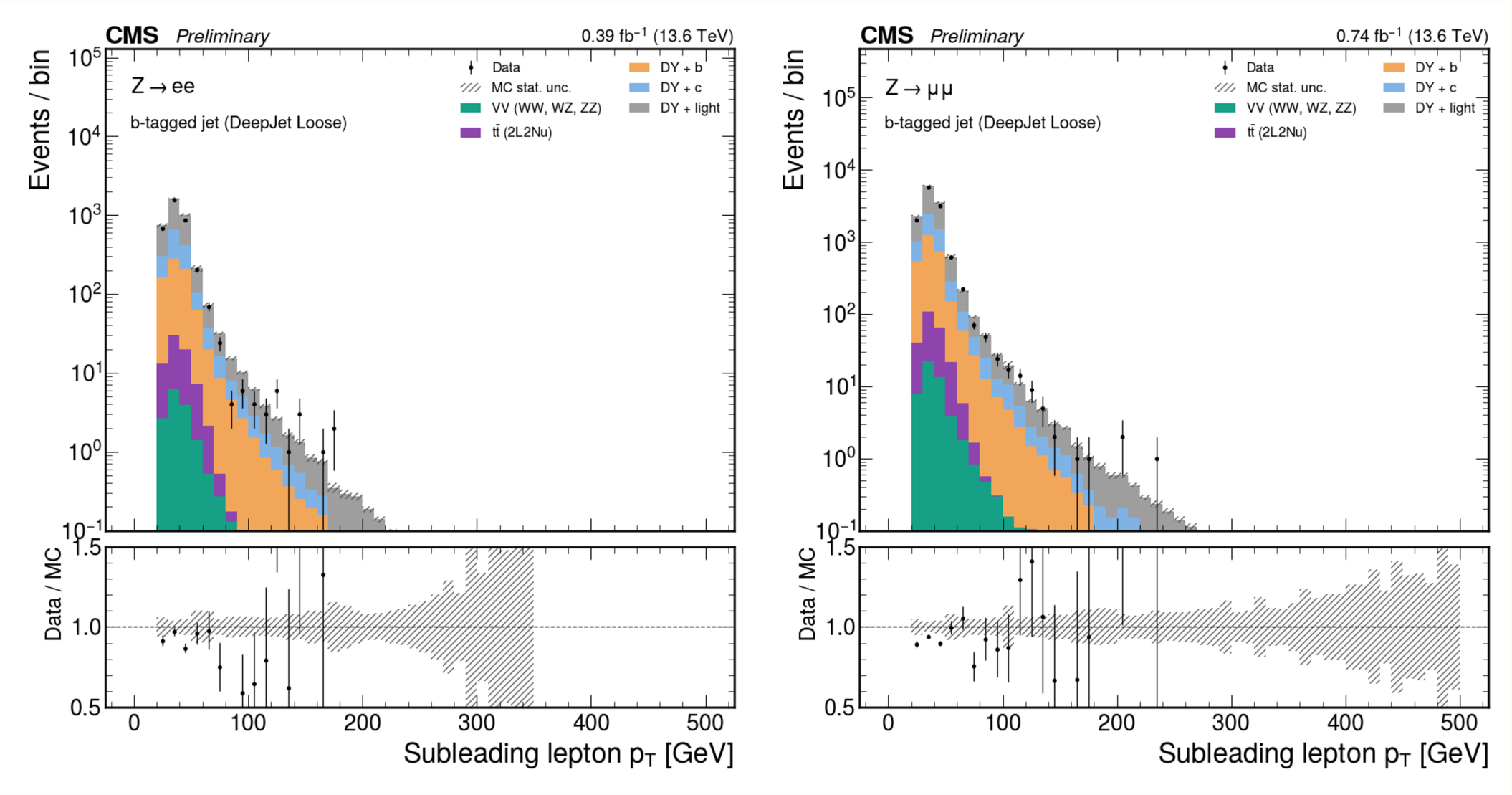
Leading-lepton η



DeepJet Loose b-tag (DeepJet B > 0.0583)

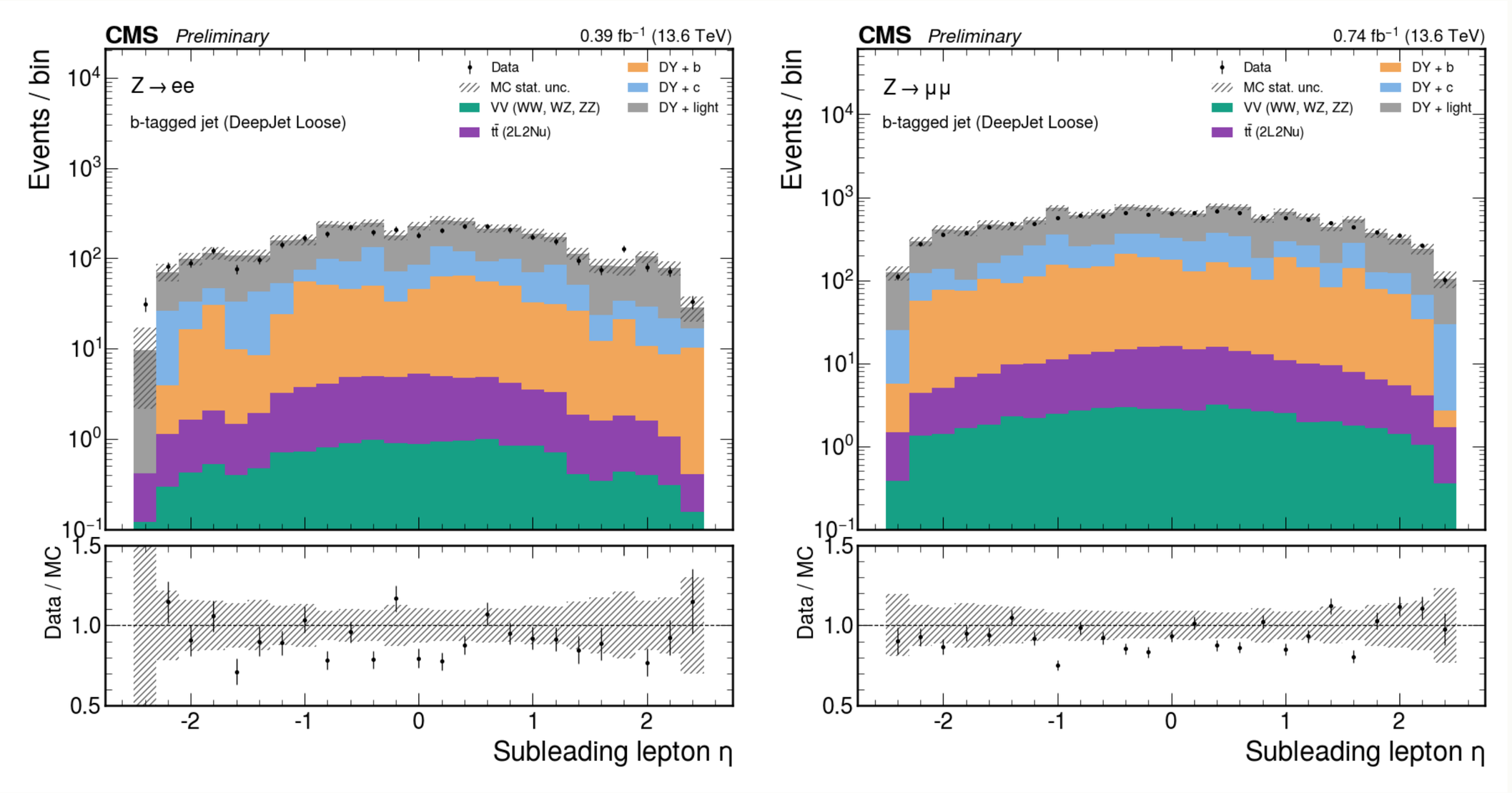
Second-lepton pT

Second lepton pT > 25 GeV (electrons), > 26 GeV (muons, new: trigger SF range).



DeepJet Loose b-tag (DeepJet B > 0.0583)

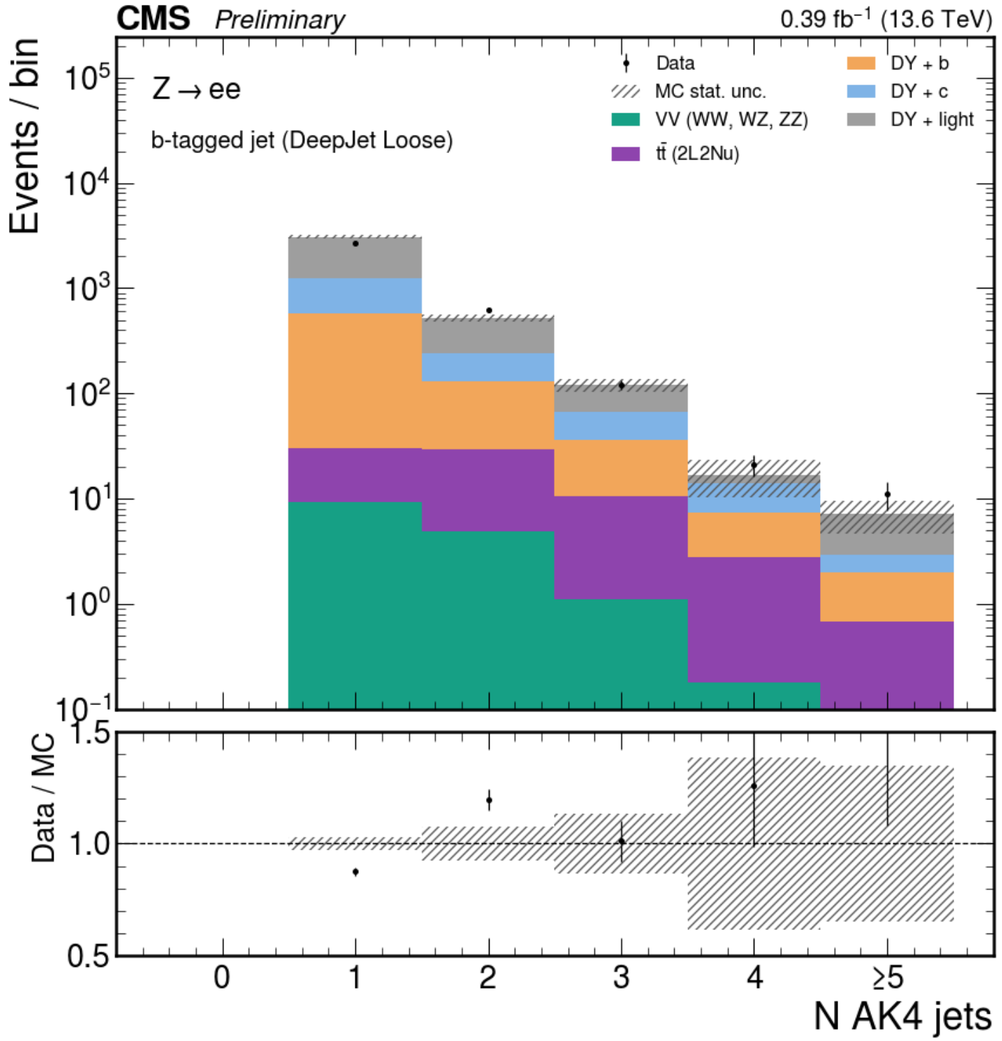
Second-lepton η

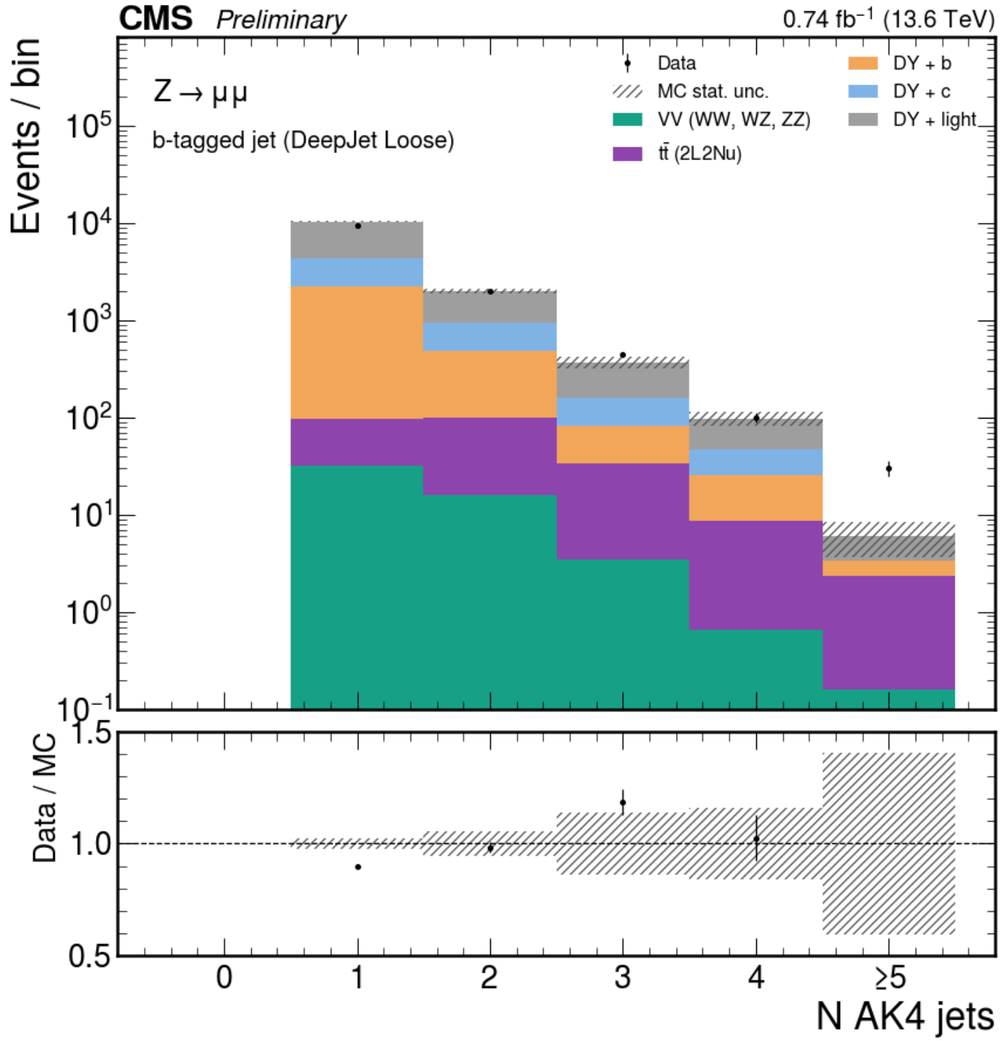


DeepJet Loose b-tag (DeepJet B > 0.0583)

Number of jets

AK4 jets with pT > 30 GeV, |η| < 2.5, away from the leptons.

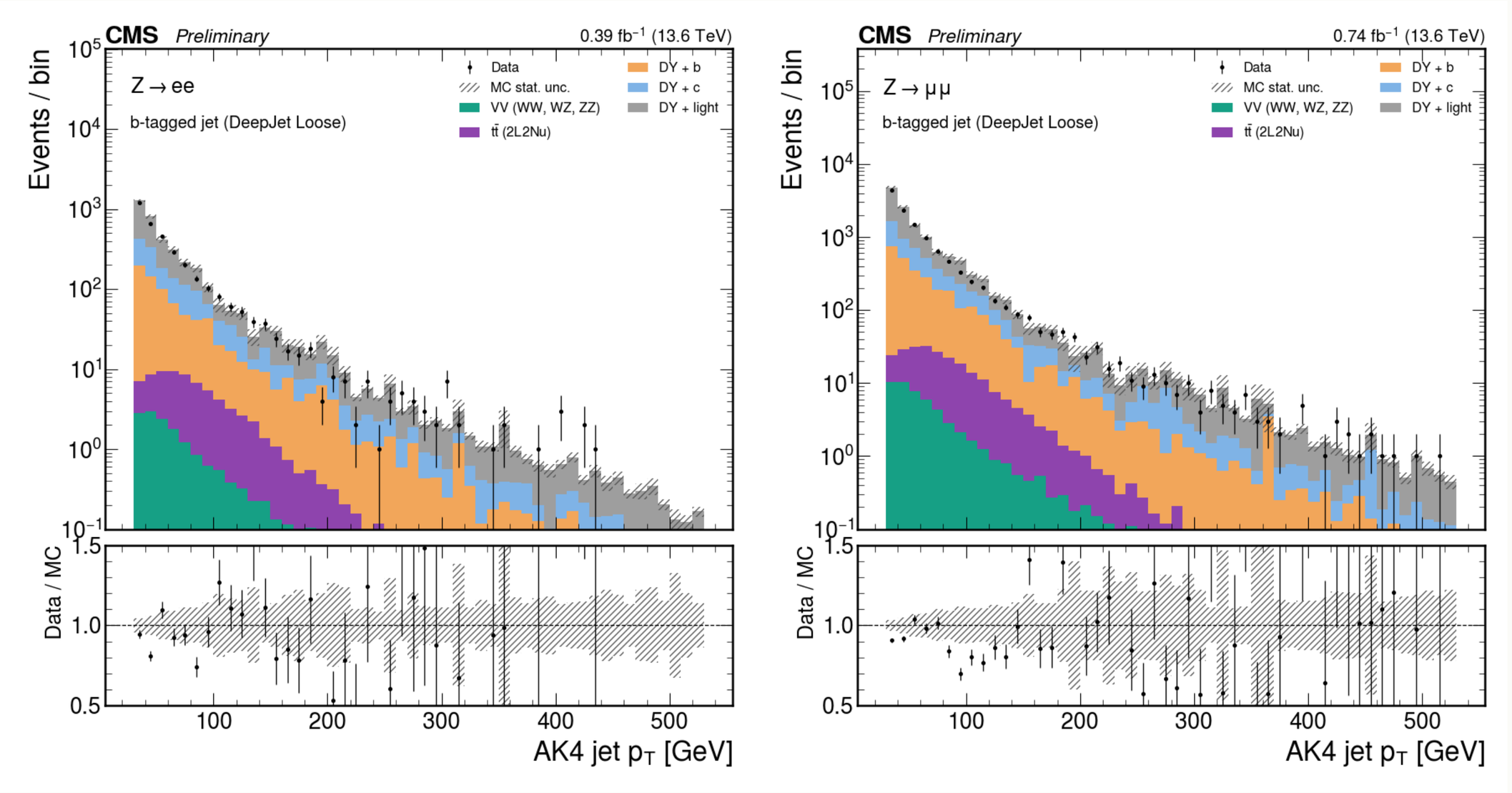




DeepJet Loose b-tag (DeepJet B > 0.0583)

Leading-jet pT

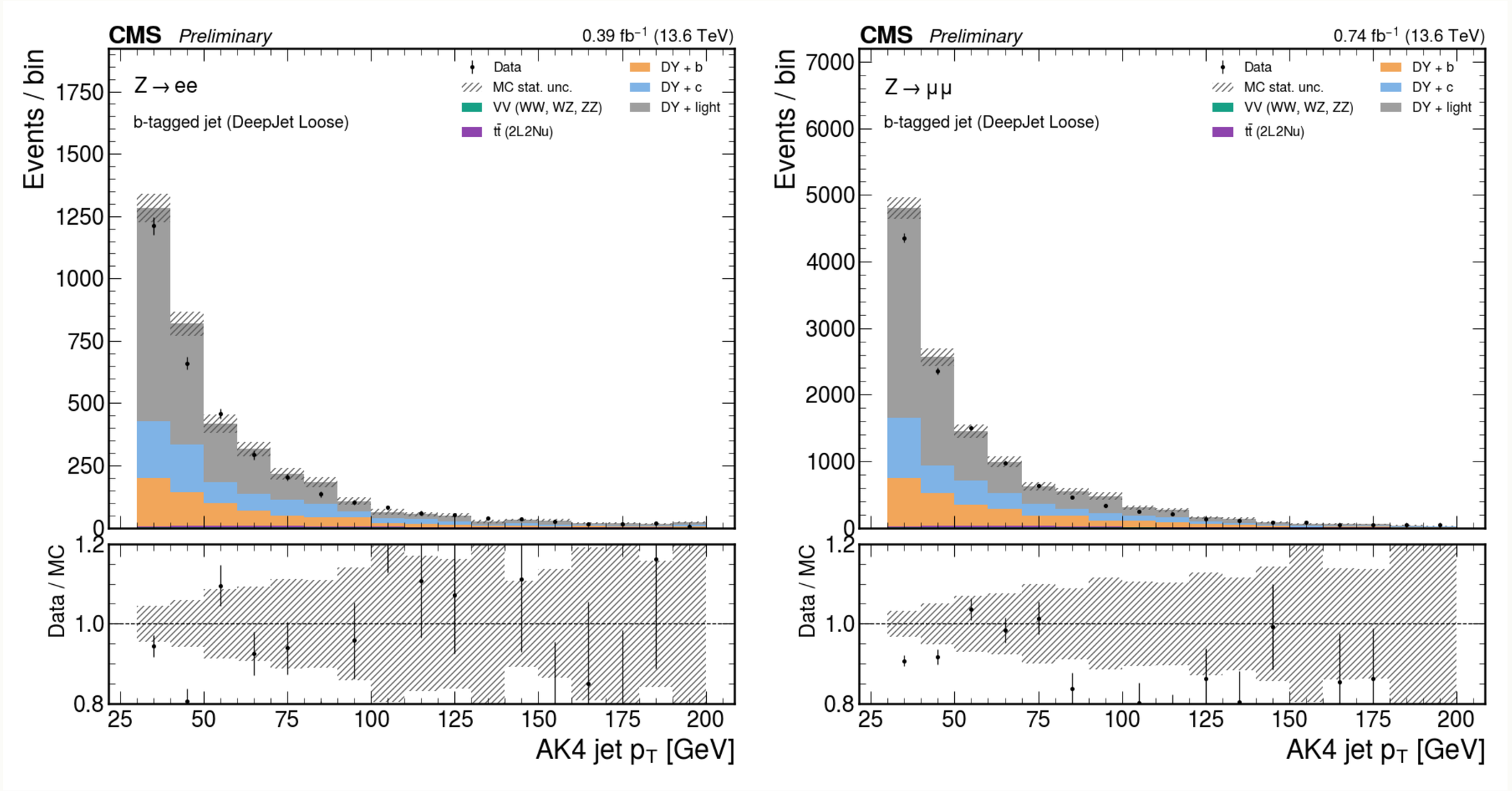
The b-tagged jet.



DeepJet Loose b-tag (DeepJet B > 0.0583)

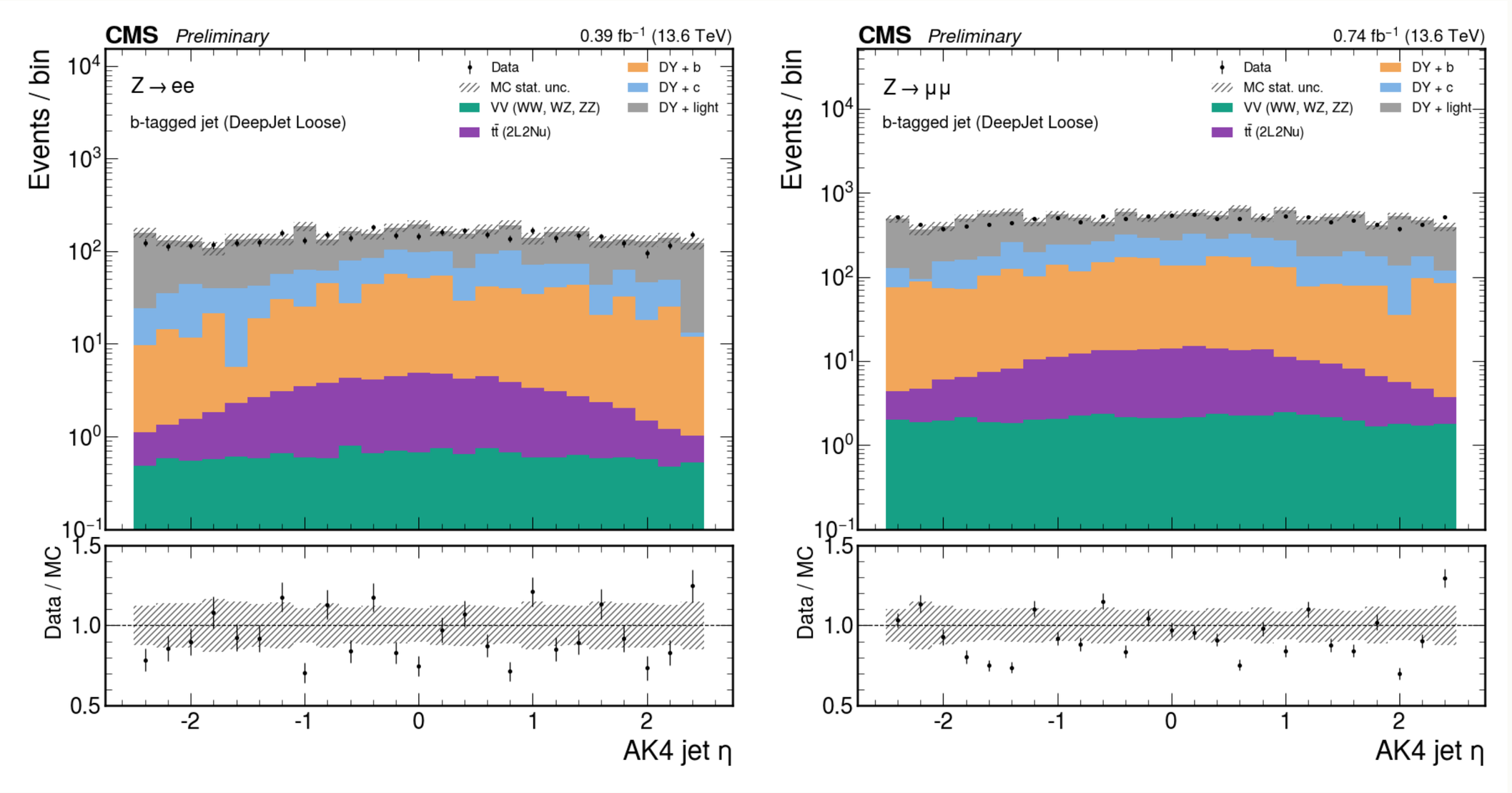
Leading-jet pT, 30–200 GeV

Zoom: linear scale, ratio 0.8–1.2.



DeepJet Loose b-tag (DeepJet B > 0.0583)

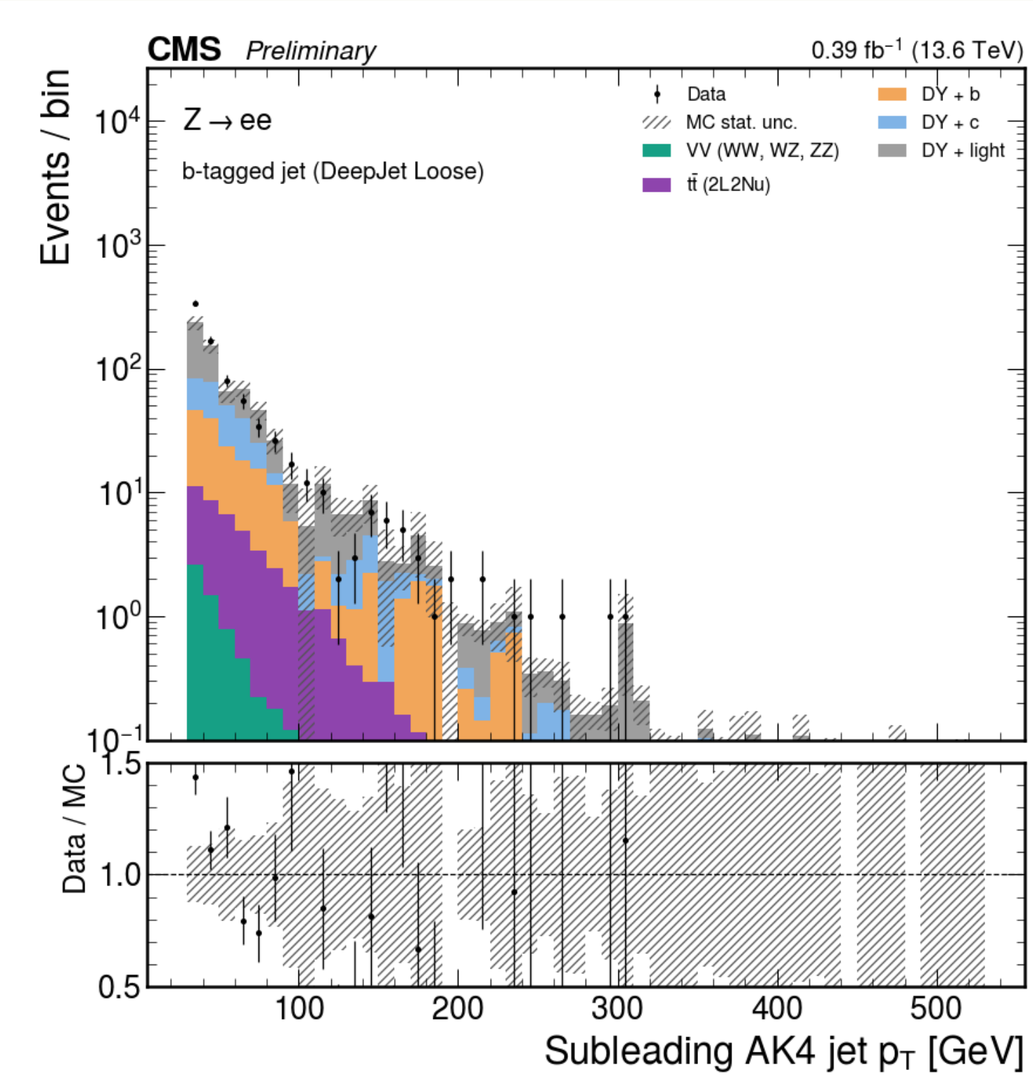
Leading-jet η

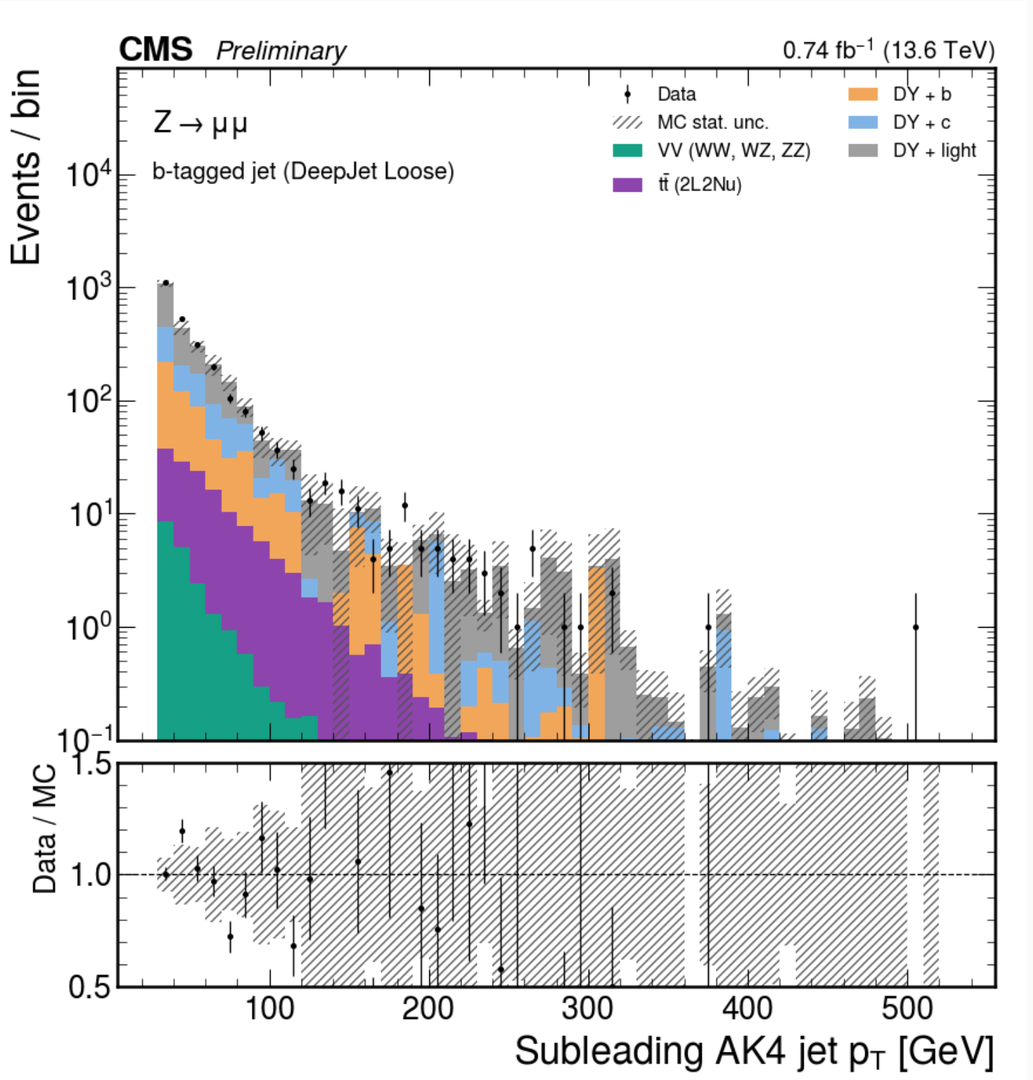


DeepJet Loose b-tag (DeepJet B > 0.0583)

Second-jet pT

Events with at least 2 jets.

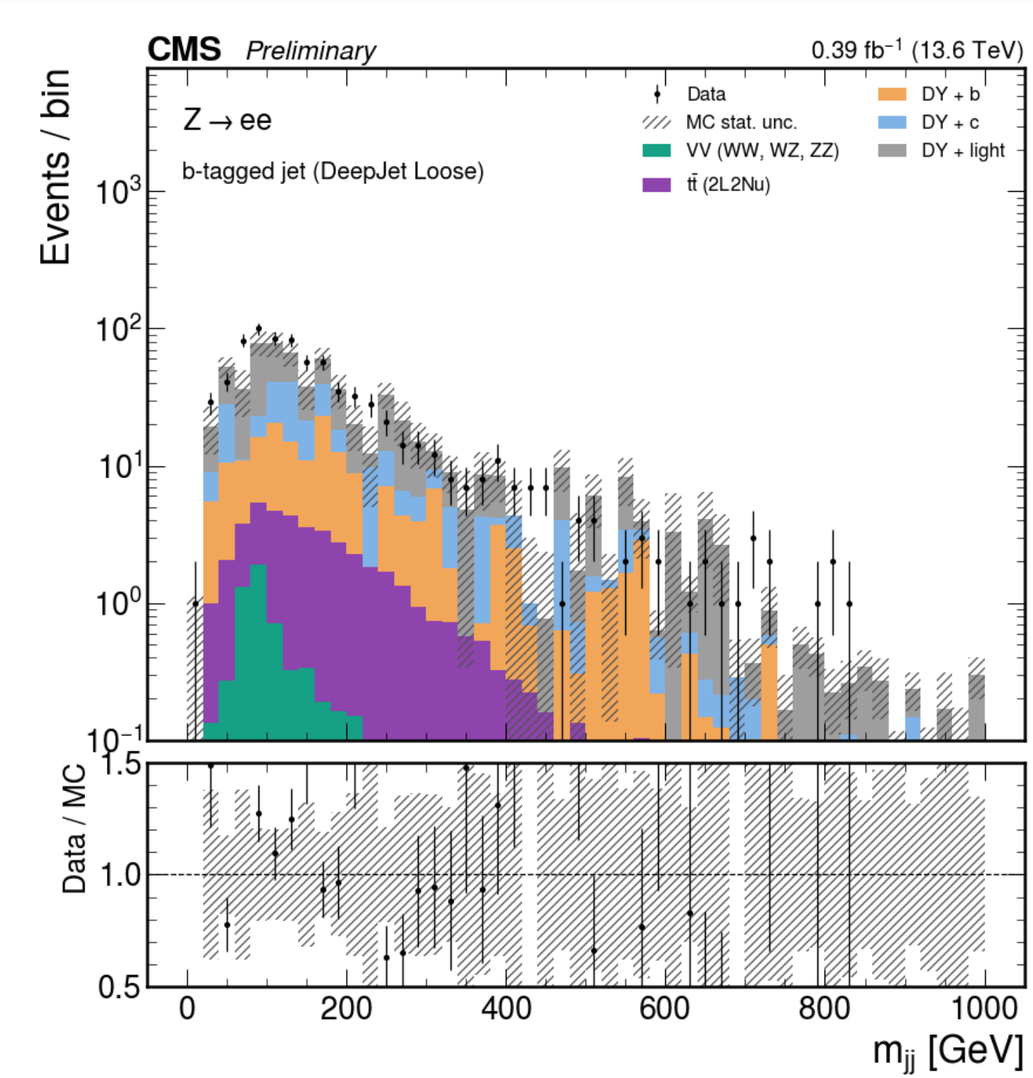


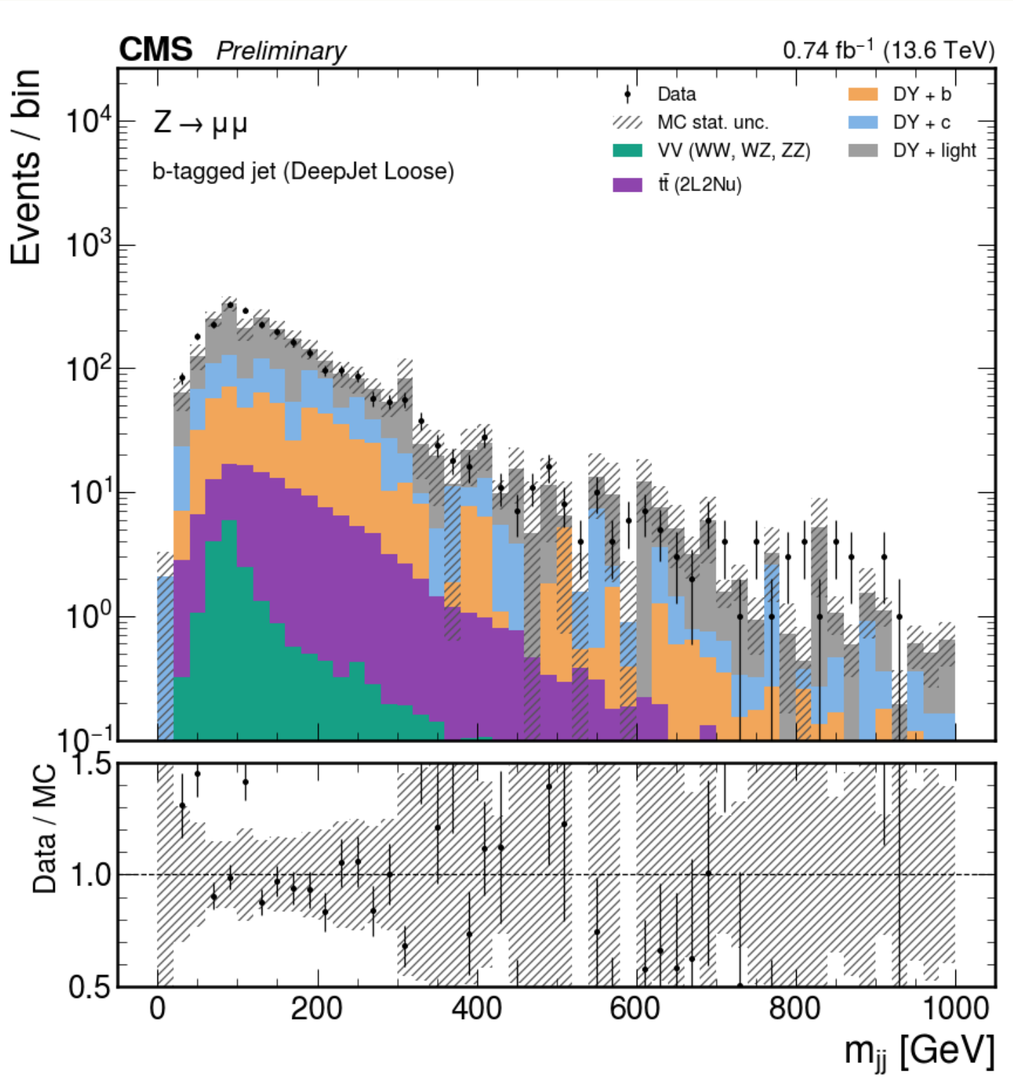


DeepJet Loose b-tag (DeepJet B > 0.0583)

Dijet mass

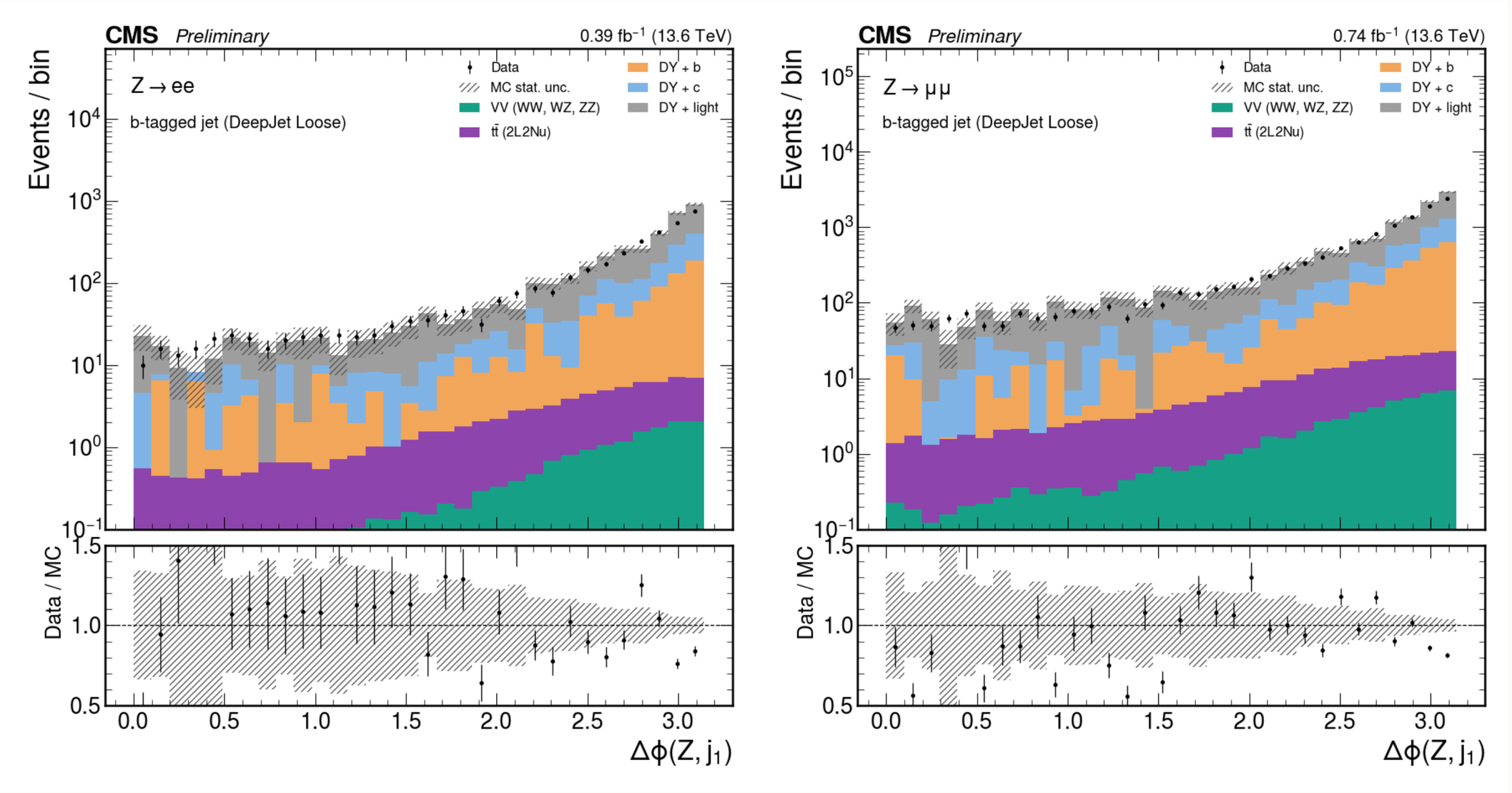
Events with at least 2 jets.





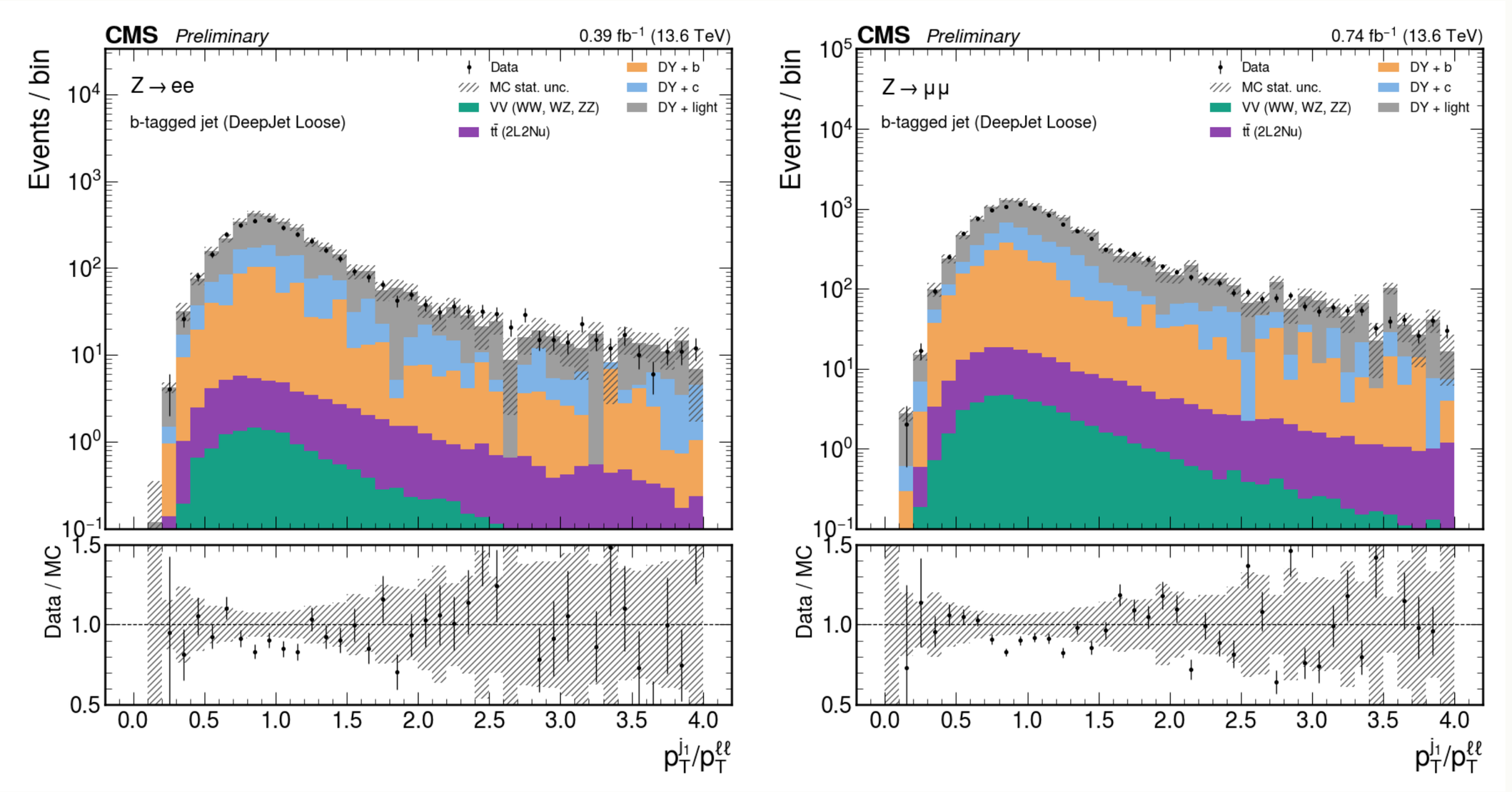
DeepJet Loose b-tag (DeepJet B > 0.0583)

Δφ(Z, leading jet)



DeepJet Loose b-tag (DeepJet B > 0.0583)

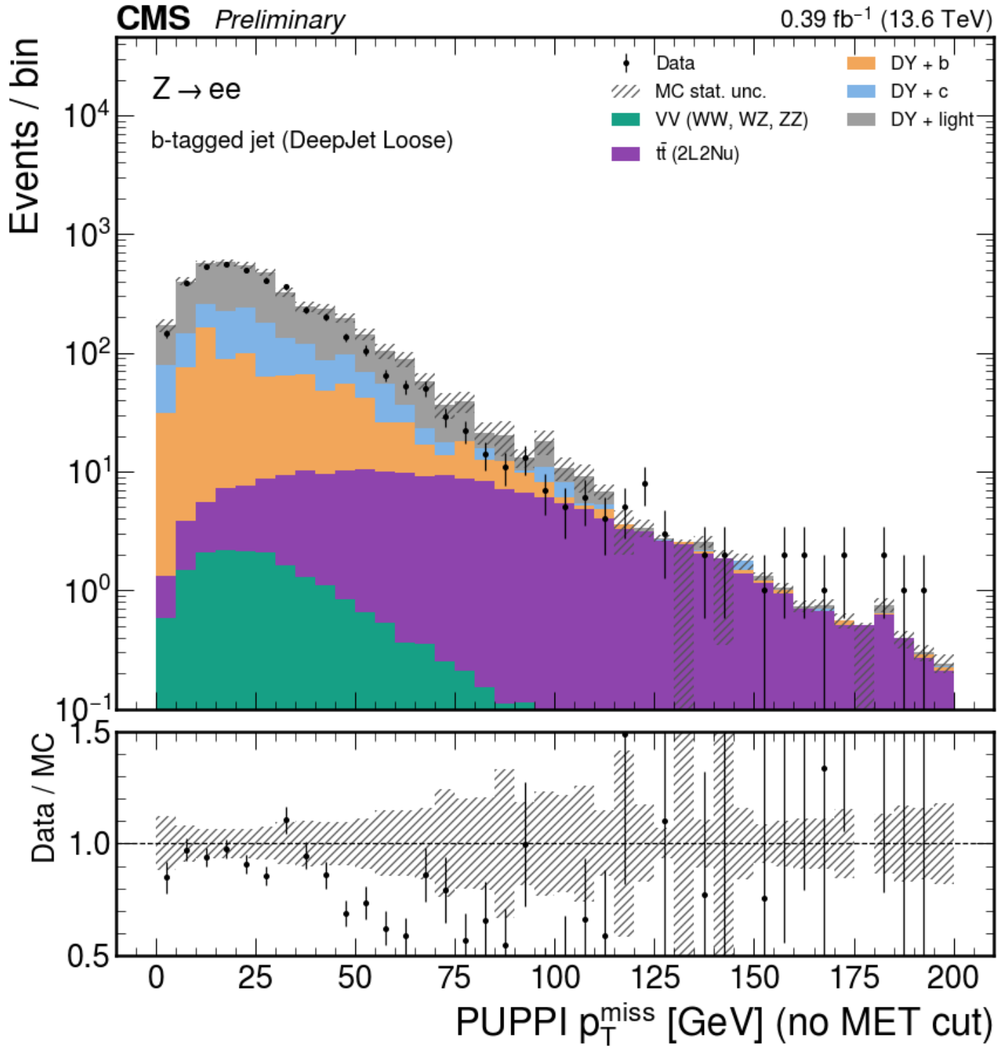
pT balance: leading-jet pT / Z pT

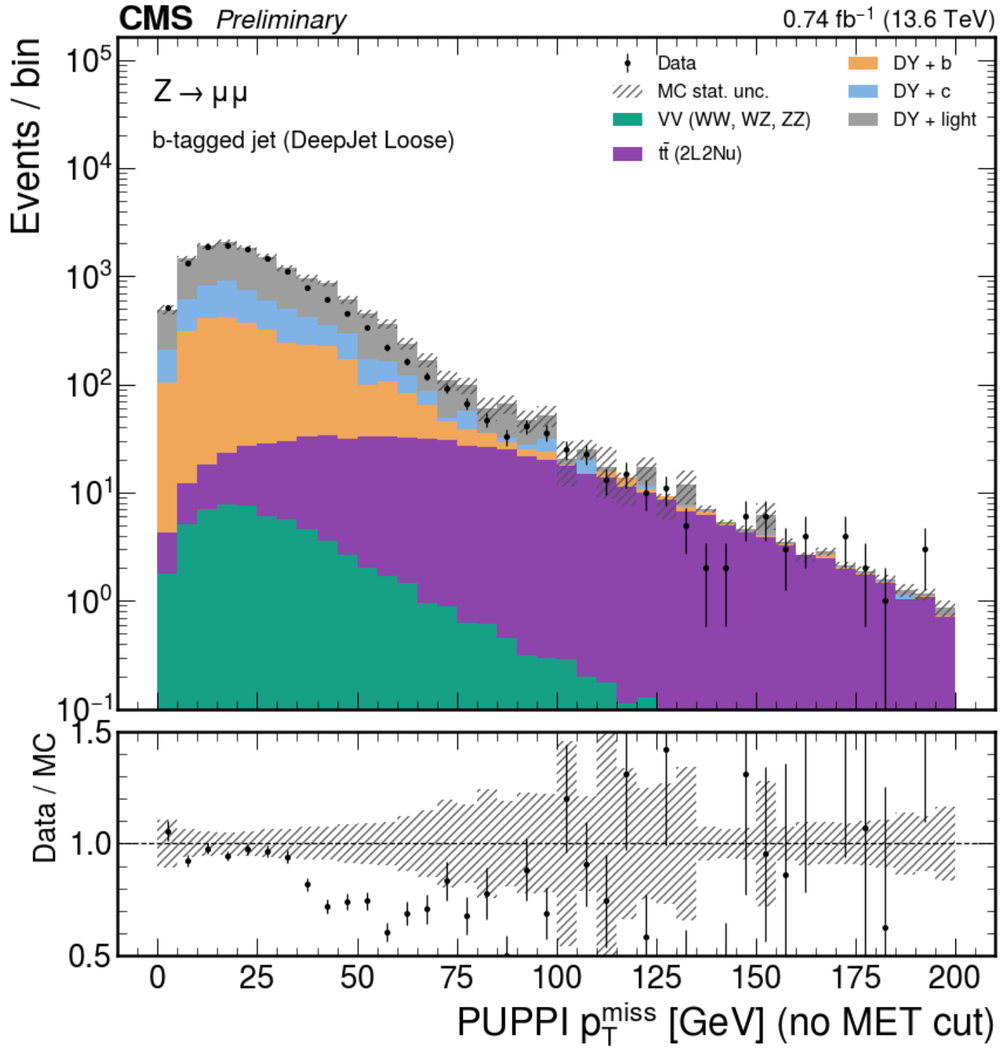


DeepJet Loose b-tag (DeepJet B > 0.0583)

PUPPI MET without the MET cut, up to 200 GeV

All other cuts applied; the selection cuts at 50 GeV.

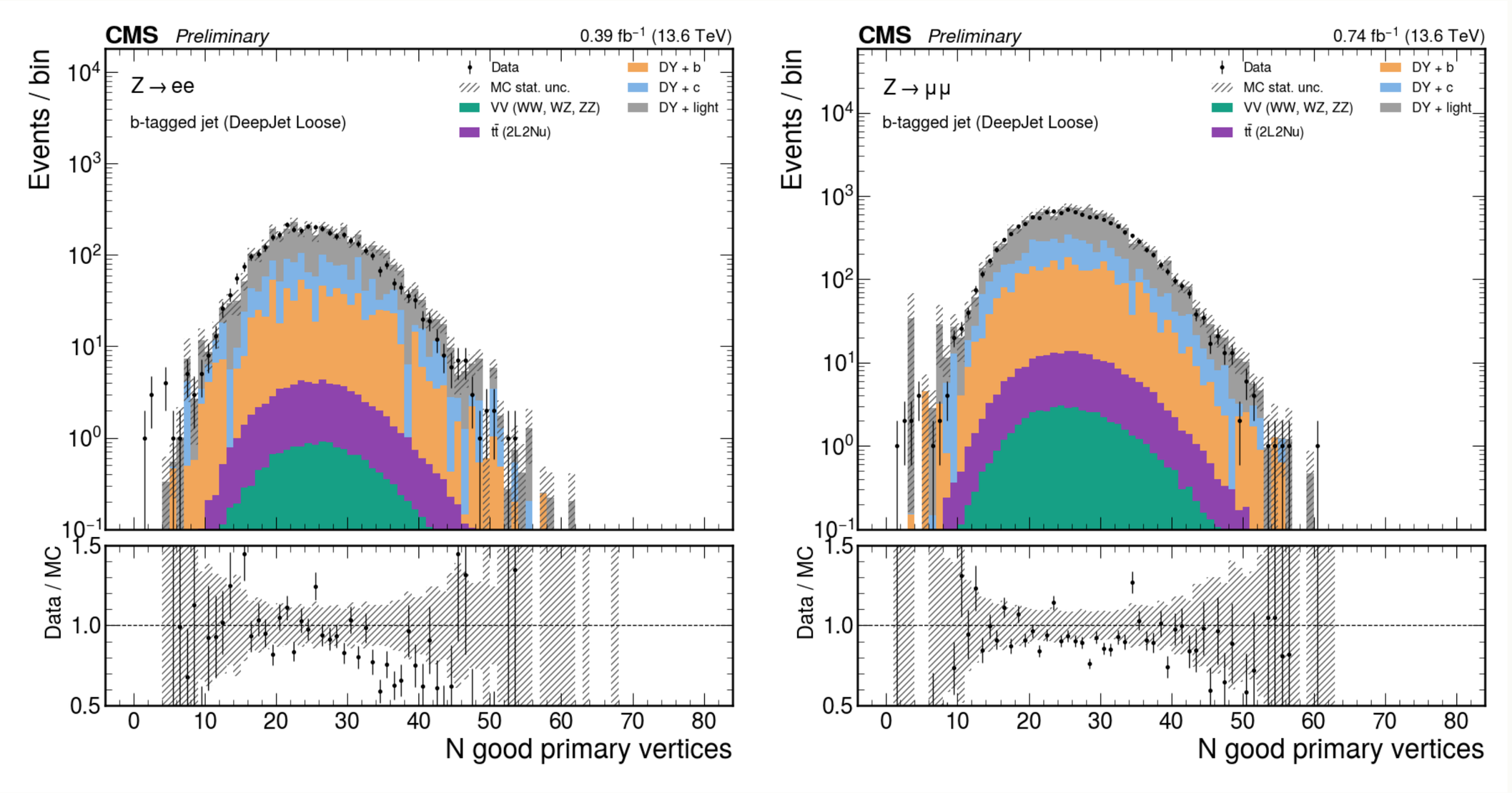




DeepJet Loose b-tag (DeepJet B > 0.0583)

Number of good primary vertices

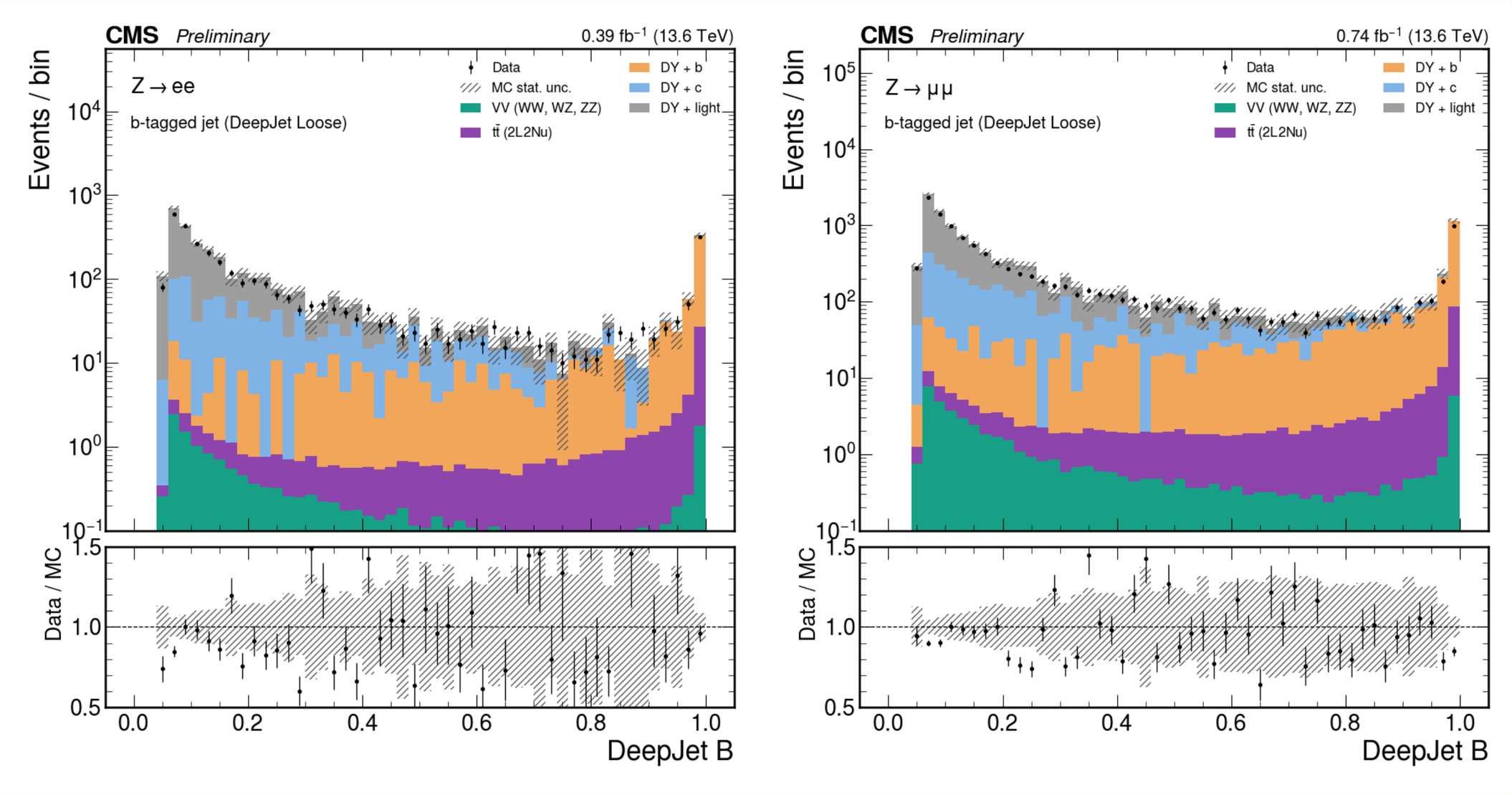
After pileup weights.



DeepJet Loose b-tag (DeepJet B > 0.0583)

Leading-jet DeepJet B score

Only jets above 0.0583 enter this region.



RobustParT, Loose b-tag

Leading jet RobustParT B > 0.0849, with the robustParticleTransformer BTV SFs

	data/MC	DY+b	DY+c	DY+light	tt	tagged MC	tagged data
Z→ee	0.962 ± 0.031	20%	22%	55%	2%	17.2%	17.4%
Z→μμ	0.981 ± 0.023	22%	21%	54%	2%	17.2%	17.9%

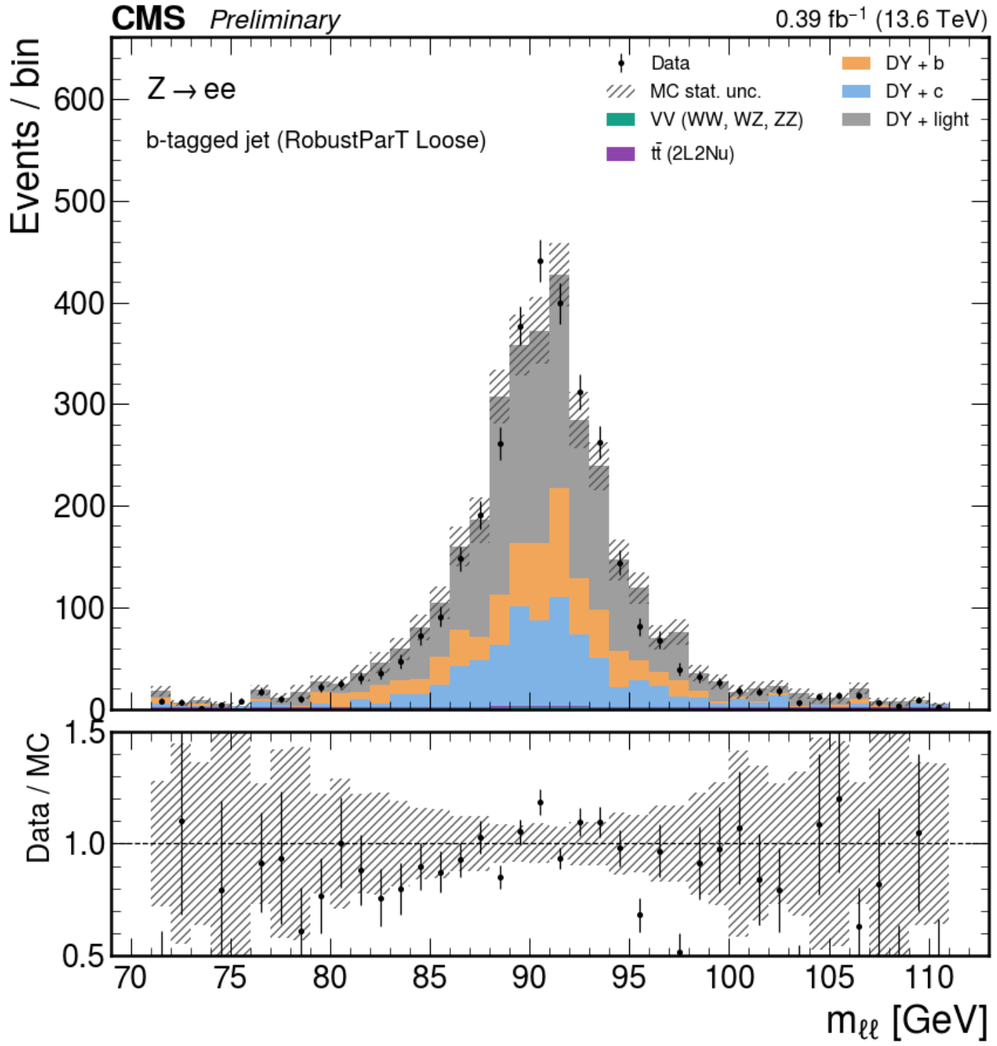
Flavour shares are of all MC. Tagged = fraction of Z + jet events with a tagged leading jet. Errors are statistical only.

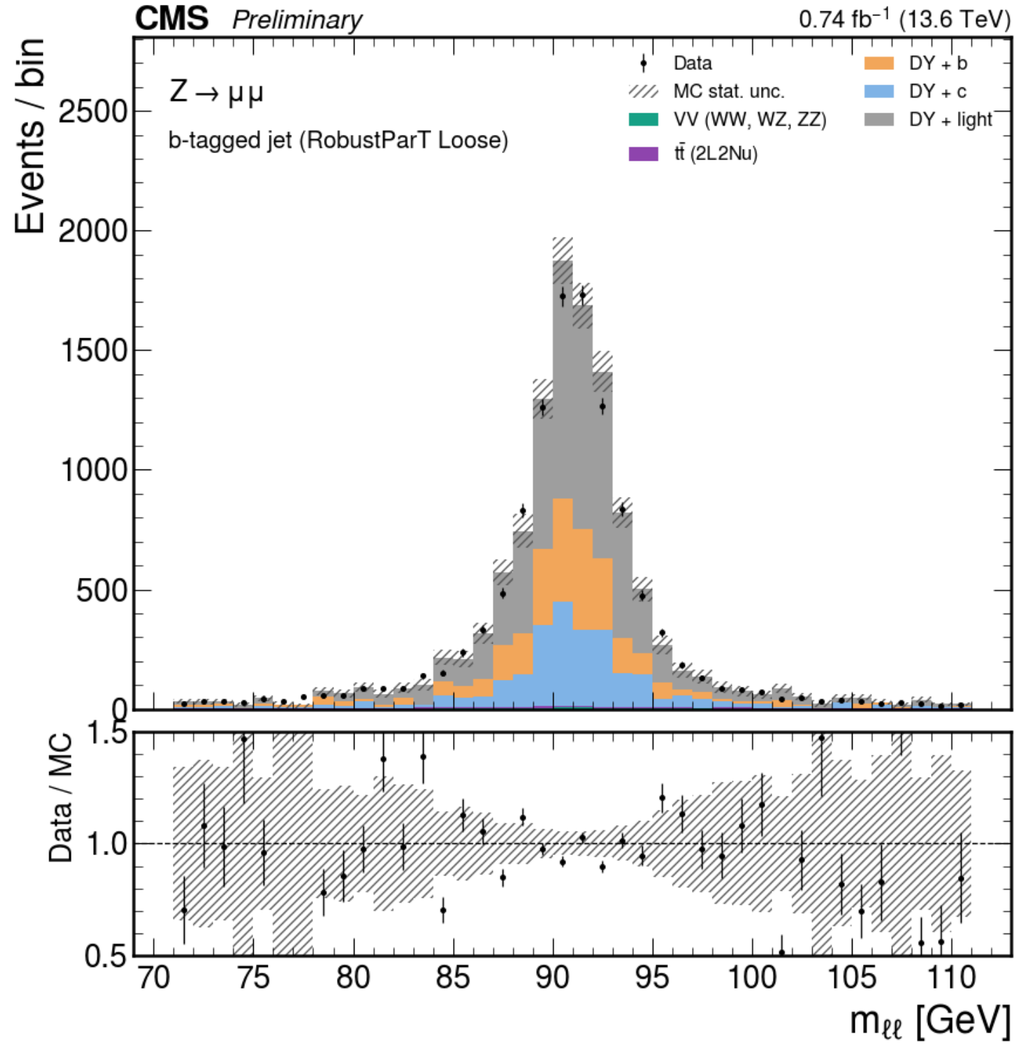
19 plots follow. Left Z→ee, right Z→μμ.

RobustParT Loose b-tag (RobustParT B > 0.0849)

Dilepton mass

71 < m(ℓℓ) < 111 GeV, linear scale.

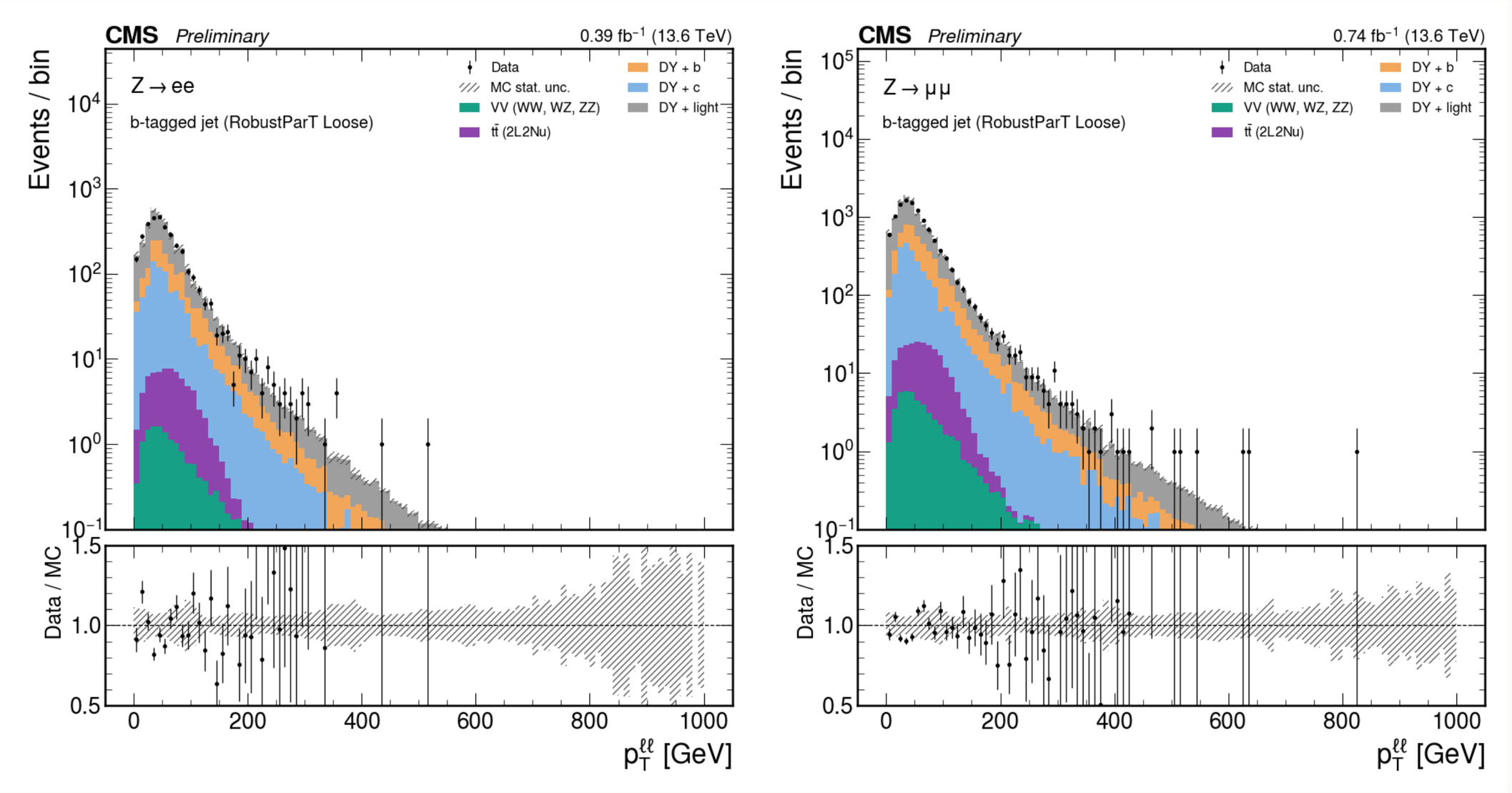




RobustParT Loose b-tag (RobustParT B > 0.0849)

Z pT

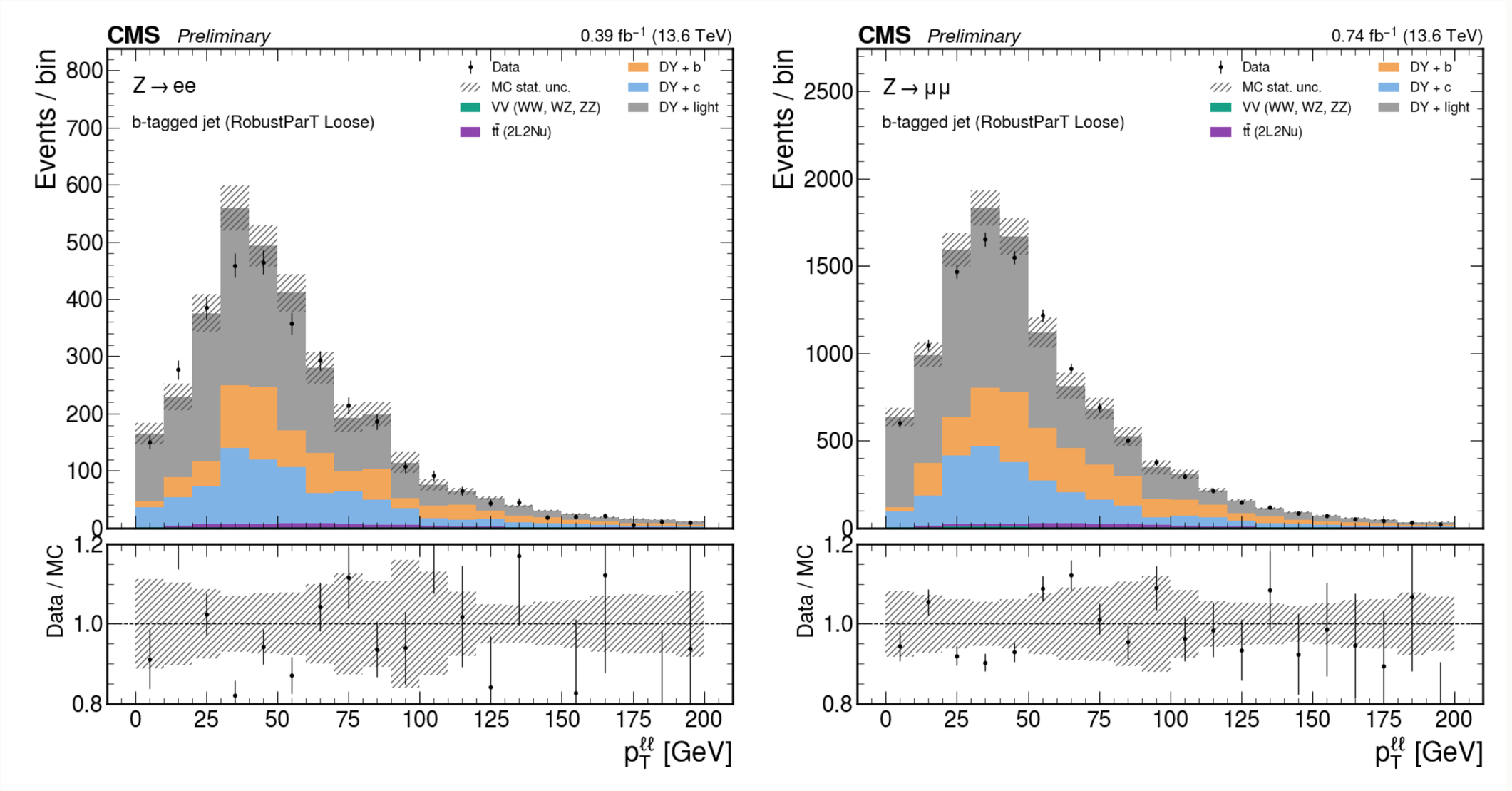
pT of the lepton pair.



RobustParT Loose b-tag (RobustParT B > 0.0849)

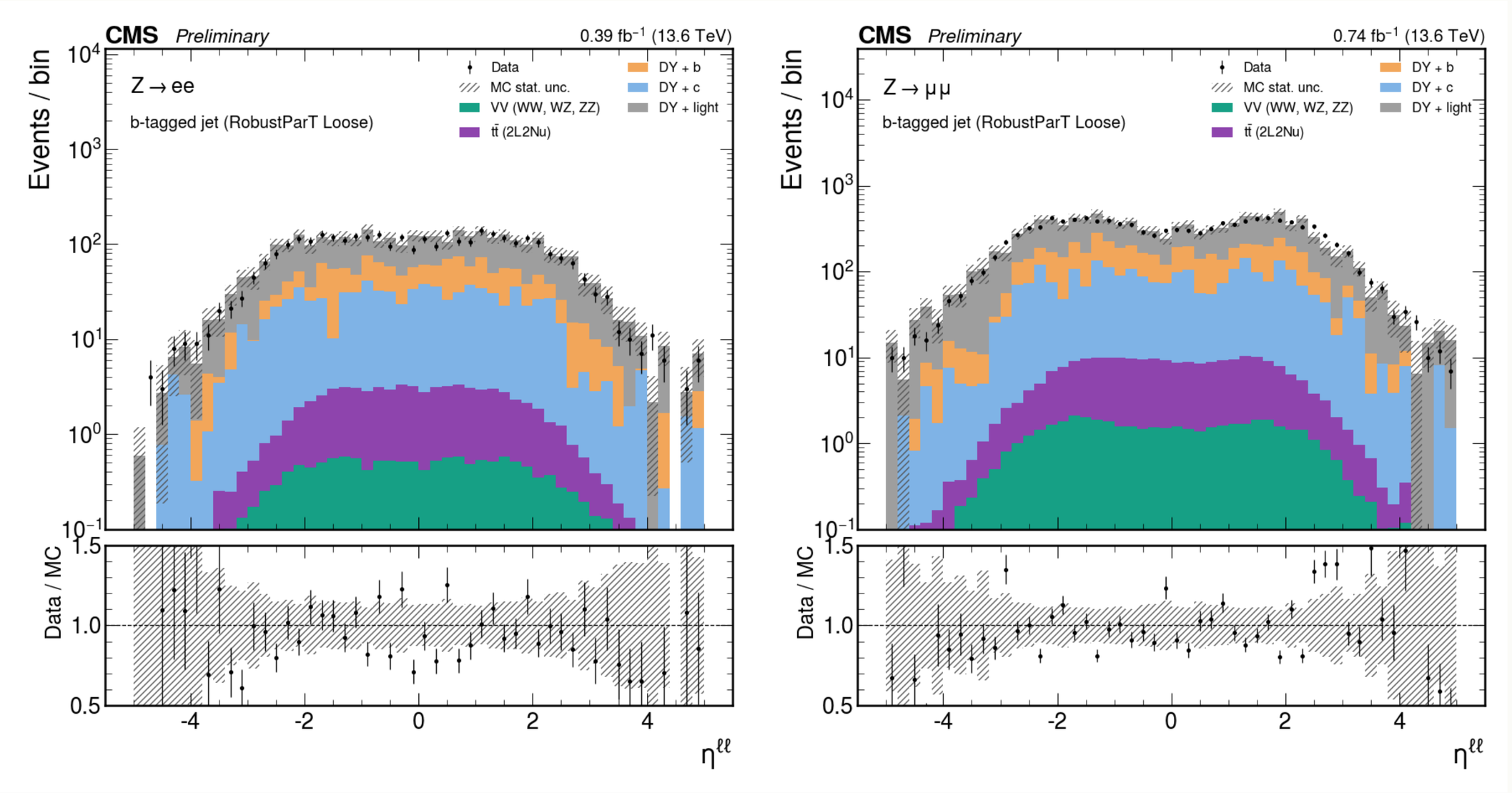
Z pT, 0–200 GeV

Zoom: linear scale, ratio 0.8–1.2.



RobustParT Loose b-tag (RobustParT B > 0.0849)

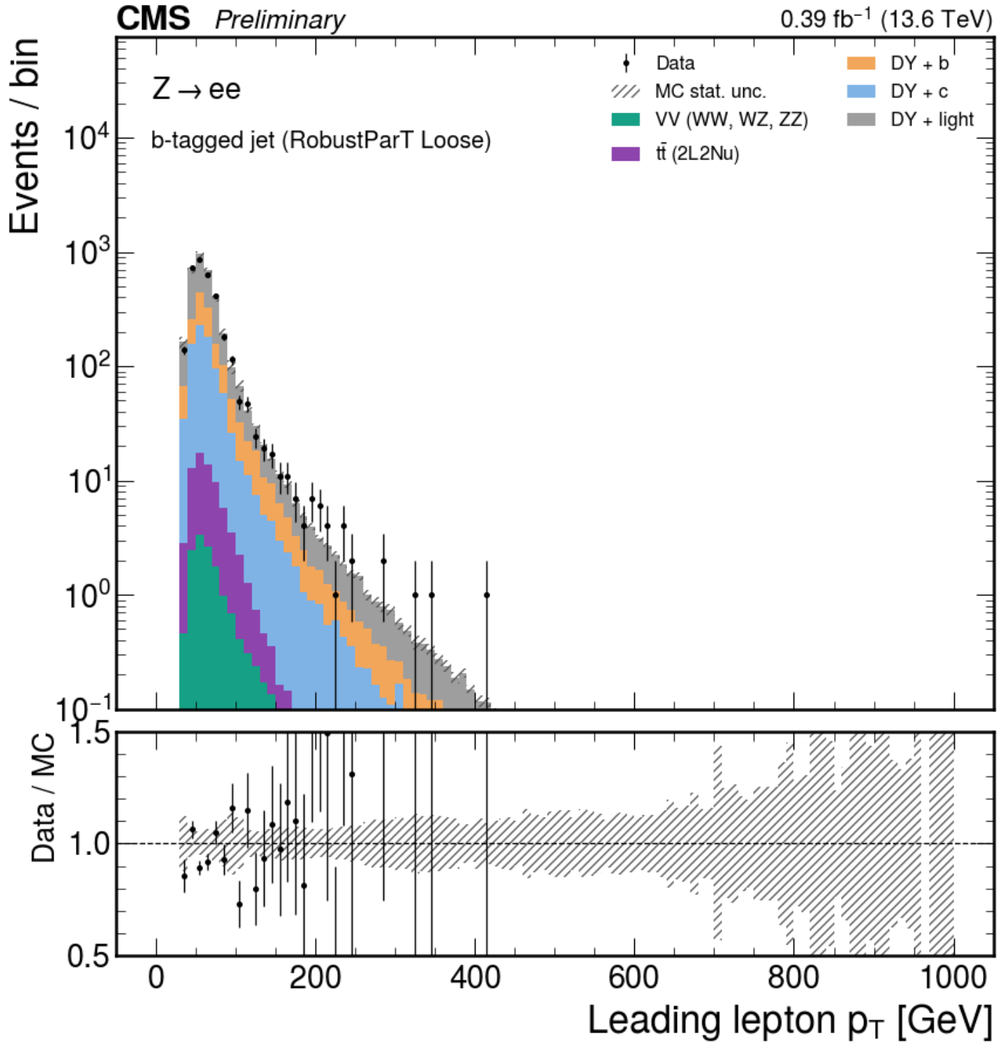
Z η

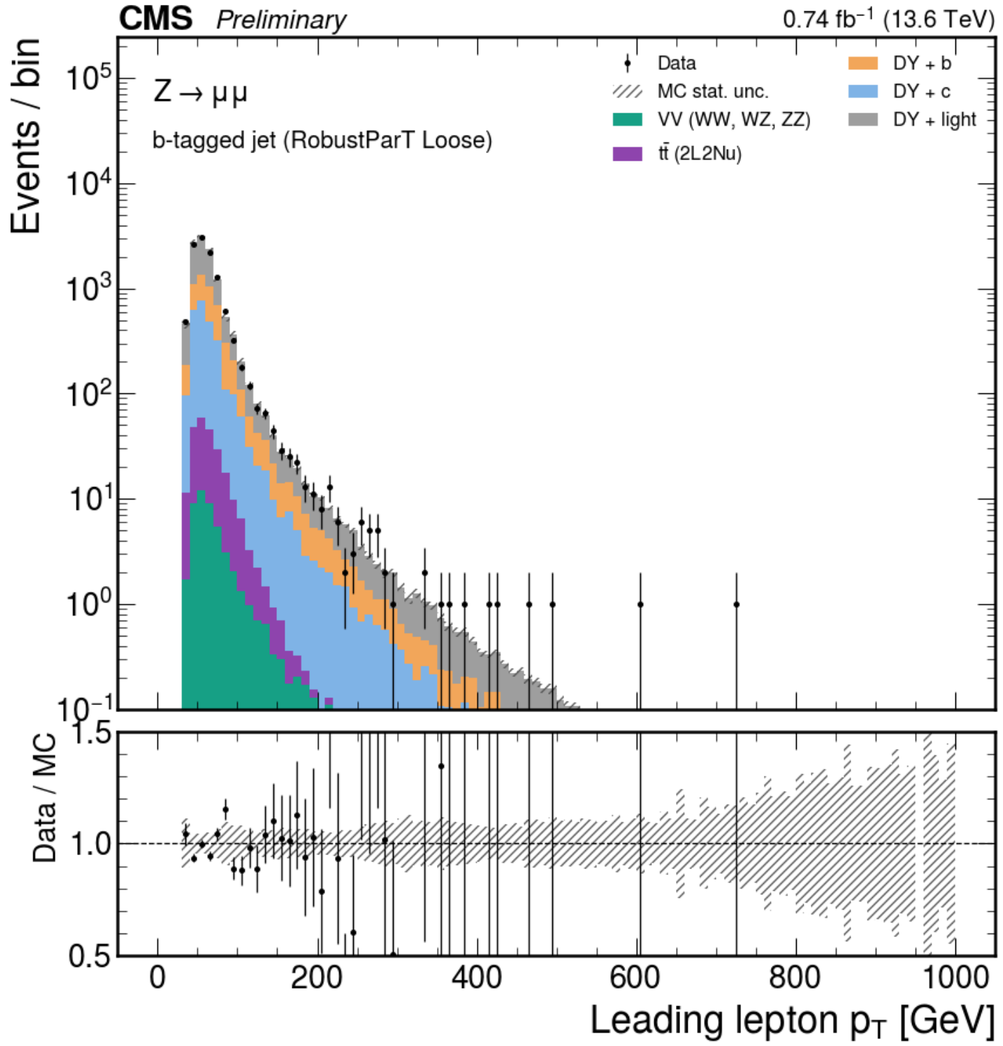


RobustParT Loose b-tag (RobustParT B > 0.0849)

Leading-lepton pT

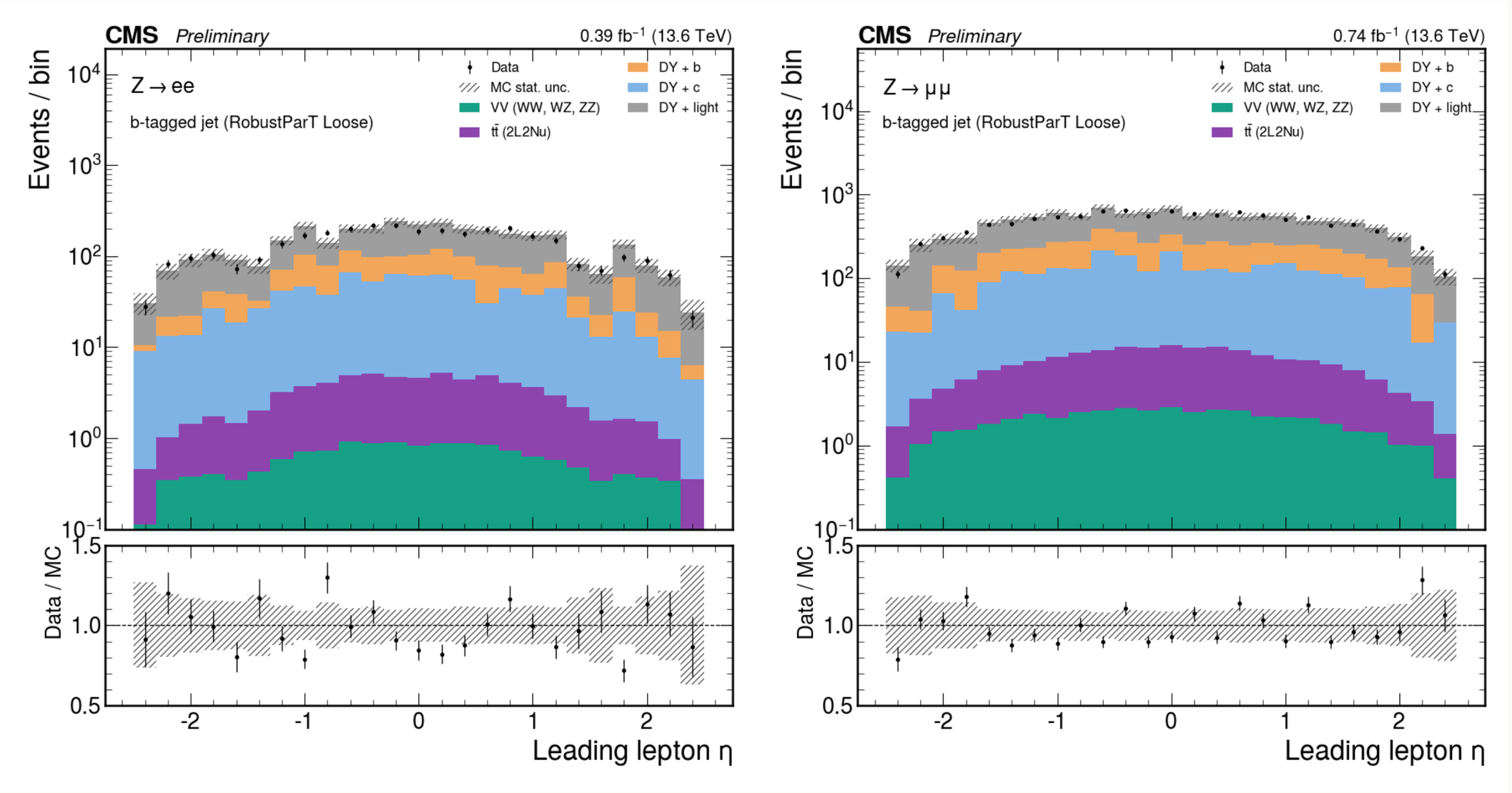
Leading lepton pT > 35 GeV.





RobustParT Loose b-tag (RobustParT B > 0.0849)

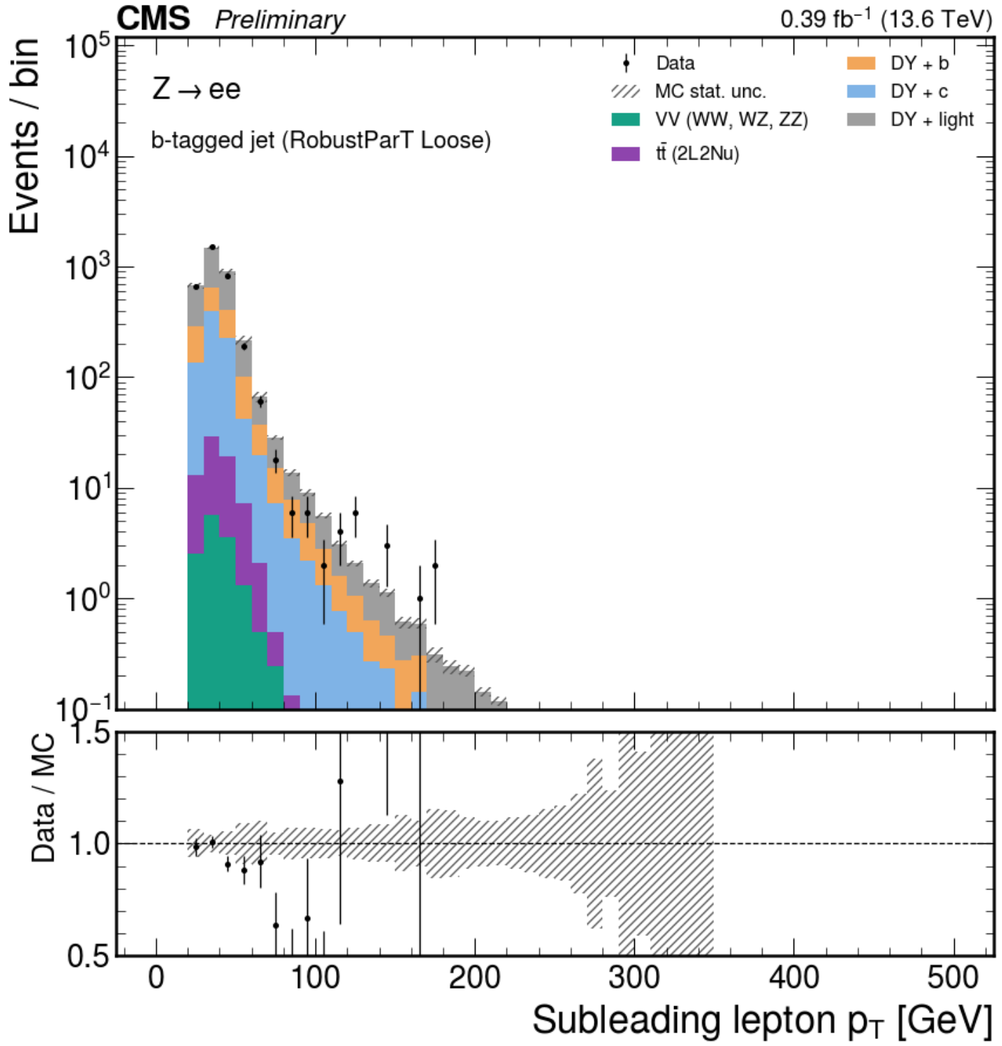
Leading-lepton η

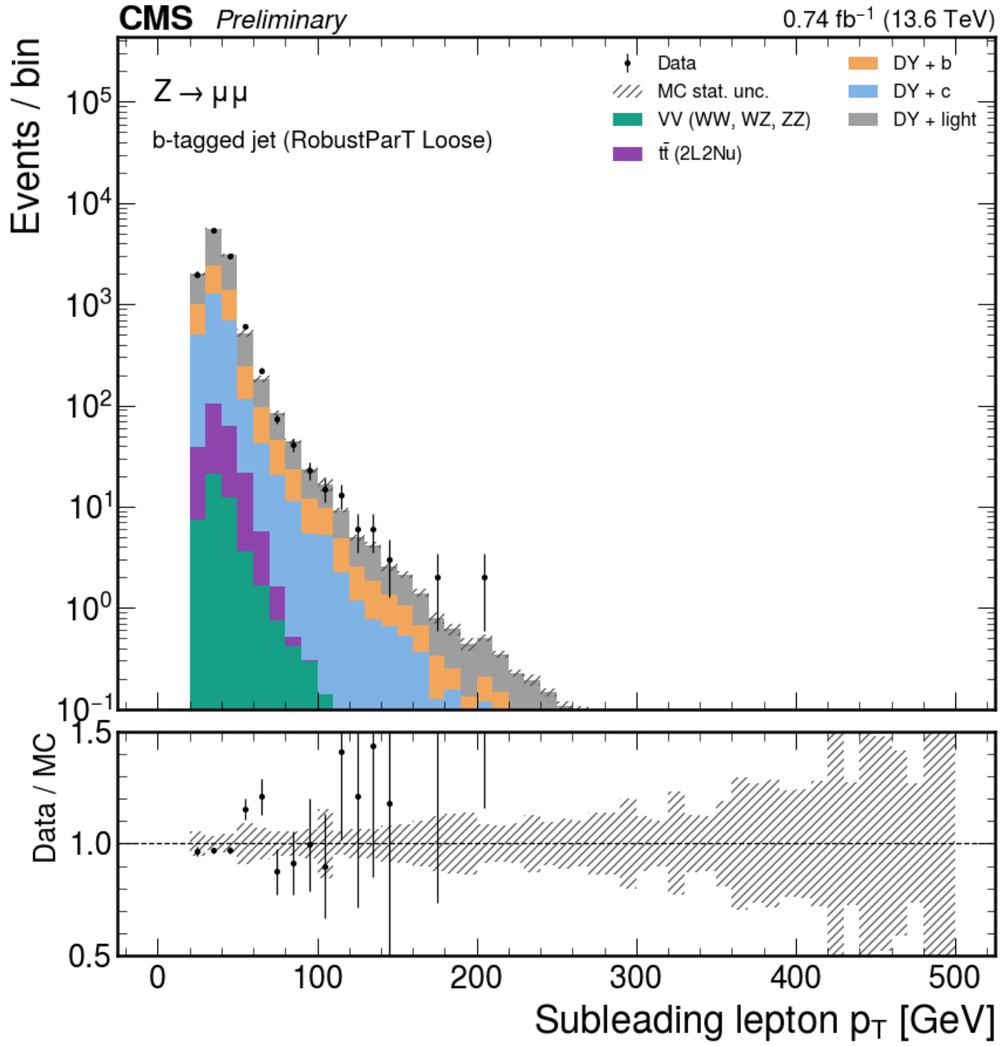


RobustParT Loose b-tag (RobustParT B > 0.0849)

Second-lepton pT

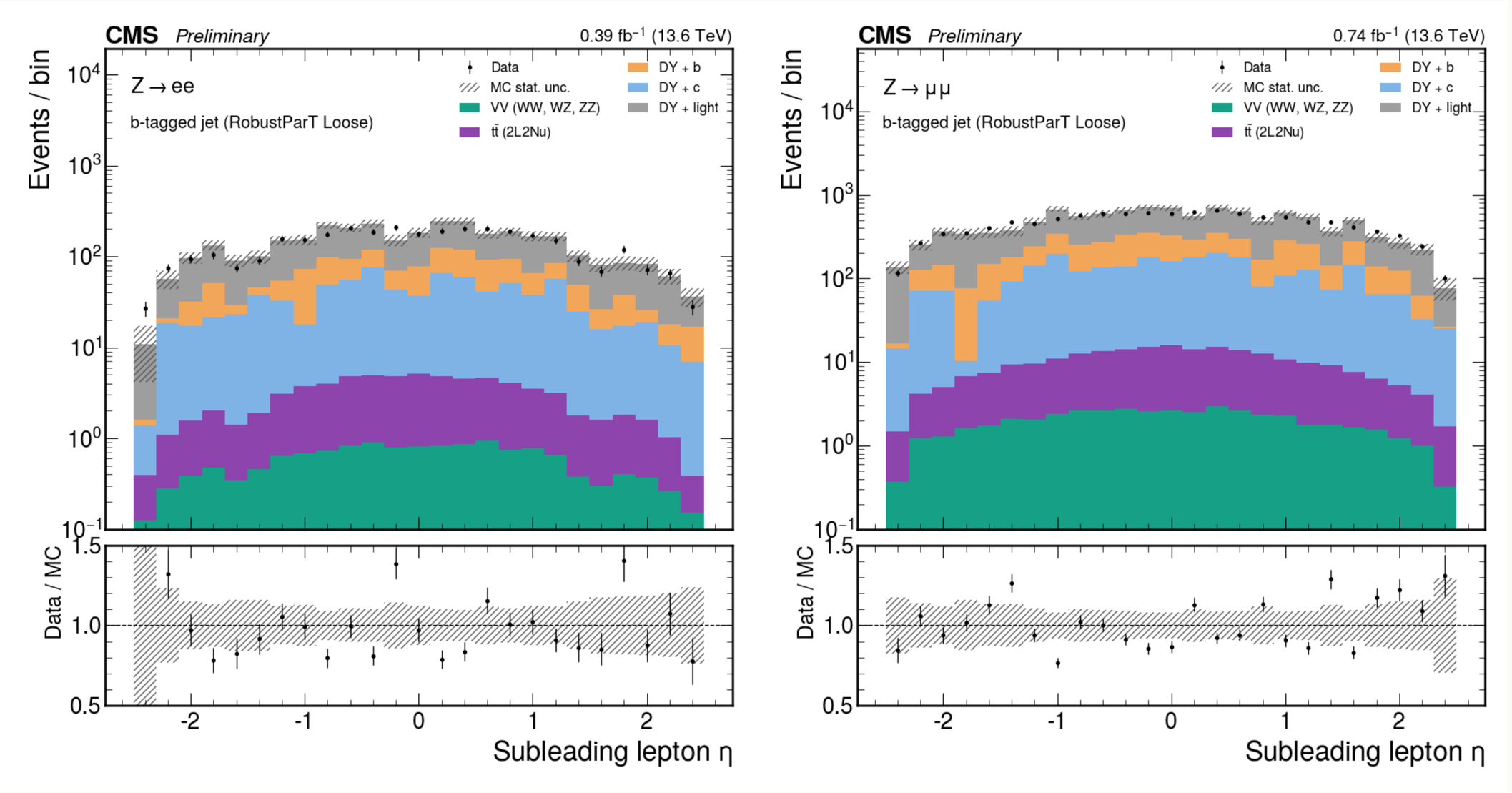
Second lepton pT > 25 GeV (electrons), > 26 GeV (muons, new: trigger SF range).





RobustParT Loose b-tag (RobustParT B > 0.0849)

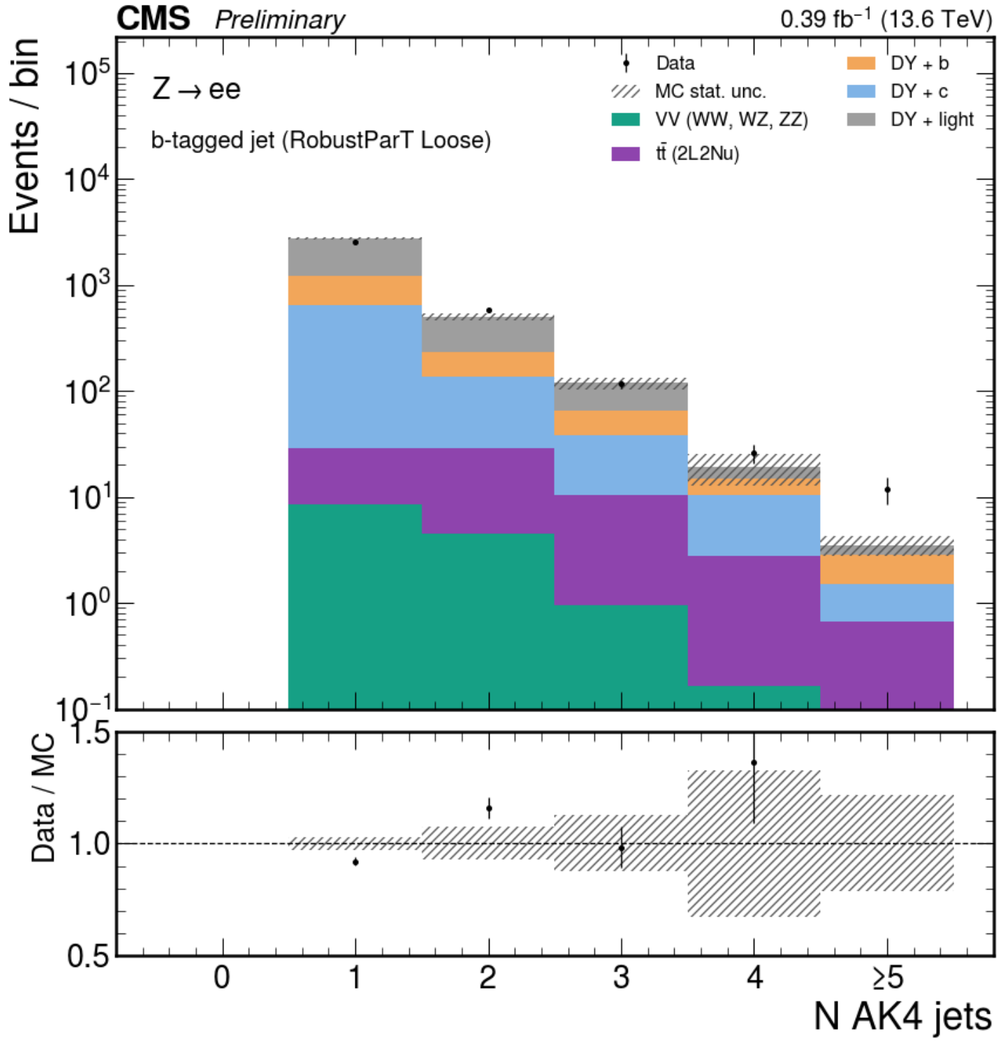
Second-lepton η

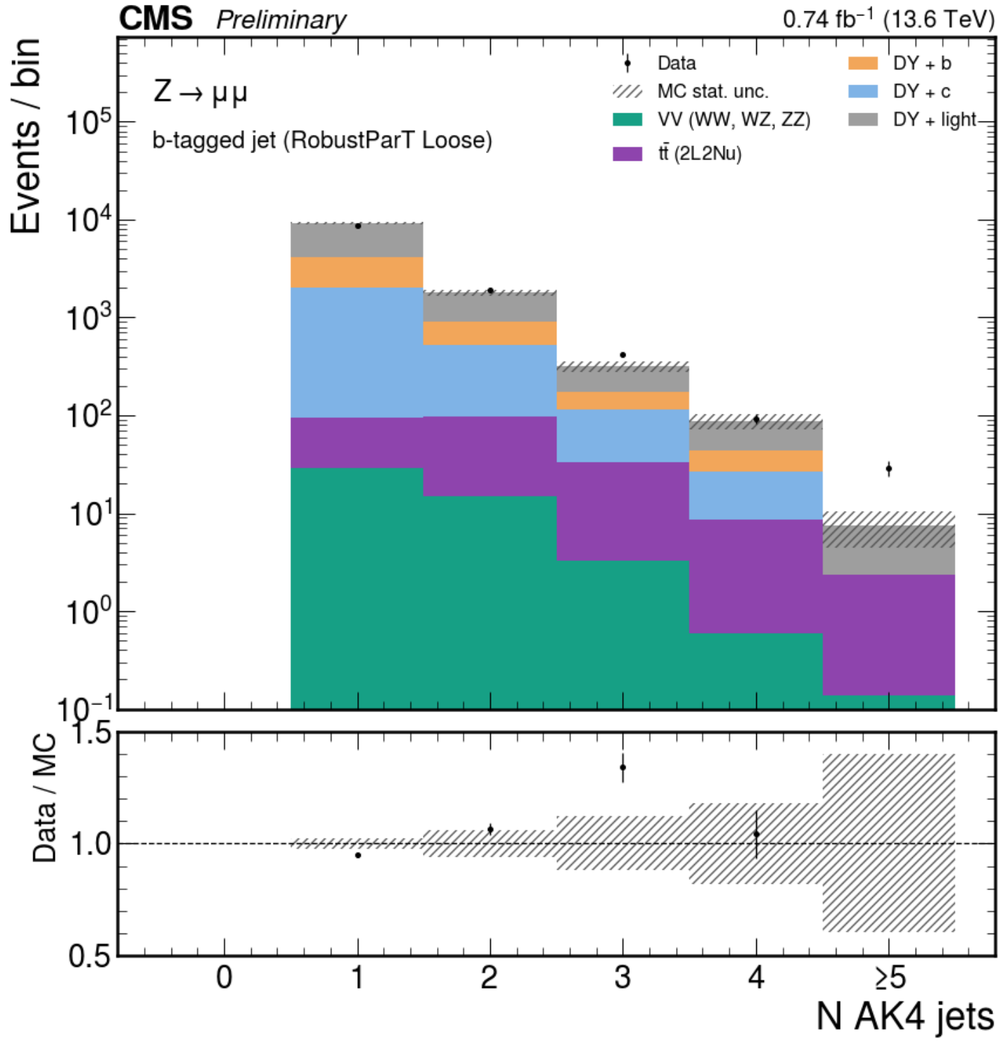


RobustParT Loose b-tag (RobustParT B > 0.0849)

Number of jets

AK4 jets with pT > 30 GeV, |η| < 2.5, away from the leptons.

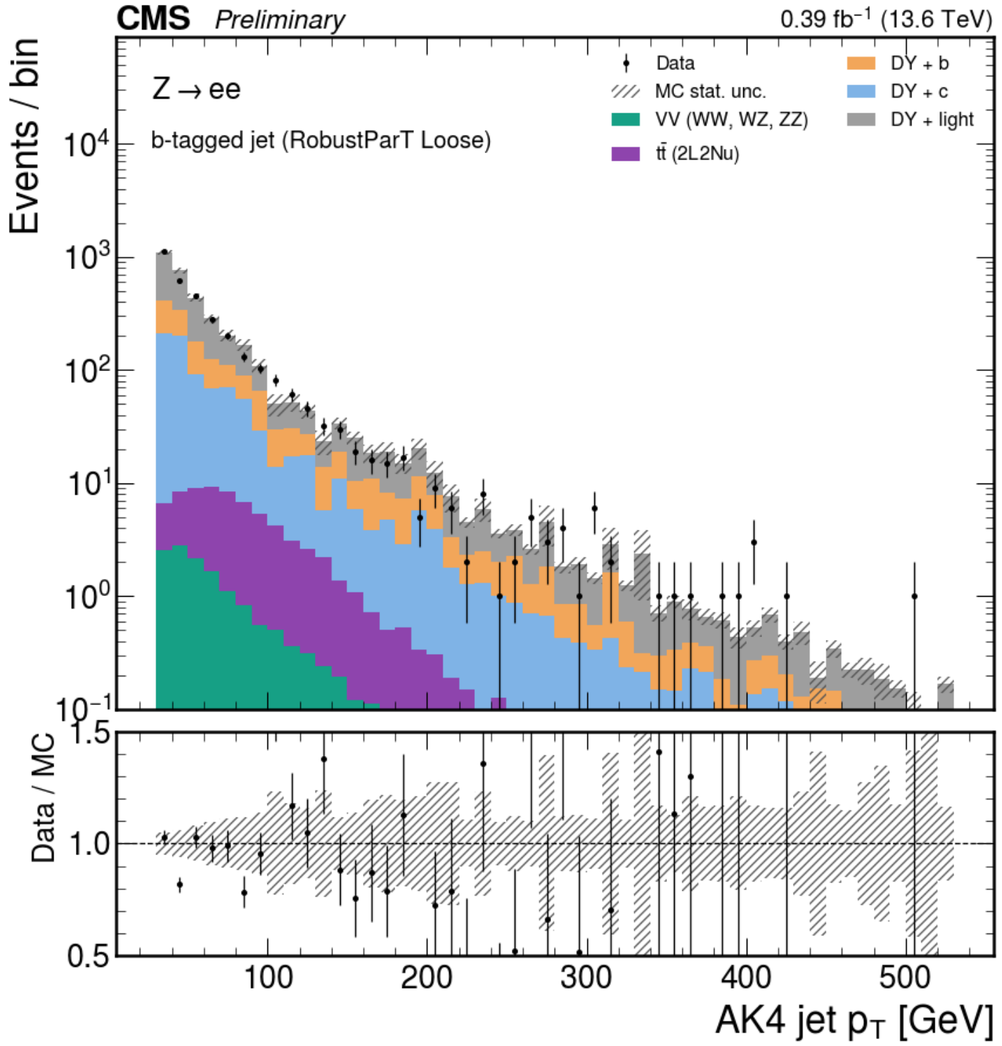


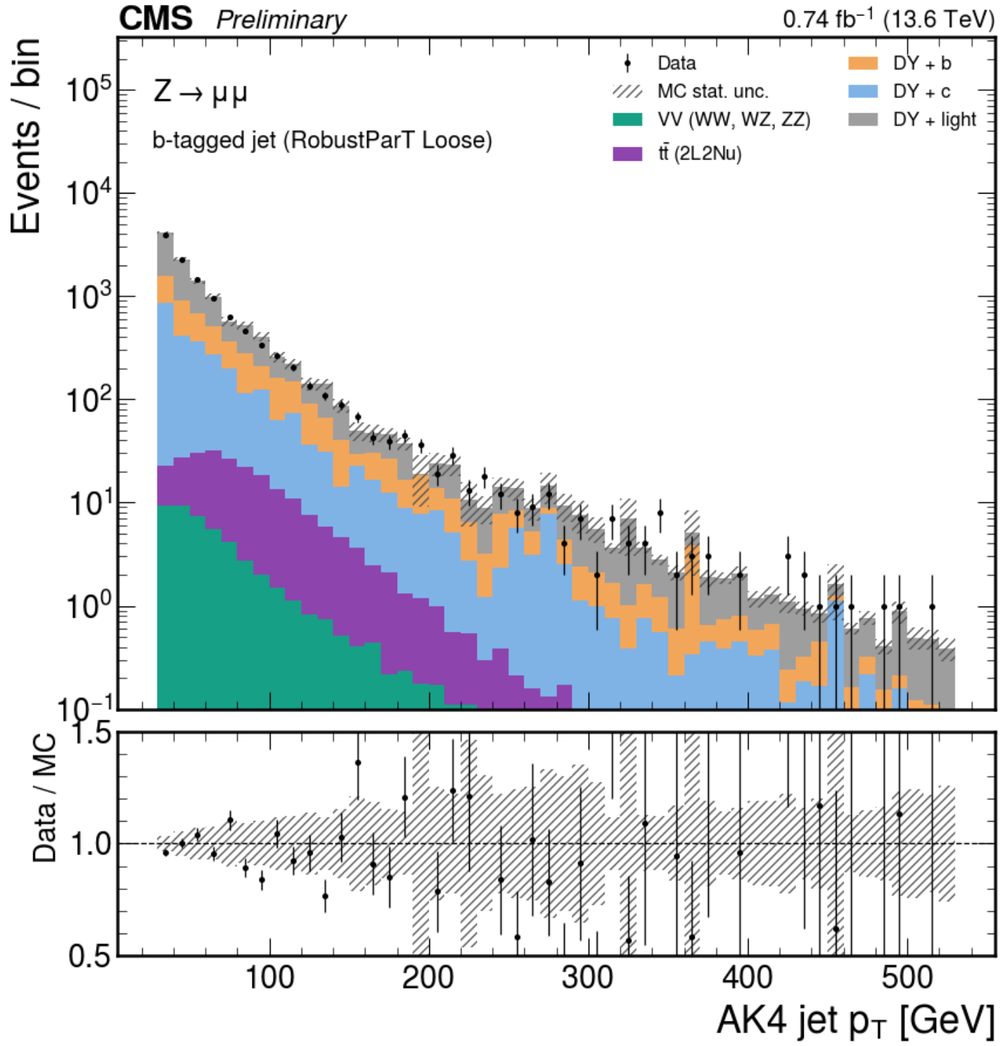


RobustParT Loose b-tag (RobustParT B > 0.0849)

Leading-jet pT

The b-tagged jet.

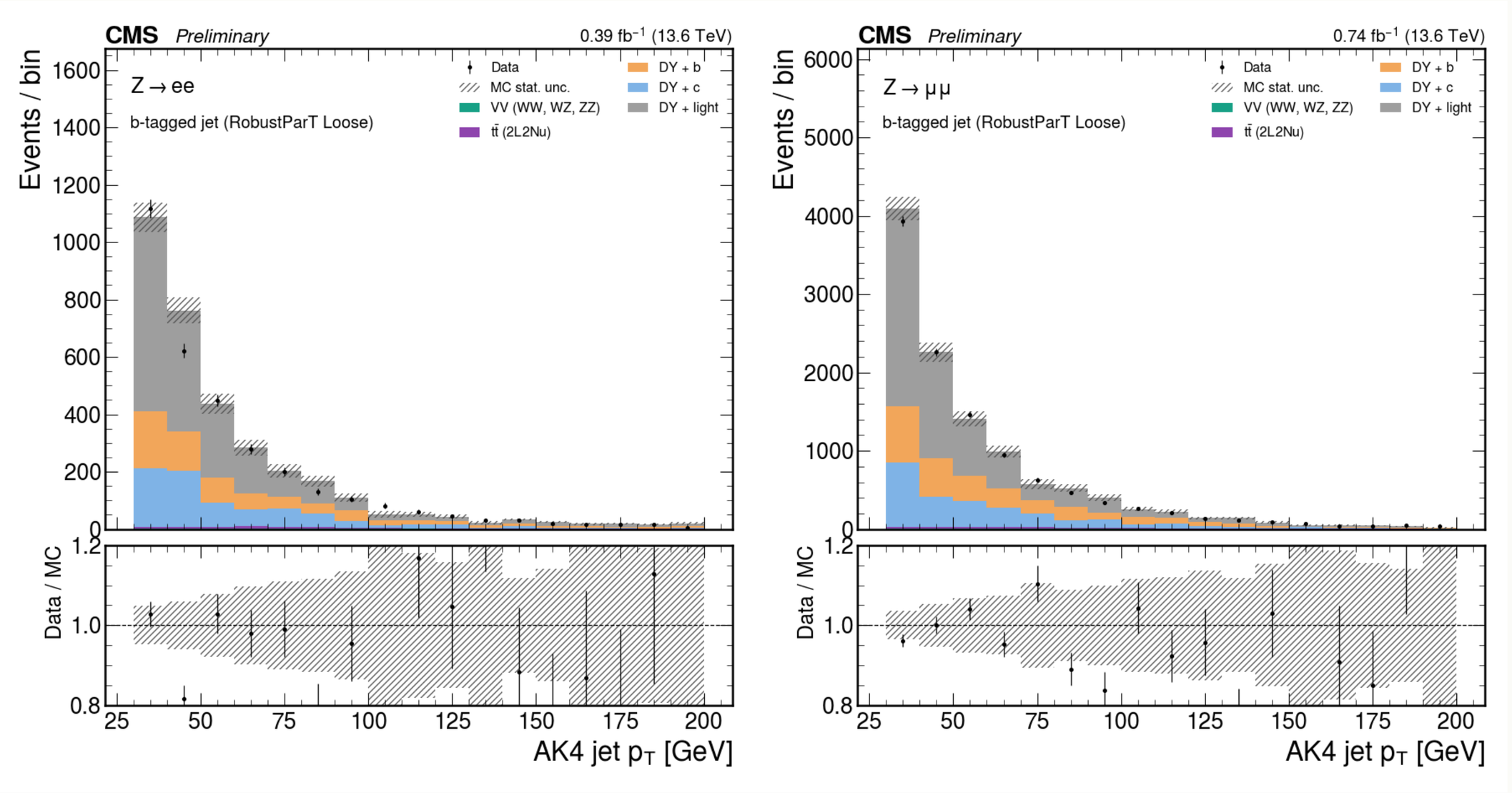




RobustParT Loose b-tag (RobustParT B > 0.0849)

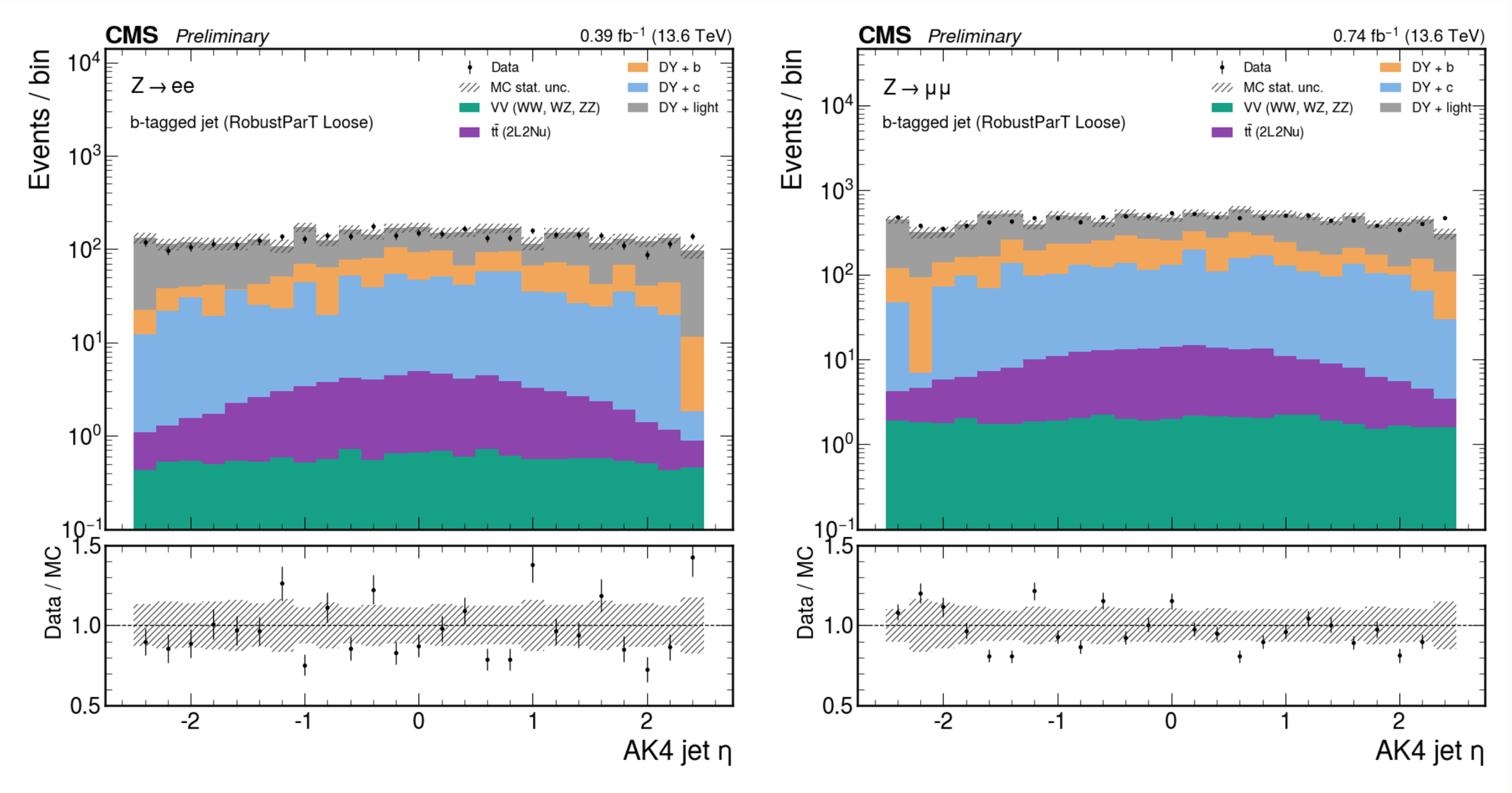
Leading-jet pT, 30–200 GeV

Zoom: linear scale, ratio 0.8–1.2.



RobustParT Loose b-tag (RobustParT B > 0.0849)

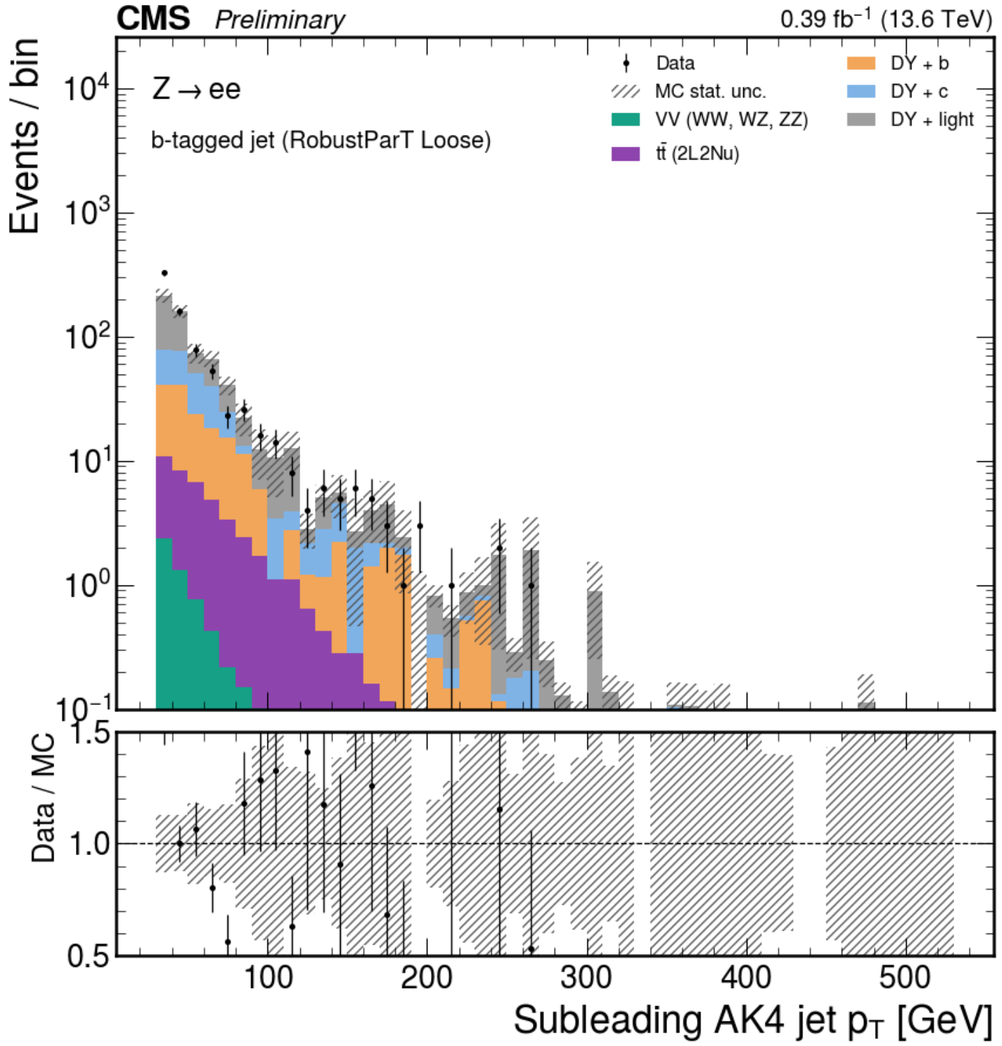
Leading-jet η

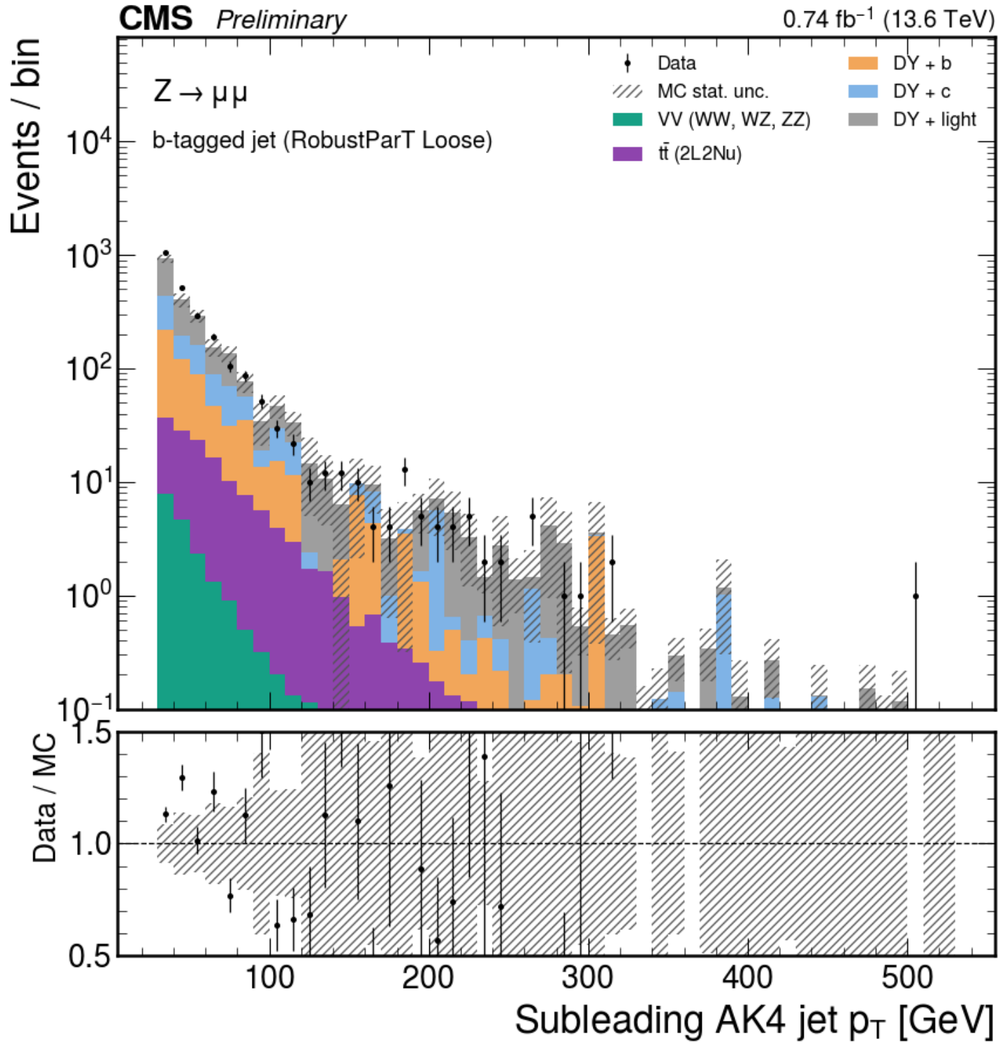


RobustParT Loose b-tag (RobustParT B > 0.0849)

Second-jet pT

Events with at least 2 jets.

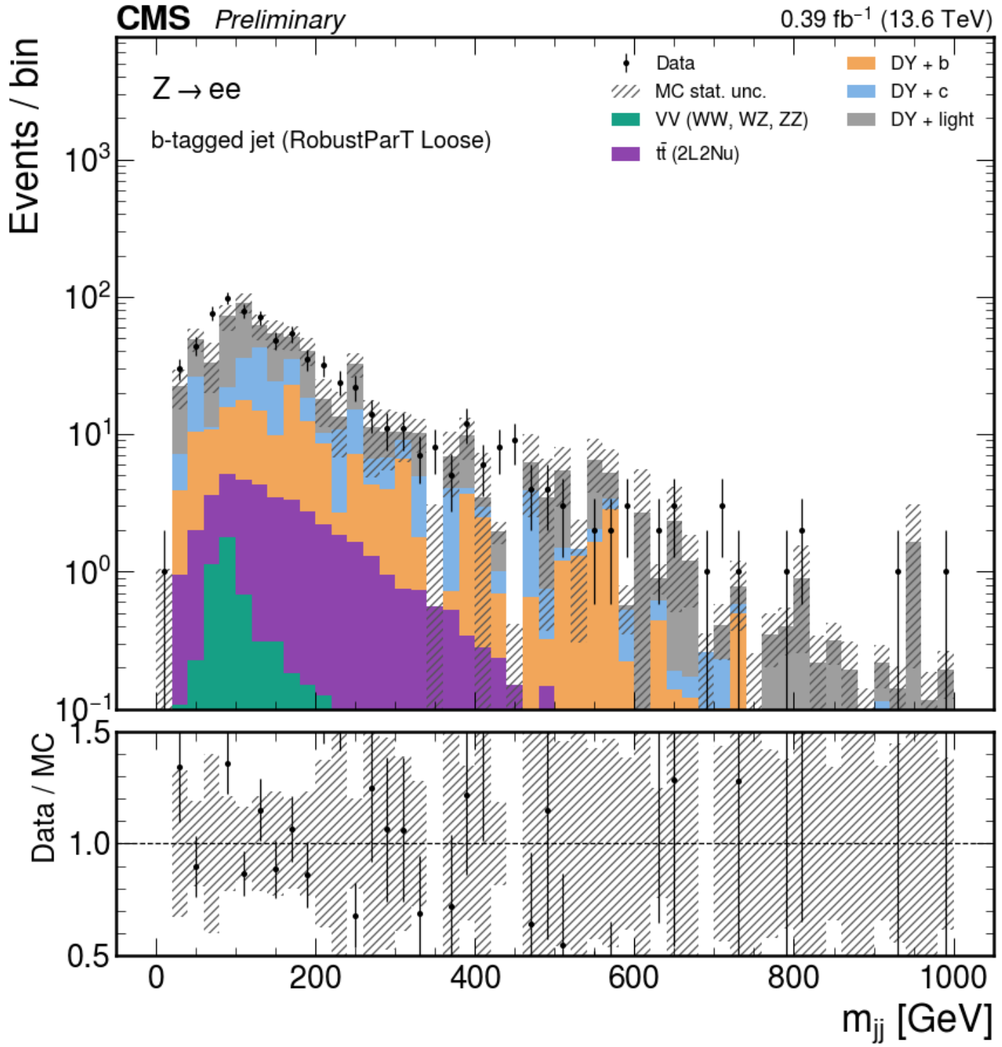


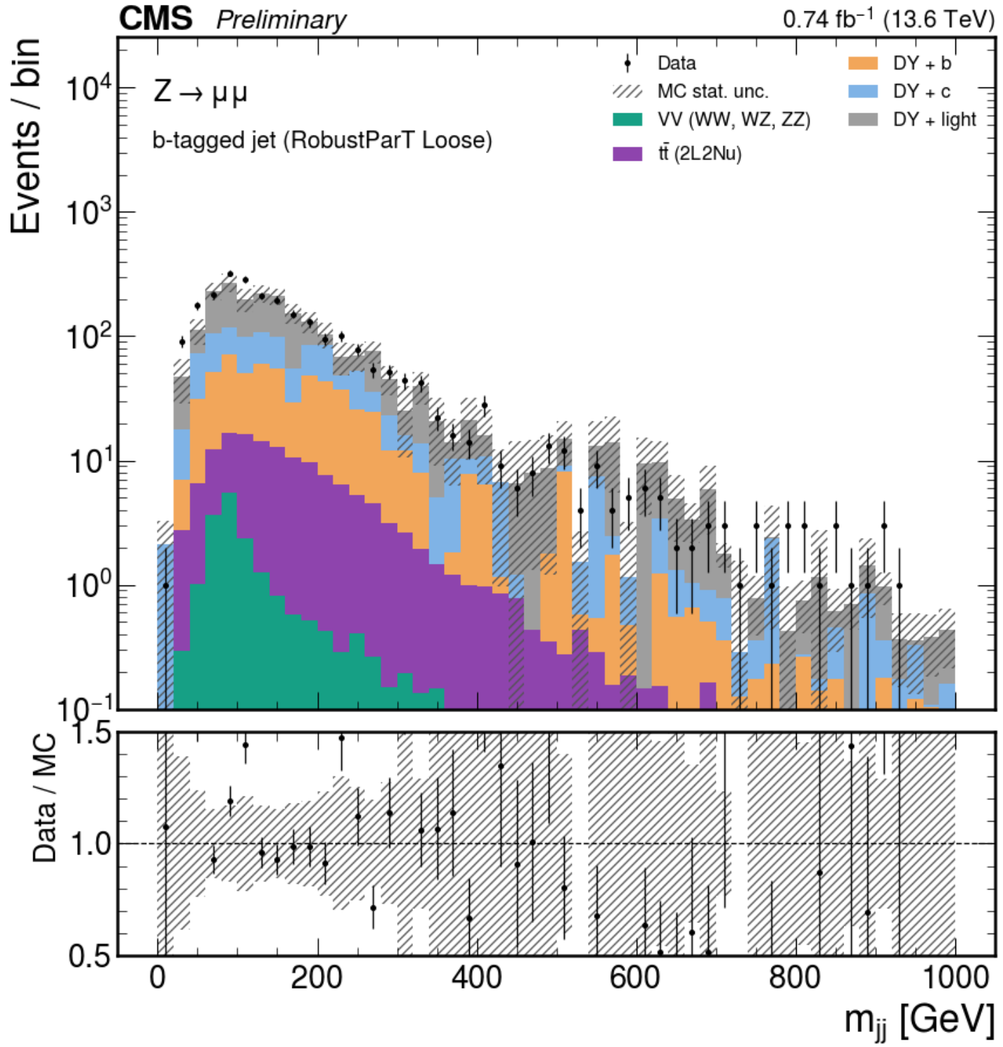


RobustParT Loose b-tag (RobustParT B > 0.0849)

Dijet mass

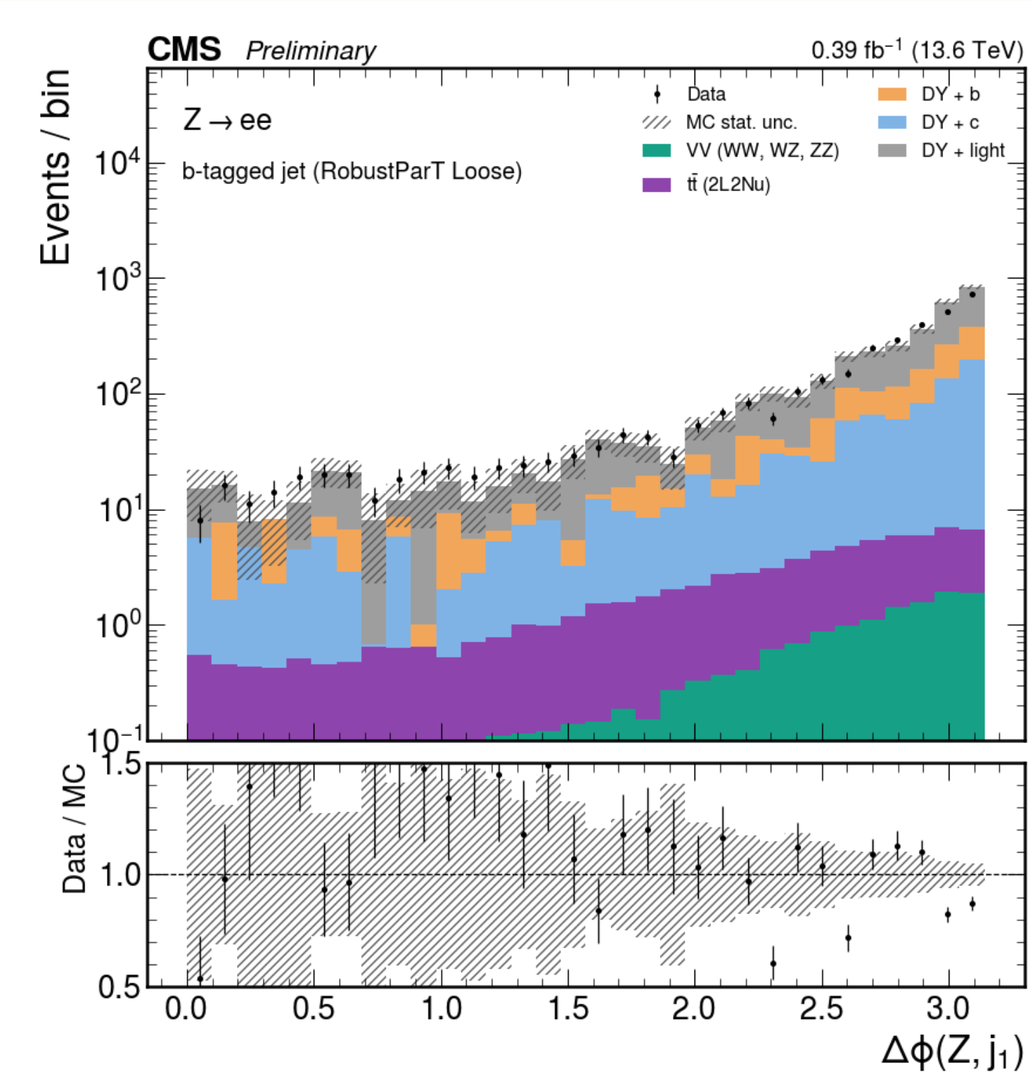
Events with at least 2 jets.

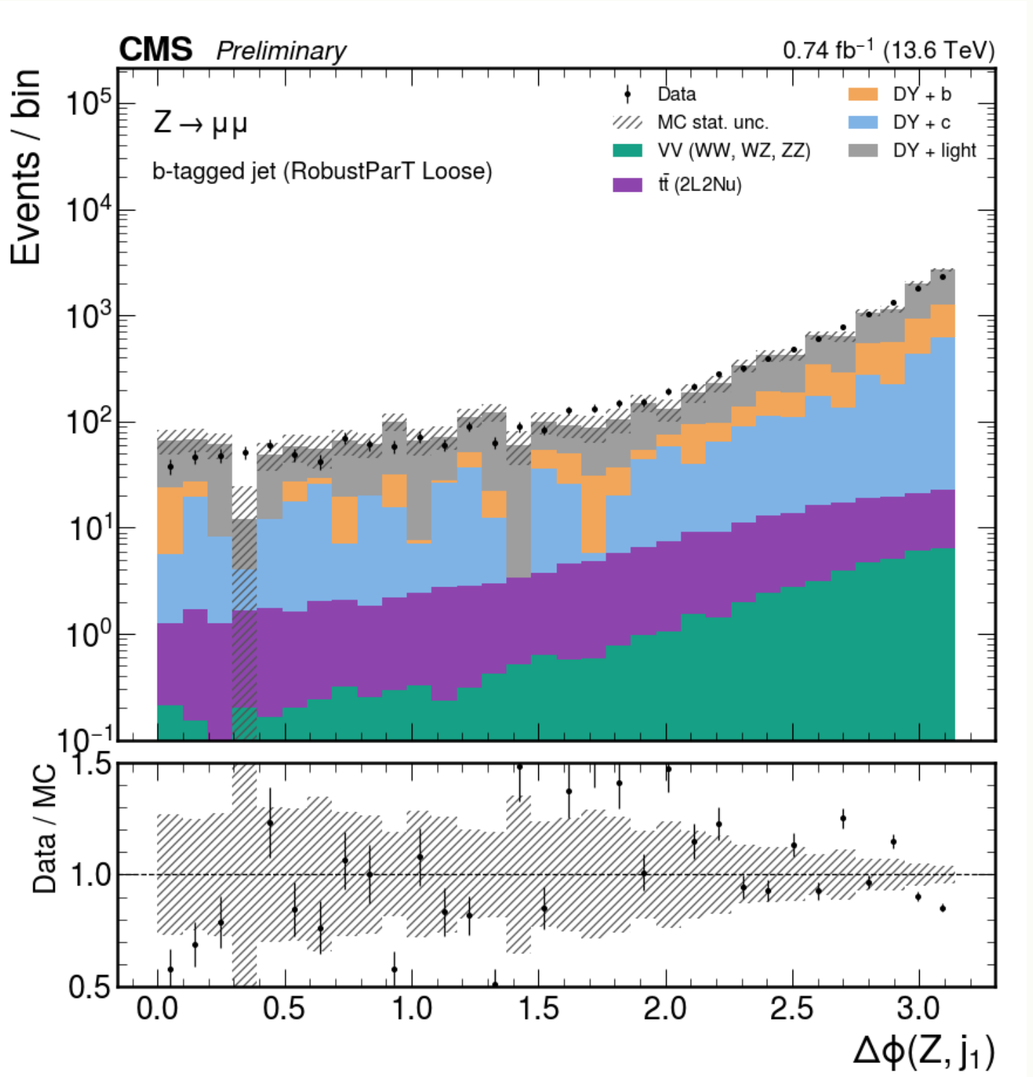




RobustParT Loose b-tag (RobustParT B > 0.0849)

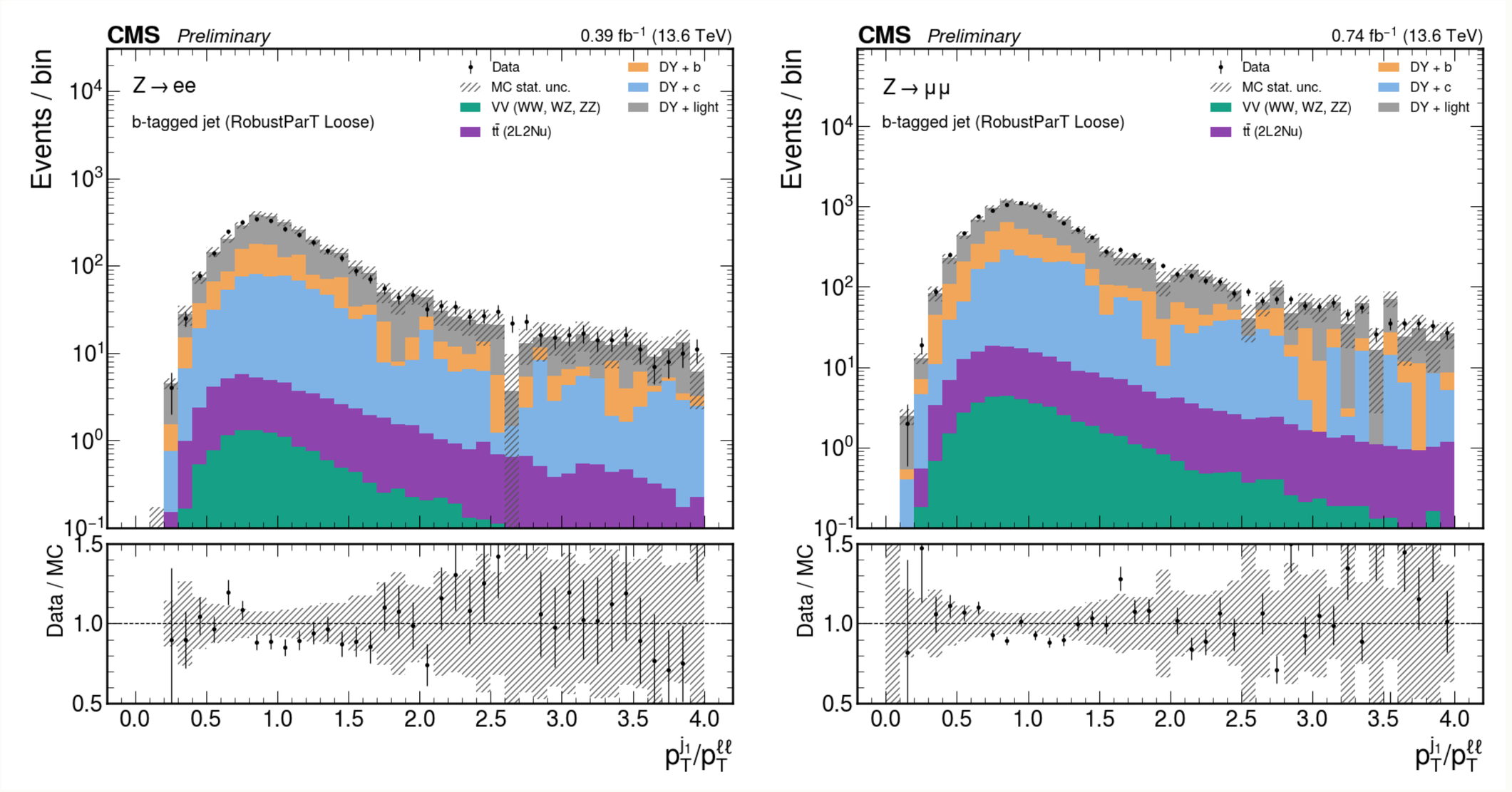
Δφ(Z, leading jet)





RobustParT Loose b-tag (RobustParT B > 0.0849)

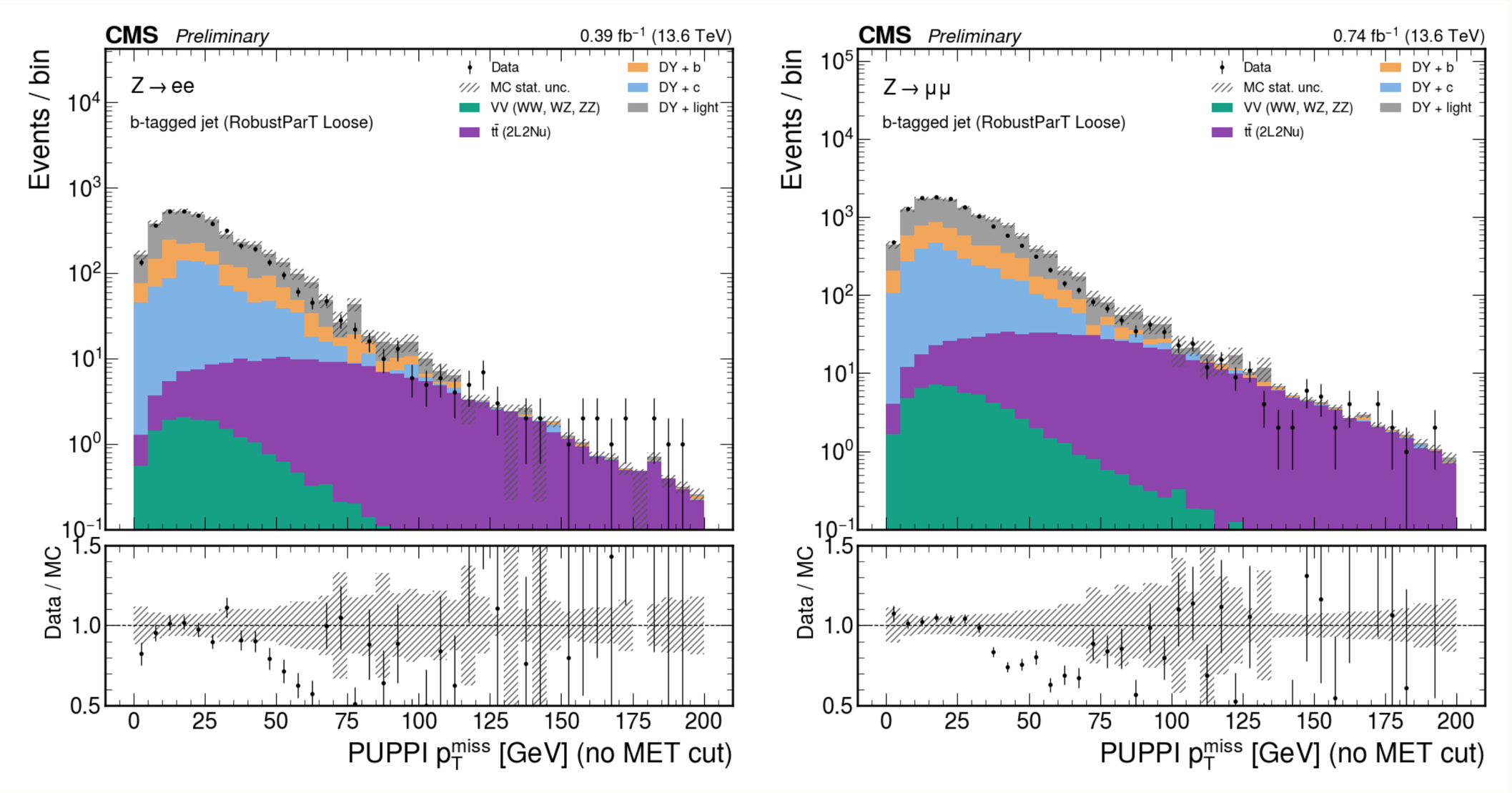
pT balance: leading-jet pT / Z pT



RobustParT Loose b-tag (RobustParT B > 0.0849)

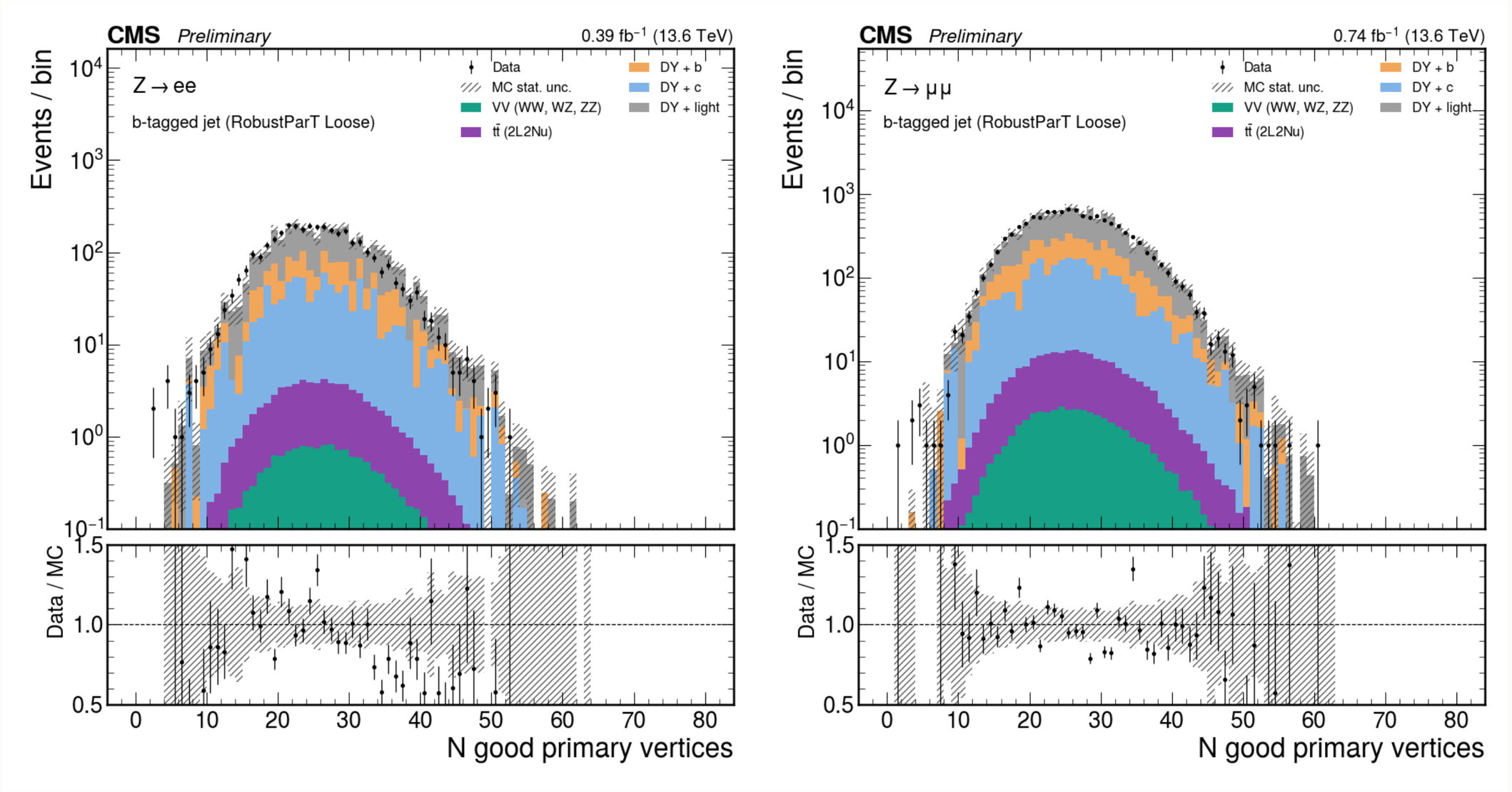
PUPPI MET without the MET cut, up to 200 GeV

All other cuts applied; the selection cuts at 50 GeV.



Number of good primary vertices

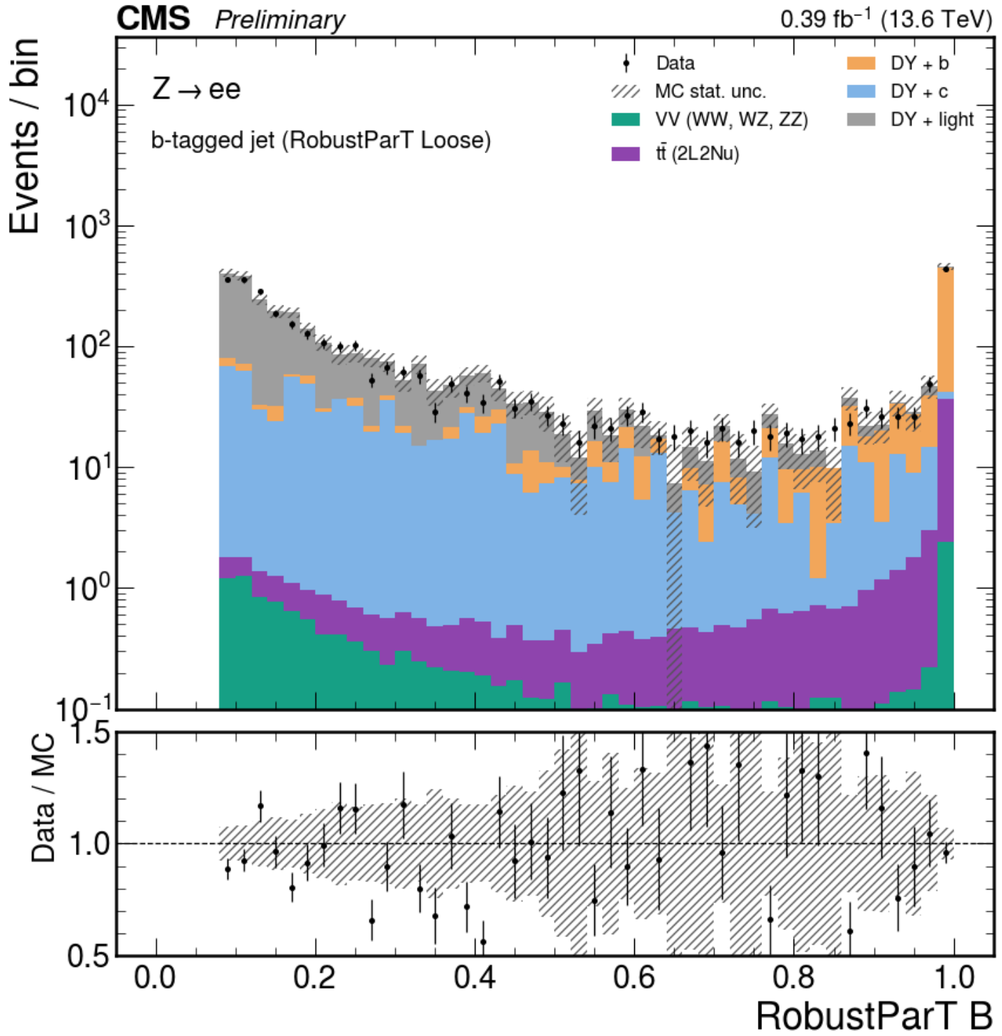
After pileup weights.

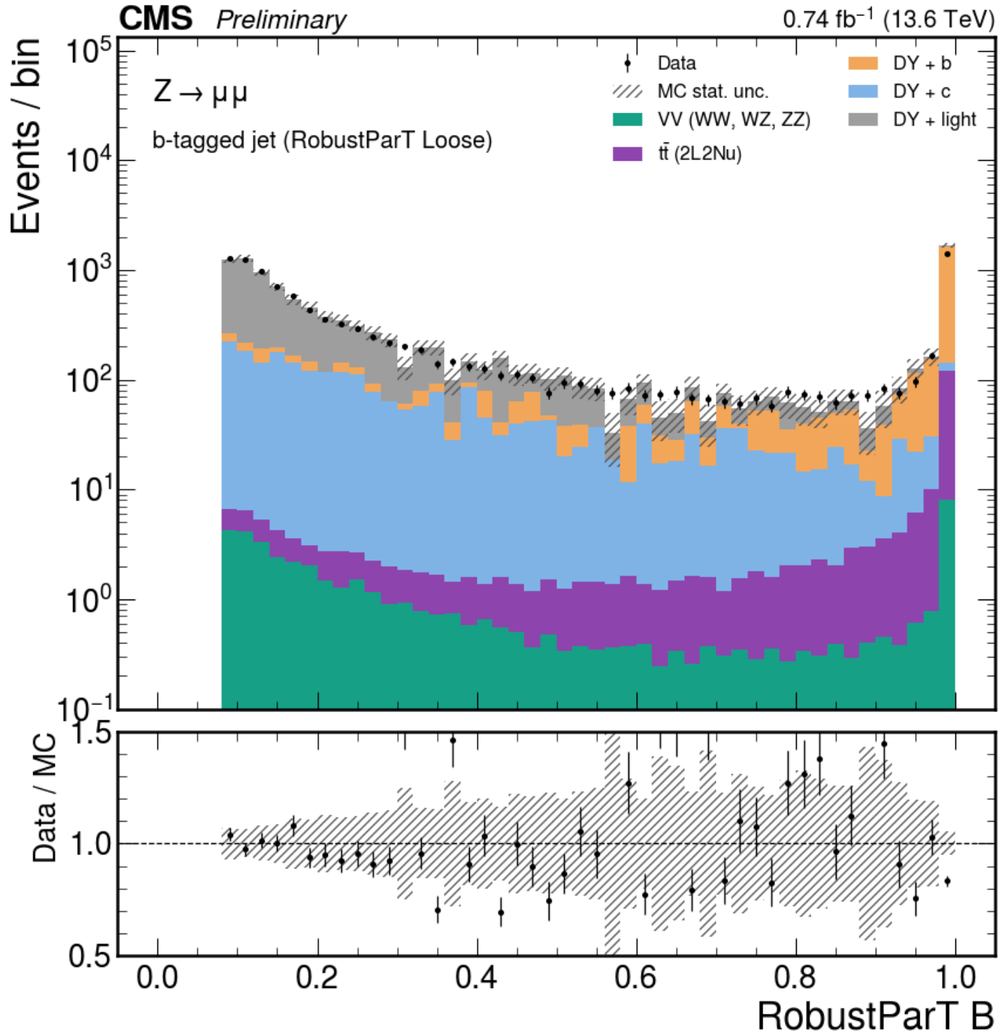


RobustParT Loose b-tag (RobustParT B > 0.0849)

Leading-jet RobustParT B score

Only jets above 0.0849 enter this region.





Summary

All three taggers select a similar sample at Loose.

DY+b share of all MC: PNet 21% / 22%, DeepJet 18% / 20%, RobustParT 20% / 22% (Z→ee / Z→μμ). About 1 in 6 Z + jet events has a Loose b-tagged leading jet; most of them are still c or light jets.

Data and MC tag the same fraction of jets within the stat. errors.

Tag ratio (data / MC): PNet 0.998 ± 0.031 / 0.980 ± 0.022; DeepJet 0.976 ± 0.029 / 0.979 ± 0.022; RobustParT 1.013 ± 0.031 / 1.040 ± 0.024 (Z→ee / Z→μμ).

Light jets are mistagged more in data than in MC (mean light SF 1.18–1.21).

The SFs take care of it; the SF uncertainties are not in these plots yet.

Not included yet

SF up/down uncertainties, the loose tau veto, the full dataset (these are 20 files per sample).